Dermatologic Agents, Topical

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The following section lists many of the active ingredients and examples of corresponding preparations used topically for their local action in veterinary medicine. The list includes both veterinary-labeled products and some useful human-labeled dermatologic products. In the marketplace, the drug sponsor, availability, and formulation of these products tend to change rapidly, so this listing should be used as a basic guide; always refer to the specific product label. Active ingredients are listed by therapeutic class. Products that are applied topically but are absorbed systemically and used primarily for their systemic effects are found in the systemic monograph section.

Portions of this section are adapted from Koch SN, Torres SMF, Plumb, DC. *Canine and Feline Dermatology Drug Handbook*. John Wiley & Sons, 2012. This reference provides additional detail on these and other compounds and products.

Anti-Inflammatory/Anti-Pruritic Agents, Topical

Corticosteroid Agents, Topical

NOTE: There are at least 20 chemical entities (plus a variety of salts) used in humans for topical corticosteroid therapy. The following section includes many veterinary topical products and some human products that may be of use in veterinary medicine. See also ***Otic Agents*** for more products.

Betamethasone, Topical

(bet-ah-***meth***-ah-zone)

For systemic use, refer to ***Betamethasone****.*

For otic use, refer to ***Corticosteroid/Antimicrobial Agents, Otic****.*

Indications/Actions

Betamethasone is considered a high-potency topical corticosteroid. It may be beneficial for adjunctive treatment of localized pruritic or inflammatory conditions. Because risks associated with betamethasone (eg, HPA axis suppression, systemic corticosteroid effects, skin atrophy) are higher than with hydrocortisone, betamethasone products are generally reserved for severe localized pruritic conditions or conditions for which hydrocortisone is not efficacious. Most veterinary-labeled topical betamethasone products also contain gentamicin, and labeled indications are for treating infected superficial lesions caused by bacteria sensitive to gentamicin. Additional otic products that contain an antimicrobial (eg, gentamicin) and an antifungal (eg, clotrimazole) are available. See ***Corticosteroid/Antimicrobial Agents, Otic***. These products can be used extra-label for treating localized mixed bacterial and yeast infections when potent anti-inflammatory activity is also desired. Topical dosage forms containing betamethasone as a sole ingredient are only available in products labeled for humans.

Glucocorticoids are nonspecific anti-inflammatory agents. They likely act by inducing phospholipase A2 inhibitory proteins (lipocortins) in cells, thereby reducing the formation, activity, and release of endogenous inflammatory mediators (eg, histamine, prostaglandins, kinins). Glucocorticoids also mitigate DNA synthesis via an antimitotic effect on epidermal cells. Topically applied corticosteroids also inhibit the migration of leukocytes and macrophages to the area, thereby reducing erythema, pruritus, and edema.

Ointments are the most potent vehicle, as they are the most occlusive, whereas creams tend to be less potent than ointments of the same medication/strength, followed by lotions and gels.[1]

Suggested Dosages/Uses

Betamethasone formulations are best suited for focal (eg, pedal) or multifocal lesions for relatively short durations (ie, less than 2 months). See ***Precautions/Adverse Effects***. Initially, topical glucocorticoids are used sparingly 1 to 4 times per day, and then the frequency is tapered when control is achieved. Long-term, frequent use can cause HPA axis suppression; risk can be reduced by treating for only as long as necessary on as small an area as possible. Refer to the individual product label for specific dosing recommendations.

Precautions/Adverse Effects

Several veterinary topical products list pregnancy as a contraindication. Systemic glucocorticoids can be teratogenic or induce parturition during the third trimester of pregnancy in animals. If considering use during pregnancy, weigh the respective risks with treating versus the potential benefits. Owners should wash their hands after application or wear gloves when applying. Avoid contact with eyes. Do not allow the animal to lick or chew at affected sites for at least 20 to 30 minutes after application. Occlusive dressings increase glucocorticoid penetration and can increase the risk for systemic side effects.

- Reactions to allergen **intradermal skin tests** may be suppressed by glucocorticoids.[2,3] A 2-week (14-day) withdrawal time before intradermal skin testing is recommended for dogs and cats[4,5]; no withdrawal time is required before serum IgE testing to identify sensitizing allergens in cats.[5]

Use care when treating large areas or when using on small-sized patients. Higher potency, increased concentration, or long duration of use increases risks for HPA axis suppression, immune suppression, systemic glucocorticoid effects (eg, PU/PD, polyphagia, vomiting, diarrhea), and complications caused by skin atrophy (eg, skin fragility, alopecia, localized pyoderma, comedones, weakened barrier function). Betamethasone may delay wound healing, primarily if used longer than 7 days. Vomiting and diarrhea have been reported after ingestion of topical products containing betamethasone. Local skin reactions (eg, burning, itching, redness) are possible but uncommon.

Veterinary-Labeled Products

NOTE: There are no veterinary-labeled topical products in the United States that contain betamethasone as a single active ingredient.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Gentamicin 0.57 mg/mL, betamethasone valerate 0.284 mg/mL (spray) Other ingredients: propylene glycol, isopropyl alcohol, parabens	Labeled for dogs	Rx; depending on the product: 60 mL, 72 mL, 120 mL, and 240 mL	*Gentocin*®; *GentaCalm*®; *GentaSpray*®; *Betagen*®; *Gentamicin*®; *Gentaved*®; *GentaSoothe*®; *GenOne*®; *Relifor*®

Human-Labeled Products

NOTE: This is a partial listing. Do not confuse products containing *augmented* betamethasone dipropionate (eg, *Diprolene*®) with betamethasone dipropionate. Augmented betamethasone dipropionate is more potent than betamethasone dipropionate. For more information on human-labeled betamethasone products, refer to a comprehensive human drug reference or contact a pharmacist.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Betamethasone dipropionate 0.05% (ointment)	Rx; 15 g, 45 g	generic
Betamethasone dipropionate 0.05% (cream)	Rx; 15 g, 45 g	generic
Betamethasone dipropionate 0.05% (lotion)	Rx; 60 mL	generic
Betamethasone valerate 0.1% (ointment)	Rx; 30 g, 45 g	generic
Betamethasone valerate 0.1% (cream)	Rx; 15 g	generic
Betamethasone valerate 0.1% (lotion)	Rx; 60 mL	generic
Betamethasone valerate 0.12% (foam)	Rx; 300 g	*Luxic*®; generic
Clotrimazole 1%, betamethasone dipropionate 0.05% (lotion)	Rx; 30 mL	generic
Clotrimazole 1%, betamethasone dipropionate 0.05% (cream)	Rx; 15 g, 45 g	*Lotrisone*®

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Hydrocortisone, Topical

(hye-droe-***kor***-ti-zone)

Indications/Actions

Hydrocortisone is a topical corticosteroid that may be useful for adjunctive treatment of localized pruritic and/or inflammatory conditions. Due to its low potency, hydrocortisone is a reasonable first choice for larger focal areas or for small-sized patients. Some products contain additional ingredients (eg, Burow's solution [aluminum acetate], pramoxine) that may have additional antipruritic effects.

Topical glucocorticoids are nonspecific anti-inflammatory agents. They are thought to act by inducing phospholipase A_2 inhibitory proteins (lipocortins) in cells, thereby reducing the formation, activity, and release of endogenous inflammatory mediators (eg, histamine, prostaglandins, kinins). Glucocorticoids also reduce DNA synthesis via an antimitotic effect on epidermal cells. Topically applied glucocorticoids also inhibit the migration of leukocytes and macrophages to the area, reducing erythema, pruritus, and edema.

Suggested Dosages/Uses

Hydrocortisone formulations are best suited for focal (eg, pedal) or multifocal lesions for relatively short durations (ie, less than 2 months; see ***Precautions/Adverse Effects***).

Initially, topical glucocorticoid creams and ointments are applied 1 to 4 times daily, and then the frequency is tapered after control is achieved. In contrast to some more potent topical corticosteroids, hydrocortisone can be applied more frequently and over a greater surface area, with less risk for local or systemic adverse effects; however, long-term frequent use can still cause HPA axis suppression. Risk can be reduced by treating patients for only as long as necessary on the smallest area possible.

Shampoos containing hydrocortisone are generally used from once a week up to once a day. They typically should remain in contact with the skin for at least 10 minutes before rinsing.

Precautions/Adverse Effects

Several veterinary topical products list pregnancy as a contraindication. Owners should wash their hands after application or wear gloves when applying. Avoid contact with eyes. Do not allow the animal to lick or chew at affected sites for at least 20 to 30 minutes after application. Occlusive dressings increase glucocorticoid penetration and can increase the risk for systemic adverse effects. The risks associated with hydrocortisone are significantly less when compared with higher potency corticosteroids (eg, betamethasone).

Local skin reactions are possible but uncommon. Atrophy associated with skin fragility, superficial follicular cysts (milia), and comedones may be seen with long-term frequent use, but these conditions occur more commonly when using more potent topical corticosteroids (eg, betamethasone). Although systemic absorption is rare with hydrocortisone, long-term use may lead to HPA axis suppression. Accidental ingestion may cause GI signs such as nausea, vomiting, or diarrhea.[1]

Reactions to allergen intradermal skin tests may be suppressed by glucocorticoids.[2,3] A 2-week (14-day) withdrawal time before intradermal skin testing is recommended for dogs and cats[4,5]; no withdrawal time is required before serum IgE testing to identify sensitizing allergens in cats.[5]

Veterinary-Labeled Products

Refer to the individual product labeling for specific dosing recommendations and inactive ingredients.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Hydrocortisone 0.5% (lotion)	Labeled for dogs	OTC; 4 oz	*Dogswell®*
Hydrocortisone 1% (lotion)	Labeled for dogs, cats, and horses	OTC; 8 oz	*Vetergen Hydrocortisone; AtopiCream HC*
Hydrocortisone 0.5% (cream)	Labeled for dogs and cats	OTC; 1 oz	*Zymox®*
Hydrocortisone 1% (cream)	Labeled for dogs and cats	OTC; 1 oz	*Zymox®*
Hydrocortisone 0.5% (spray)	Labeled for dogs	OTC; 5 oz	*Hartz®*
Hydrocortisone 1% (spray)	Labeled for dogs and cats	OTC; 2 oz	*Zymox®*
Hydrocortisone 1%, chlorhexidine 4%, (spray)	Labeled for dogs, cats, and horses	OTC; 8 oz	*TrizCHLOR® 4HC*
Hydrocortisone 1%, pramoxine 1% (spray)	Labeled for dogs and cats	OTC; 4 oz, 8 oz	*Comfort HC®*; *Vetraseb®* HC
Hydrocortisone 1%, acetic acid 1%, boric acid 2%, ketoconazole 0.15% (spray)	Labeled for dogs, cats, and horses	OTC; 8 oz	*MalAcetic® ULTRA*
Hydrocortisone 1%, Burow's solution 2% (solution)	Labeled for dogs and/or cats	OTC or Rx; 1 oz, 2 oz, and 16 oz bottle	*Cort/Astrin®; Corti-Derm®; Bur-O-Cort 2:1®; Hydro-B 1020®; Hydro-Plus®*
Hydrocortisone 0.5%, chlorhexidine 2%, ketoconazole 1% (salve)	Labeled for dogs, cats, and horses	OTC; 4 oz	*EquiShield® CK HC*
Hydrocortisone 1%, chlorhexidine 4% (shampoo)	Labeled for dogs, cats, and/or horses	OTC; 8 oz, 16 oz	*TrizCHLOR 4HC; Chlorhexidine 4% HC*
Hydrocortisone 1%, chlorhexidine 2%, ketoconazole 1% (shampoo)	Labeled for dogs, cats, and horses	OTC; 12 oz	*EquiShield® CK HC*
Hydrocortisone 1%, acetic acid 1%, boric acid 2%, ketoconazole 0.15% (shampoo)	Labeled for dogs and cats	OTC; 8 oz	*MalAcetic® ULTRA*
Hydrocortisone 1%, acetic acid 1%, boric acid 1% (wipes)	Labeled for dogs and cats	OTC; 25 count jar	*MalAcetic® HC*

Human-Labeled Products

NOTE: Partial listing; there are many branded products available with hydrocortisone. For more information on human-labeled hydrocortisone products, refer to a comprehensive human drug reference (eg, *Facts and Comparisons*) or contact a pharmacist.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Hydrocortisone 0.5%, hydrocortisone 1% (ointment)	OTC/Rx; 25 g, 28 g, 28.35 g, 28.4 g, 30 g, 56 g, 110 g, 430 g, and 454 g	Status determined by labeling
Hydrocortisone 2.5% (ointment)	Rx; 20 g, 28.35 g, 435.6 g, and 454 g	
Hydrocortisone 0.5%, hydrocortisone 1% (cream)	OTC/Rx; 1 g, 1.5 g packets, 14 g, 14.2 g, 15 g, 20 g, 26 g, 28 g, 28.3 g, 28.35 g, 28.4 g, 30 g, 42 g, 56 g, 120 g, and 454 g	Status determined by labeling
Hydrocortisone 2.5% (cream)	Rx; 15 g, 20 g, 28 g, 28.35 g, 30 g, and 454 g	
Hydrocortisone 1% (lotion)	OTC/Rx; 30 mL, 59 mL, 60 mL, 88.7 mL, 99 mL, 113 mL, 114 mL, 118 mL, and 120 mL	Status determined by labeling
Hydrocortisone 2%, hydrocortisone 2.5% (lotion)	Rx; 29.6 mL (2% only), 59 mL and 118 mL (2.5% only)	*Ala Scalp®; Scalacort®;* generic
Hydrocortisone 1% (gel)	OTC; 28 g, 30 g, 42.53 g, and 56 g	
Hydrocortisone 2.5% (gel)	Rx; 43 g	
Hydrocortisone 10% (gel compounding kit)	Rx; 60 g jar	*First-Hydrocortisone®*
Hydrocortisone 1%, hydrocortisone 2.5% (solution)	OTC/Rx; 30 mL, 44 mL, 59 mL, and 120 mL	
Hydrocortisone 1% (foam)	OTC; 88.7 mL	*Itch-X®*
Hydrocortisone 1% (applicator)	OTC; 36 mL	*Cortizone-10 Easy Relief®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Isoflupredone Combinations, Topical

(eye-soe-***floo***-preh-dohn)

Indications/Actions

Isoflupredone is a high-potency topical corticosteroid. In combination with neomycin and tetracaine, isoflupredone is indicated in dogs as a dressing for minor cuts and abrasions, as a postsurgical therapy, and for treating anal gland infections and moist dermatitis.[1] It is indicated in cats and horses as a dressing for minor cuts and abrasions and as postsurgical therapy.[1,2]

Because risks associated with isoflupredone (eg, HPA axis suppression, systemic corticosteroid effects, skin atrophy) are higher than with hydrocortisone, these products are generally reserved for severe localized pruritic conditions or for conditions when hydrocortisone is not efficacious.

Corticosteroids are nonspecific anti-inflammatory agents. They are thought to act by inducing phospholipase A2 inhibitory proteins (lipocortins) in cells, thereby reducing the formation, activity, and release of endogenous inflammatory mediators (eg, histamine, prostaglandins, kinins). Corticosteroids also mitigate DNA synthesis via an anti-mitotic effect on epidermal cells. Topically applied corticosteroids also inhibit the migration of leukocytes and macrophages to the area, reducing erythema, pruritus, and edema.

Suggested Dosages/Uses

Clean the affected area prior to application of isoflupredone. Apply a small amount of ointment or powder 1 to 3 times daily as needed and continue based on clinical response.[1,2]

Precautions/Adverse Effects

Topical corticosteroids are often considered contraindicated in patients that are pregnant or have tuberculosis of the skin. Systemic corticosteroids can be teratogenic or induce parturition during the third trimester of pregnancy in animals. If isoflupredone is being considered for use during pregnancy, weigh the risks for treating versus the potential benefits.

Residual activity may affect intradermal or allergy serum tests; it is recommended to discontinue use of isoflupredone 2 weeks before allergy testing.[3] Isoflupredone may delay wound healing, particularly when used longer than 7 days.

Isoflupredone products may be harmful if swallowed; prevent treated animals from licking or chewing at affected sites for at least 20 to 30 minutes after application. Powdered formulations may cause allergy or asthma symptoms if inhaled.[2]

Wear gloves or wash hands after application. Avoid contact with eyes.

Use isoflupredone cautiously on large treatment areas and on small-sized patients. Treat the smallest area possible for the minimum duration. The prolonged use of corticosteroid products increases the risk for HPA axis suppression, systemic corticosteroid effects (eg, polydipsia, polyuria, iatrogenic hyperadrenocorticism, skin atrophy). Local skin reactions (eg, burning, itching, redness) or hypersensitivity reactions are possible.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Isoflupredone acetate 0.1%, neomycin sulfate 0.5%, tetracaine HCl 0.5% (ointment)	Labeled for dogs, cats, and horses	Rx; 10 g tube	*Neo-Predef® with Tetracaine; Tritop®*
Isoflupredone acetate 0.1%, neomycin sulfate 0.5%, tetracaine HCl 0.5% (powder)	Labeled for dogs, cats, and horses	Rx; 15 g insufflator bottle	*Neo-Predef® with Tetracaine* Store in a dry place. Do not allow the tip of the bottle to contact moisture.

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Mometasone, Topical

(moe-***met***-a-zone)

For otic use, see ***Corticosteroid/Antimicrobial Agents, Otic****.*

Indications

Mometasone is a high-potency topical corticosteroid. It may be beneficial for adjunctive treatment of pruritic and inflammatory conditions. Mometasone products are generally reserved for severe pruritic conditions or for hydrocortisone treatment failures, as the risks associated with mometasone (eg, HPA axis suppression, skin atrophy) are higher than with hydrocortisone.

Mometasone is found in several veterinary-labeled otic suspensions in combination with antibiotics and antifungals. These otic combination products have been used topically extra-label for yeast and bacterial skin infections when a potent anti-inflammatory effect is needed but should only be used when there is evidence of bacterial or yeast infections.

Corticosteroids are non-specific anti-inflammatory agents. They are thought to induce phospholipase A2 inhibitory proteins (lipocortins) in cells, thereby reducing the formation, activity, and release of endogenous inflammatory mediators (eg, histamine, prostaglandins, kinins). Corticosteroids also reduce DNA synthesis via an anti-mitotic effect on epidermal cells. Topically applied corticosteroids also inhibit the migration of leukocytes and macrophages to the area, reducing erythema, pruritus, and edema.

Suggested Dosages/Uses

Mometasone should be applied sparingly, initially 1 to 2 times daily then less frequent as clinical signs improve. Mometasone formulations are best suited for focal (eg, pedal) or multifocal lesions that require relatively short treatment durations (ie, less than 2 months). Frequency of application and duration of treatment should be based on the severity of clinical signs. For veterinary-labeled products, refer to the specific product label for additional usage information.

Precautions/Adverse Effects

Safe use of mometasone during pregnancy is not established; it should only be used when the benefits outweigh the risks. Use cautiously on large treatment areas or in small-sized patients due to risk for systemic absorption. Treat only for as long as necessary on as small of an area as possible to reduce the risk for toxicities such as HPA axis suppression, systemic corticosteroid effects (eg, polydipsia, polyuria, iatrogenic Cushing's, adverse GI effects), and cutaneous atrophy associated with skin fragility. Other potential adverse effects that occur with increases in product concentrations and prolonged duration of use include alopecia, localized pyoderma, superficial follicular cysts (milia), and comedones. Local skin reactions (eg, burning, pruritus, erythema) are possible but unlikely. Mometasone may delay wound healing, particularly if used longer than 7 days. Do not let treated animals lick or chew at affected areas for at least 20 to 30 minutes after application; ingestion may result in GI upset. Occlusive dressings increase steroid penetration and can increase the risk for systemic adverse effects.

Reactions to allergen intradermal skin tests may be suppressed by glucocorticoids. A 2-week (14-day) withdrawal time before intradermal skin testing is recommended for dogs and cats[1]; no withdrawal time is required before serum IgE testing to identify sensitizing allergens in cats.[2]

Wash hands after handling or wear gloves during application. Avoid contact with eyes.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Mometasone 1 mg/g, gentamicin 3 mg/g, clotrimazole 10 mg/g (suspension)	Labeled for otic use in dogs	Rx; 7.5 g, 15 g, 30 g, and 215 g tubes	*Mometamax®; Malmetazone®; Mometavet®; Maxi-Otic®; MotaZol®* Topical use is extra-label.
Mometasone 1 mg/g, orbifloxacin 10 mg/g, posaconazole 1 mg/g (suspension)	Labeled for otic use in dogs	Rx; 7.5 g, 15 g, and 30 g tubes	*Posatex®* Topical use is extra-label.

Human-Labeled Products

NOTES:

1. Mometasone furoate is the salt generally used in pharmaceutical products.
2. Many commercial products exist. Below is a representative list of available products. For more information on human-labeled mometasone products, refer to a comprehensive human drug reference or contact a pharmacist.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Mometasone furoate 0.1% (cream)	Rx; 15 g and 45 g	generic
Mometasone furoate 0.1% (ointment)	Rx; 15 g and 45 g	generic
Mometasone furoate 0.1% (lotion)	Rx; 30 mL and 60 mL	*Elocon®*; generic
Mometasone furoate 0.1% (solution)	Rx; 30 mL and 60 mL	generic
Mometasone furoate 50 µg/100 mg (nasal spray)	OTC; 10 mL and 17 mL bottle	*Nasonex®* Extra-label when used topically

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Triamcinolone, Topical

(trye-am-***sin***-oh-lone)

*For systemic use, see **Triamcinolone.***

*For otic use, see **Corticosteroid/Antimicrobial Agents, Otic**.*

Indications/Actions

Triamcinolone is generally considered a medium potency corticosteroid at concentrations less than 0.5%. It may be beneficial for adjunctive treatment of pruritic or inflammatory conditions. Triamcinolone products are typically reserved for severe pruritic conditions or for conditions when hydrocortisone is not efficacious, as the risk for HPA axis suppression and skin atrophy are higher than with hydrocortisone.

Triamcinolone can be found as a sole agent in some veterinary creams or sprays, although most of these contain an antibiotic and/or antifungal agent. These products can be used for treating localized mixed bacterial and yeast infections when potent anti-inflammatory activity is also desired. Additional otic products that contain an antibiotic and an antifungal are available. See ***Corticosteroid/Antimicrobial Agents, Otic***.

Corticosteroids are non-specific anti-inflammatory agents. They are thought to induce phospholipase A2 inhibitory proteins (lipocortins) in cells, thereby reducing the formation, activity, and release of endogenous inflammatory mediators (eg, histamine, prostaglandins, kinins). Corticosteroids also mitigate DNA synthesis via an anti-mitotic effect on epidermal cells. Topically applied corticosteroids also inhibit the migration of leukocytes and macrophages to the area, reducing erythema, pruritus, and edema.

Suggested Dosages/Uses

Topical triamcinolone is best suited for focal (eg, pedal) or multifocal lesions for relatively short durations (ie, less than 2 months). Triamcinolone ointment should be applied sparingly, initially 1 to 4 times daily and then tapered when control is achieved. Triamcinolone spray is labeled for use twice daily for 7 days, then once daily for 7 days, then every other day for 14 days. For veterinary products, refer to the specific product label for dosing recommendations.

Precautions/Adverse Effects

Systemic corticosteroids can be teratogenic or induce parturition during the third trimester of pregnancy in animals. If considering use during pregnancy, weigh the risks of treating versus potential benefits. Wear gloves when applying and wash hands after application. Spray should be applied in a well-ventilated area. Avoid contact with eyes. Do not allow animals to lick or chew at affected sites for at least 20 to 30 minutes after application; oral ingestion may result in GI upset.

Reactions to allergen intradermal skin tests may be suppressed by glucocorticoids. A 2-week (14-day) withdrawal time before intradermal skin testing is recommended for dogs and cats[1]; no withdrawal time is required before serum IgE testing to identify sensitizing allergens in cats.[2]

Long-term, frequent use can cause HPA axis suppression. Treat for only as long as necessary on as small of an area as possible using the lowest effective concentration to reduce the risk for HPA axis suppression, cutaneous effects (eg, skin atrophy, skin fragility, alopecia, localized pyoderma, comedones), and other systemic corticosteroid effects (eg, polydipsia, polyuria, iatrogenic Cushing's, GI effects). Use with caution when treating small-sized patients or large areas. Triamcinolone may delay wound healing, mainly if used longer than 7 days. Local skin reactions (eg, burning, itching, redness) are also possible.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Triamcinolone acetonide 0.015% (spray)	Labeled for dogs	Rx; 8 oz, 16 oz bottle	*Genesis®*
Triamcinolone acetonide 0.1% (cream)	Labeled for dogs	Rx; 7.5 g, 15 g	*Medalone®*; *Vetalog®* Although these products are still listed in the FDA Green Book of approved animal drugs, they may no longer be commercially available.
Triamcinolone acetonide 1 mg/g, neomycin 2.5 mg/g, thiostrepton 2500 units/g, nystatin 100,000 units/g (cream)	Labeled for dogs and cats	Rx; 7.5 g, 15 g tube	*Animax®; Derma-Vet®; Panalog®* Although these products are still listed in the FDA Green Book of approved animal drugs, they may no longer be commercially available.
Triamcinolone acetonide 1 mg/mL, neomycin 2.5 mg/mL, thiostrepton 2500 units/mL, nystatin 100,000 units/mL, (ointment)	Labeled for dogs and cats	Rx; 7.5 mL, 15 mL, 30 mL, and 240 mL	*Animax®; Derma-Vet®; Dermalog®; Dermalone®; EnteDerm®; Quadritop®; Quadruple®; Resortin®; VetaDerm Tetrafect®*

Human-Labeled Products

NOTE: For more information on human-labeled triamcinolone products, refer to a comprehensive human drug reference or contact a pharmacist.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Triamcinolone acetonide 0.025% (ointment)	Rx; 15 g, 80 g, 454 g	generic
Triamcinolone acetonide 0.05% (ointment)	Rx; 430 g	*Trianex®*; generic
Triamcinolone acetonide 0.1% (ointment)	Rx; 15 g, 80 g, 454 g	generic
Triamcinolone acetonide 0.5% (ointment)	Rx; 15 g	generic
Triamcinolone acetonide 0.025% (cream)	Rx; 15 g, 85.2 g, 454 g	*Triderm®*; generic
Triamcinolone acetonide 0.1% (cream)	Rx; 28.4 g, 85.2 g	*Triderm®*; generic
Triamcinolone acetonide 0.5% (cream)	Rx; 15 g, 454 g	*Triderm®*; generic
Triamcinolone acetonide 0.025% (lotion)	Rx; 60 mL	generic
Triamcinolone acetonide 0.1% (lotion)	Rx; 60 mL	generic
Triamcinolone acetonide 0.147 mg/g (spray)	Rx; 63 g, 100 g	*Kenalog®*; generic 10.3% alcohol
Triamcinolone acetonide 0.1%, nystatin 100,000 units/g (cream)	Rx; 15 g, 30 g, 60 g	generic
Triamcinolone acetonide 0.1%, nystatin 100,000 units/g (ointment)	Rx; 15 g, 30 g, 60 g	generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Non-Corticosteroid Agents, Topical

Aluminum Acetate Solution, Topical
Burow's Solution

(ah-***loo***-mi-num ***ass***-ih-tate)

Astringent

For otic use, see ***Corticosteroid Agents, Otic.***

Indications/Actions

Aluminum acetate solution (Burow's solution, modified Burow's solution) is an astringent antipruritic agent that can be used for adjunctive treatment of minor skin irritations, such as insect bites and localized inflamed and exudative skin conditions (eg, acute moist dermatitis, fold dermatitis [intertrigo], contact dermatitis). This solution can also be used to reduce inflammation associated with otitis externa (see ***Corticosteroid Agents, Otic***). In addition to astringent and antipruritic actions, aluminum acetate solution also has acidifying effects and it is mildly antiseptic. The exact mechanisms of action for each of these effects is not fully understood; it was shown to have an antibacterial effect against Staphylococcus aureus and Pseudomonas aeruginosa through alterations and damage to the cell wall and bacteriolysis.[1]

Suggested Dosages/Uses

Topical use of aluminum acetate as a single agent is usually via a wet compress, dressing, or soak. Application for 15 to 30 minutes is generally recommended, and the affected area is air-dried between applications. Use can be as often as necessary, but every 4 to 6 hours is often employed. As aluminum acetate solution products come in various dosage forms (powder or tablets for dissolving, liquid), refer to package directions for proper dilutions; dilutions of 1:40, 1:20, or 1:10 are commonly used. Various generic products exist. **NOTE**: The commercially prepared, veterinary-labeled products that also contain hydrocortisone may be directly applied without dilution. See ***Hydrocortisone, Topical***.

Precautions/Adverse Effects

Do not cover application site with plastic or occlusive dressing to try to prevent product evaporation. Use cool or warm[2] water for dissolving powder/tablets. Avoid contact with eyes. Pet owners should wash their hands after application. Aluminum acetate solution may cause dry skin or skin irritation in some patients. For products containing hydrocortisone, prolonged use should be avoided or limited to 1 to 2 weeks at a time to prevent potential skin atrophy and systemic absorption. A minimum of a 2-week withdrawal period is recommended for these products before intradermal or allergy serum testing. See ***Hydrocortisone, Topical***.

Veterinary-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Aluminum acetate 2%, hydrocortisone 1% (solution) Other ingredients: water, propylene glycol	OTC; 1 oz, 2 oz, 16 oz	*Cort/Astrin®; Corti-Derm®; Hydro-B 1020*

Human-Labeled Products

NOTE: In some products, aluminum acetate solution may be listed as aluminum sulfate and calcium acetate combined in water.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Aluminum sulfate tetradecahydrate, 1347 mg; calcium acetate monohydrate, 952 mg (powder)	OTC; packets of various concentrations (12/box)	*Domeboro®* Dilute as directed. When combined with water, the active ingredient (aluminum acetate) is formed.
Aluminum acetate 0.5% (gel)	OTC; 3 oz	*Domeboro®*
Aluminum acetate 0.2% (gel) Other ingredients: dehydroacetic acid, ethylhexylglycerin, glycine, hexylene glycol, malic acid	OTC; 2 oz	*TriCalm®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Camphor/Menthol/Phenol, Topical

(***kam***-for; ***men***-thol; ***fee***-nol)

Indications/Actions

When used in low concentrations, camphor/menthol/phenol products can be used as counterirritants; they may be added to proprietary or compounded products, primarily as antipruritics. Camphor and phenol may also have some antiseptic properties. Camphor and menthol act by stimulating and desensitizing nociceptors.[1] Camphor produces a warming sensation, whereas menthol produces a cooling effect.[2]

Precautions/Adverse Effects

These compounds may cause local irritation and should not be used around or in the eyes. Products containing phenol should NOT be used on cats.

Veterinary-Labeled Products

NOTE: There are also several over-the-counter (OTC) products containing menthol, phenol, or camphor; these products are not listed below and are used primarily on equine patients for overexertion, soreness, or stiffness. These include a variety of liniments (eg, white liniment, Choate's liniment) or gels (eg, *Cool Gel*®, *Ice-O-Gel*®, *Shin-O-Gel*®).

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Menthol 7.5 mg/mL, oil of camphor 7.5 mg/mL, oil of eucalyptus 7.5 mg/mL, oil of pine 7.5 mg/mL, oil of thyme 2.8 mg/mL, Peru balsam 1.5 mg/mL, phenol 7.5 mg/mL, biebrich scarlet red 100 ppm (spray)	Labeled for superficial cuts, wounds, and burns for horses and mules	OTC; 500 mL	Many trade names Shake well

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Colloidal Oatmeal, Topical

(ko-***loyd***-al ***ote***-meel)

Indications/Actions

Colloidal oatmeal is made from oat grains (*Avena sativa*) and is used topically as an anti-inflammatory, antipruritic, and emollient agent. It ameliorates and provides soothing effects in pruritic, inflamed, and dry and/or scaly skin disorders, such as atopic dermatitis and irritant skin reactions. Although colloidal oatmeal's mechanism of action is not completely understood, avenanthramides, one of its antioxidant phenolic compounds, have been shown to exert anti-inflammatory effects in the skin by inhibiting activities of pro-inflammatory cytokines in human epidermal keratinocytes.[1] Avenanthramides were also shown to inhibit histamine release, prostaglandin biosynthesis, and cleavage of arachidonic acid from phospholipids in keratinocytes. In animal models, avenanthramides also reduced scratching and inflammatory cytokines implicated in pruritus; the antipruritic effect appears to increase as the oatmeal concentration increases. Colloidal oatmeal also forms a hydrophilic film that retains and replenishes moisture in the stratum corneum, and it also helps to normalize the skin's pH. Moreover, colloidal oatmeal was found to improve the epidermal skin barrier by inducing and upregulating the expression of important skin barrier markers in human equivalent keratinocytes.

Suggested Dosages/Uses

Spray formulations can be used 2 to 3 times per day or as needed for itching or pain. Shampoo or conditioner is usually used once a day to once a week. Shampoo should be in contact with the skin for at least 5 to 10 minutes and then rinsed well. Refer to each product's label for further details.

Precautions/Adverse Effects

Colloidal oatmeal has been known to be very safe. In humans, there are some reports of contact dermatitis associated with its use; however, most data indicate a low potential for oatmeal-containing products to cause irritation or allergic sensitization. Some animals might develop sensitivity (eg, pruritus, erythema) to some of these topical agents, in which case the product should be discontinued. Because colloidal oatmeal has a potential antihistamine effect, a minimum of a 2-week withdrawal period is recommended before intradermal testing.

Veterinary-Labeled Products

NOTE: Products listed are representative of those containing only colloidal oatmeal as the principal active ingredient.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Colloidal oatmeal (leave-on spray) Other ingredients: ceramides, safflower oil	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, and 1 gallon	*Dermallay*®

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Colloidal oatmeal 1% (cream rinse) Other ingredients: lactic acid, propylene glycol	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 1 gallon	*Epi-Soothe®*
Colloidal oatmeal (conditioner); other ingredients vary	Labeled for dogs and cats	OTC; 16 oz, 17 oz, 1 gallon	*Aloe & Oatmeal Skin and Coat; Aloe & Oatmeal®*
Colloidal oatmeal (shampoo); other ingredients vary	Labeled for dogs and cats; some products labeled for horses	OTC; 4 oz, 8 oz, 12 oz, 16 oz, 17 oz, 1 gallon	*Aloe & Oatmeal®; Dermallay®*(soap free); *Epi-Soothe®; Vet Solutions Aloe & Oatmeal®; Simply Pure®; Aloe and Oatmeal* (soap and alcohol-free); *Aloe Oatmeal* (soap-free)

Human-Labeled Products

NOTE: There are several human products available containing colloidal oatmeal, including creams, lotions, and products, to be added to the bath. Common trade names include *Aveeno®*, *Geri Silk®*, and *Actibath®*.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Diphenhydramine, Topical

(dye-fen-***hye***-dra-meen) *Benadryl®*

For systemic use, see ***Diphenhydramine.***

Indications/Actions

Diphenhydramine is a first-generation antihistamine that has some local anesthetic activity. This activity is probably its primary antipruritic mechanism of action. Diphenhydramine may be absorbed in small amounts transdermally but is unlikely to cause systemic adverse effects with typical use.

Suggested Dosages/Uses

Topical creams, ointments, gels, lotions, or sprays are usually applied 1 to 3 times a day as needed. The shampoos and conditioners are generally used once a week to once a day after bathing, respectively. Shampoos should remain in contact with the skin for at least 10 minutes before being rinsed. Check product label for specific instructions.

Precautions/Adverse Effects

Avoid contact with eyes or mucous membranes. Do not apply to blistered or oozing areas of the skin. Fatalities have rarely been reported in children after excessive exposure (ie, application to a large percentage of body surface area with open sores); veterinary significance is uncertain. Pet owners should wash their hands after application or wear gloves when applying.

Prolonged use could potentially cause local irritation and/or hypersensitization. Residual activity may affect intradermal or allergy serum tests. It has been suggested to stop using this product 2 weeks before allergy intradermal testing.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Diphenhydramine 1%, pramoxine 1%, (conditioner spray)	Labeled for dogs and cats	OTC; 8 oz	*Benasoothe®* Other ingredients: ceramides, phytosphingosine, safflower oil, *Aloe barbadensis*, glycerin, lanolin cholesterol, glycerin, lanolin

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Diphenhydramine 2% (spray), diphenhydramine 2% (gel), diphenhydramine 1% (cream), diphenhydramine 2% (cream), diphenhydramine 2% (ointment), diphenhydramine 2% (stick), diphenhydramine 12.5 mg (strip)	OTC; various sizes available	*Benadryl®; Banophen®; CareOne®; Allergy Relief®* Other ingredients may include calamine, zinc oxide, acetate, pyrilamine, tripelennamine, menthol, camphor.

Fatty Acids (Omega), Topical

For systemic use, see ***Fatty Acids, Omega-3***

Indications/Actions

Essential fatty acids are polyunsaturated fatty acids that are not synthesized by the body and must be obtained from external sources. The 2 main types of essential fatty acids are omega-3 and omega-6 fatty acids. Essential fatty acids are indicated primarily for pruritic and inflammatory conditions (eg, atopic dermatitis, sebaceous adenitis) and keratinization disorders (eg, seborrhea). They may also be used to improve coat quality and dry skin. Some products may contain other active ingredients and natural oils, and therefore, may have additional indications.

The exact mechanism of action of essential fatty acids is not well understood; however, essential fatty acids affect the arachidonic acid concentrations in plasma lipids and platelet membranes. They decrease the production of inflammatory prostaglandins in the body, thereby reducing inflammation and pruritus. Essential fatty acids may also play a role in restoring the skin barrier.

Suggested Uses/Dosages

Sprays, wipes, and mousses are typically applied up to 2 to 3 times daily as needed. Spot-on treatments vary from weekly to every 2 weeks. Shampoos and crème rinses can be applied daily to weekly. Shampoos should generally be left in contact with the skin for 5 to 10 minutes before rinsing.

Precautions/Adverse Effects

No specific precautions or adverse effects are described for topical essential fatty acids; however, sensitivity reactions are possible. If signs such as pruritus or erythema occur, discontinue use.

Veterinary-Labeled Products

Below is a representative sample of the products available. **NOTE:** Refer to specific product labels for details on inactive ingredients and instructions for use.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Essential fatty acids from safflower oil (shampoo)	Labeled for dogs and cats; some products labeled for horses	OTC; 4 oz, 12 oz, 16 oz, 1 gallon	*HyLyt®; DermaLyte®; EFA Essential Fatty Acid®* Soap-free
Essential fatty acids from safflower oil with oatmeal (shampoo)	Labeled for dogs, cats, and horses	OTC; 12 oz, 1 gallon	*DermAllay®* Soap-free
Essential fatty acids from safflower oil (creme rinse)	Labeled for dogs and cats; some products labeled for horses	OTC; 8 oz, 1 gallon	*HyLyt®*; generic
Essential fatty acids from hemp seed oil (spot-on)	Labeled for dogs, cats, horses, and small mammals	OTC; boxes of 4 pipettes in the following sizes: Dogs: Under 10 kg (22 lb): 0.6 mL 10 - 20.5 kg (22-45 lb): 1.2 mL 20.5 - 40.9 kg (45-90 lb): 2.4 mL Cats: 0.4 mL Small mammals: 0.6 mL Horses: 30 mL bottles	*Essential 6®* Directions: Initially, 1 dose weekly for 2 months, then 1 dose every 2 weeks for maintenance. Do not bathe within 2 days before and 2 days after application. Brush application site 24 hours after applying to remove product residue.
Essential fatty acids from hemp seed oil and tamanu oil (spot-on)	Labeled for dogs	OTC; boxes of 4 pipettes in the following sizes: **Dogs:** under 10 kg (22 lb): 0.6 mL Dogs 10 - 20.4 kg (22-45 lb): 1.2 mL 20.4 - 40.8 kg (45-90 lb): 2.4 mL	*PYOspot®* Directions: 1 pipette weekly Do not bathe during 2 days before and 2 days after application. Brush application site 24 hours after applying to remove product residue.
Essential fatty acids from hemp seed oil (mousse)	Labeled for dogs, cats, and small mammals	OTC; 150 mL	*Essential 6®* Soap-free and rinse-free. Shake before use.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Essential fatty acids from soybean oil (balm)	Labeled for dogs	OTC; 1.67 oz jar	*BIO BALM*® Occlusive and water-resistant
Essential fatty acids from hemp seed oil (wipes)	Labeled for dogs and cats	OTC; pack of 20 wipes	*PYOClean*® Soap-free and alcohol-free
Essential fatty acids from safflower oil (spray)	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, and 1 gallon	*Dermallay*® Soap-free, oatmeal spray conditioner

Human-Labeled Products

NOTE: Several human over-the-counter products are available in the United States but may contain additional ingredients and generally are not used in dogs and cats.

Lidocaine/Lidocaine Combinations, Topical

(***lye***-doe-kane)

For systemic use of lidocaine, please refer to ***Lidocaine (Intravenous; Systemic)****. For local anesthetic use, please refer to* ***Lidocaine, Local Anesthetic****.*

Indications/Actions

Lidocaine is used topically as a dermal anesthetic or antipruritic. It is included in several products used for acute moist dermatitis, pruritic lesions, and painful skin conditions. When combined with prilocaine, it may provide dermal anesthesia before painful or invasive procedures such as catheter placements.

Lidocaine exerts its anesthetic properties by altering sodium ion permeability across the cell membrane, thereby inhibiting conduction from sensory nerves.

Suggested Dosages/Uses

Lidocaine can be applied in a thin layer every 3 to 4 hours as needed. For veterinary-labeled products, refer to the specific product label for details on use.

Precautions/Adverse Effects

Although topical lidocaine may be absorbed systemically, systemic toxicity is unlikely. Applying to a significant percentage of body area, using for prolonged times, using high concentrations, applying to irritated or broken skin, or covering the application site may increase the risk for systemic absorption and subsequent toxicity. Use with caution in patients receiving Class-I antiarrhythmics such as systemic lidocaine or mexiletine. Patches can cause serious adverse effects if chewed.[1]

Possible adverse effects include hypersensitivity reactions or skin irritation such as burning or tenderness. Products containing prilocaine may be more likely to cause methemoglobinemia or systemic toxicity; however, these effects occur rarely.[2] Some products may also contain antiseptics, such as chlorhexidine and povidone-iodine.

Wear gloves when applying and wash hands after handling. Avoid contact with eyes. Do not use in ears unless the product is explicitly labeled for otic use.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Lidocaine 2.4% (spray)	Labeled for dogs and horses	OTC; 2 oz	*Silver L*®; *BioCalm*®
Lidocaine 2.4%, benzalkonium chloride 0.1% (spray)	Labeled for dogs	OTC; 4 oz	*Allercaine*® *with Bittran*® *II*
Lidocaine 2.46%, benzalkonium chloride 0.2% (spray)	Labeled for dogs, cats, and horses	OTC; 4 oz	*Hexa-Caine*®
Lidocaine 0.5%, chlorhexidine 0.2% (solution)	Labeled for dogs, cats, and horses	OTC; 4 oz	*Dermachlor Flush Plus*® These products may no longer be commercially available.
Lidocaine 2%, povidone-iodine 2% (solution)	Labeled for dogs, cats, and horses	OTC; 16 oz	*Barrier*® *II*

Human-Labeled Products

NOTE: Many commercial products exist. Below is a representative list of available products.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Lidocaine 2% (cream)	OTC; various	*Xolido*®

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Lidocaine 4% (cream)	OTC; various	*Anecream®; Bengay®; Aspercreme®; LC-4; LMX-4*
Lidocaine 5% (cream)	OTC; various	*Anecream® 5; LC-5*; generic
Lidocaine 3% (lotion)	Rx; 4 oz, 6 oz	*Lidosorb®*; generic
Lidocaine 5% (ointment)	Rx; 30 g, 35.44 g, or 50 g tube, 50 g, 150 g, or 250 g jar	*Xylocaine®*; generic
Lidocaine 1.8% (patch)	Rx; box of 30 patches	*ZTlido®*
Lidocaine 4% (patch)	OTC; various	*Aspercreme®*; generic
Lidocaine 5% (patch)	Rx; box of 30 patches	*Dermalid®; Lidoderm®*; generic
Lidocaine 0.5% (spray)	OTC; various	*Burnamycin®*; generic
Lidocaine 4% (spray)	OTC; various	*Aspercreme®*; generic
Lidocaine 2% (gel/jelly)	Rx; 6 mL or 11 mL prefilled syringes; 5 mL, 10 mL, or 20 mL *Uro-Jet®* syringes; 5 mL or 30 mL tube	*Xylocaine®*; generic
Lidocaine 4% (gel/jelly)	OTC; various	*Alocane®; Topicaine®*; generic
Lidocaine 5% (gel/jelly)	OTC; various	*Topicaine®*; generic
Lidocaine 2.5%, prilocaine 2.5% (cream)	Rx; 5 g, 15 g, and 30 g	*EMLA®*; generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Phytosphingosine, Topical

(fye-tos-***fin***-goh-seen)

Indications/Actions

Phytosphingosine products are indicated for localized and generalized inflammatory conditions of the skin associated with pruritus and dryness, including allergic diseases such as atopic dermatitis. Some products are also indicated for use as a liquid wound dressing for localized inflammation and after surgery (could be sprayed on sutures). Some products are marketed for seborrheic conditions. There is some evidence supporting the role of phytosphingosine on skin barrier restoration in dogs.[1]

Phytosphingosine acts as a ceramide precursor, and it is a salient molecule in the natural defense mechanism of the skin. Ceramides comprise 40% to 50% of the main lipids responsible for maintaining the cohesion of the stratum corneum. Therefore, ceramides are essential for restoring the skin lipid barrier, controlling local flora, and maintaining the correct moisture balance. Phytosphingosine is also anti-inflammatory; it has anti-IL-1 activity, impairing the production of PGE2, and it inhibits kinase protein C.

Suggested Dosages/Uses

Phytosphingosine spray should be applied every 3 days for 2 to 3 weeks.[2] Gel should be applied once daily as needed for itching relief.[3] Mousse should be applied to dry hair and should be used every 3 days for 2 to 3 weeks.[4] Shampoo should be used 2 to 3 times weekly for 1 to 2 weeks.[5] Leave the shampoo in contact with the skin for at least 5 to 10 minutes before rinsing well. Shake products well before use. Refer to specific product label for details on use.

Precautions/Adverse Effects

Discontinue use if skin redness or irritation occurs. Avoid contact with the eyes.

Veterinary-Labeled Products

NOTE: *Be aware when selecting products that different combinations of active ingredients may be marketed under similar trade names. For products containing other active ingredients, including* ***Chlorhexidine, Topical*** *and* ***Pramoxine, Topical****, please refer to their respective monographs.*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Phytosphingosine salicyloyl 0.2% (gel)	Labeled for dogs and cats	OTC; 2 oz	*Douxo® Calm*
Phytosphingosine salicyloyl 0.05% (micro-emulsion spray)	Labeled for dogs and cats	OTC; 6.8 oz	*Douxo® Calm*
Phytosphingosine salicyloyl 0.05%, chlorhexidine gluconate 3% (micro-emulsion spray)	Labeled for dogs and cats	OTC; 6.8 oz	*Douxo® Chlorhexidine*
Phytosphingosine hydrochloride 0.2% (mousse)	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Balance*
Phytosphingosine salicyloyl 0.05%, chlorhexidine gluconate 3%, climbazole 0.5% (mousse)	Labeled for dogs and cats	OTC; 6.8 oz	*SkinGuard® Restore*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Phytosphingosine salicyloyl 0.05%, pramoxine hydrochloride 1% (mousse)	Labeled for dogs and cats	OTC; 6.8 oz	*SkinGuard® Relieve*
Phytosphingosine salicyloyl 0.05%, chlorhexidine gluconate 3%, climbazole 0.5% (pads)	Labeled for dogs and cats	OTC; 30 pads	*SkinGuard® Restore*
Phytosphingosine hydrochloride 0.1% (shampoo)	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Balance*
Phytosphingosine hydrochloride 0.05%, benzoyl peroxide 3%, salicylic acid 2%, sodium thiosulfate 2% (shampoo)	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Clear*
Phytosphingosine salicyloyl 0.05%, chlorhexidine gluconate 3%, climbazole 0.5% (shampoo)	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Restore*
Phytosphingosine salicyloyl 0.05%, pramoxine hydrochloride 1% (shampoo)	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Relieve*

Human-Labeled Products

NOTE: There are several over-the-counter human cosmetic products containing phytosphingosine in the United States. These products mainly target lipid barrier restoration and are generally not used in dogs and cats.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Pramoxine, Topical

(pra-***moks***-een)

Indications/Actions

Pramoxine is a topical anesthetic that is often combined with other topical medications to temporarily reduce pain and itching.

Pramoxine affects peripheral nerves and has been shown to antagonize histamine-mediated pruritus; however, the mechanism of action is unknown. Pramoxine is not structurally related to procaine-type anesthetics. Peak local anesthetic effects occur within 3 to 5 minutes of application.

Suggested Dosages/Uses

Pramoxine sprays are typically labeled for daily use as needed. Shampoos are typically labeled for use up to 2 to 3 times weekly and may be used as often as once daily. Allow shampoo to remain in contact with the skin for at least 5 to 10 minutes before rinsing well. Mousse products are labeled for use every 3 days and should be applied to dry hair, massaged in, and left to air dry. Rinses may be applied daily to once weekly, separately or after baths. Ointment is labeled for use every other day. Labeled doses may differ based on active ingredient combinations; refer to specific product labels for detailed use instructions.

Precautions/Adverse Effects

Avoid contact with eyes; pramoxine is too irritating for ophthalmic use. Wear gloves when applying or wash hands after application. Adverse effects are uncommon; however, localized dermatitis is possible. Discontinue use if redness, irritation, swelling, or pain occurs or increases with use.

Pramoxine shows an antagonizing impact on histamine-mediated pruritus. Therefore, at least a 2-week withdrawal period before intradermal testing is recommended.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pramoxine HCl 1% (spray)	Labeled for dogs and cats; some products also labeled for horses	OTC; 8 oz, 12 oz	*Comfort; Micro-Pearls Advantage Dermal-Soothe®; Pramox-1®; Relief®*
Pramoxine HCl 0.5%, hydrocortisone 1% (spray)	Labeled for dogs and cats; some products labeled for horses	OTC; 4 oz, 8 oz	*Dermabliss®*
Pramoxine HCl 1%, hydrocortisone 1% (spray)	Labeled for dogs and cats; some products also labeled for horses	OTC; 4 oz, 8 oz	*Comfort HC; EquiShield® IR; Pramoxine HC; Suffusion® HC; VetraSeb® HC*
Pramoxine HCl 1%, phytosphingosine salicyloyl 0.05% (spray)	Labeled for dogs and cats	OTC; 8 oz	*VetraSeb® P; Suffusion® P +PS*
Pramoxine HCl 1%, diphenhydramine HCl 1% (spray)	Labeled for dogs and cats	OTC; 8 oz	*Benasoothe®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pramoxine HCl 1% (shampoo)	Labeled for dogs and cats; some products also labeled for horses	OTC; 8 oz, 12 oz, 16 oz, 1 gallon	*BioCalm®; CeraSoothe®; Comfort Shampoo; Dermabliss®; Micro-Pearls Advantage Dermal-Soothe®; Pramox-1®; Relief®; Suffusion® P; VetraSeb® CeraDerm® P*
Pramoxine HCl 1%, diphenhydramine HCl 1% (shampoo)	Labeled for dogs and cats	OTC; 12 oz	*Benasoothe®*
Pramoxine HCl 1%, hydrocortisone 1% (shampoo)	Labeled for dogs, cats, and horses	OTC; 12 oz	*EquiShield® IR*
Pramoxine HCl 1%, phytosphingosine salicyloyl 0.05% (shampoo)	Labeled for dogs and cats; some products also labeled for horses	OTC; 8 oz, 16 oz	*Phyto P; Suffusion® P +PS; VetraSeb® P*
Pramoxine HCl 1% (rinse)	Labeled for dogs and cats; some products also labeled for horses	OTC; 12 oz, 16 oz, 1 gallon	*Comfort; Micro-Pearls Advantage Dermal-Soothe®; Pramox*
Pramoxine HCl 1%, phytosphingosine salicyloyl 0.05% (rinse)	Labeled for dogs, cats, and horses	OTC; 8 oz	*Phyto P*
Pramoxine HCl 1% (mousse)	Labeled for dogs and cats; some products also labeled for horses	OTC; 6.8 oz	*CeraSoothe®; Suffusion® P; VetraSeb® CeraDerm® P;*
Pramoxine HCl 1%, phytosphingosine salicyloyl 0.05% (mousse)	Labeled for dogs and cats	OTC; 6.8 oz	*Phyto P; Suffusion® P +PS; VetraSeb® P*
Pramoxine HCl 1% (ointment)	Labeled for dogs and cats	OTC; 15 g	*Deter-X®*

Human-Labeled Products

NOTE: Partial listing; there are many branded combination products available that also contain pramoxine. For more information on human-labeled products, refer to a comprehensive human drug reference (eg, *Facts and Comparisons*) or contact a pharmacist.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Pramoxine 1% (lotion)	OTC; 7.5 oz, 8 oz	*Sarna Sensitive®*, generic
Pramoxine 1% (spray)	OTC; 2 oz	*Itch-X®*; generic
Pramoxine 1% (gel)	OTC; 35.4 g	*Itch-X®*
Pramoxine HCl 1% (wipes)	OTC; 30 count	generic

Zinc Gluconate (Neutralized), Topical

*For otic use, see **Cleaning/Drying Otic Agents**.*

Indications/Actions

Zinc gluconate can be used as a sole agent for mild pruritus or as an adjunct treatment for moderate to severe pruritic conditions, mild bacterial infections, or dry skin. It can also be used for minor skin irritations, such as insect bite reactions, acute moist dermatitis, acral lick dermatitis, fold dermatitis (intertrigo), feline acne, and postsurgical wounds.

The exact mechanism of action of topical zinc is unknown. Zinc plays a role in extracellular matrix remodeling, wound healing, connective tissue repair, inflammation, and cell proliferation. Zinc also has antiseptic and astringent properties.

Suggested Dosages/Uses

Zinc gluconate may be applied to affected areas as needed up to twice daily to relieve itching and soothe the skin. For veterinary products, refer to specific product label for details on use. Do not let animals lick or chew at affected areas for at least 20 to 30 minutes after application.

Precautions/Adverse Effects

Avoid contact with eyes and mucous membranes.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Zinc gluconate (spray)	Labeled for dogs, cats, horses, and some exotic pets	OTC; 2 oz, 4 oz	*Maxi/Guard® Zn7 Derm*
Zinc gluconate (gel)	Labeled for dogs, cats, horses, and some exotic pets	OTC; 1 oz, 2 oz	*Maxi/Guard® Zn7 Derm*

Human-Labeled Products

There are a variety of human-labeled OTC zinc gluconate products, some in combination with other active ingredients.

Antimicrobial Agents, Topical

Antibacterial Agents, Topical

See also:
Sulfur, Precipitated, Topical (Keratolytic Agents)

Bacitracin, Topical

(bass-ih-***trase***-in)

Antibiotic

For ophthalmic use, see ***Triple Antibiotic Ophthalmic***.

Indications/Actions

Bacitracin is used topically to treat and prevent infection after dermal lacerations, scrapes, minor burns, and surgical procedures. Bacitracin acts by inhibiting cell wall synthesis of susceptible bacteria by preventing the formation of peptidoglycan chains. It is either bactericidal or bacteriostatic, depending on drug concentration and bacterial susceptibility. Bacitracin is primarily active against gram-positive bacteria, but *Staphylococcus* spp are becoming increasingly resistant; therefore, bacitracin ideally should be used only after confirmation of bacterial skin infections in dogs and cats. Bacitracin activity is not impaired by blood, pus, necrotic tissue, or large inocula.

Suggested Dosages/Uses

Bacitracin may be applied up to 3 times daily and be covered by a suitable dressing. Administration of bacitracin is usually not recommended to continue for more than 1 week.

Precautions/Adverse Effects

Bacitracin topical ointment should not be used in or around the eyes, for the treatment of ulcerated lesions, chronic canine dermatoses, or in patients that are known to be hypersensitive to it. There have been reports of cats developing fatal anaphylactic reactions after administration of triple-antibiotic ophthalmic ointment containing bacitracin.[1] Bacitracin is also well documented as causing allergic contact dermatitis after topical administration in humans with the potential for anaphylaxis.[2] Deep puncture wounds, animal bites, or deep cutaneous infections may require systemic antibiotic therapy. Although topical administration generally results in negligible systemic levels, if bacitracin is used over large areas of the body or on serious burns or puncture wounds, measurable absorption and potential toxicity may occur. Do not let animal lick or chew at affected areas for at least 20 to 30 minutes after application. Pet owners should wash their hands after application.

Veterinary-Labeled Products

None

Human-Labeled Products

Bacitracin ointment is available alone as 500 units/g in various tube sizes. There are many OTC human products available with various combinations of antibiotics. Some products may also contain pramoxine hydrochloride for soothing and pain relief.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Bacitracin zinc 500 units/g (ointment) Other ingredients (depending on the manufacturer): white petrolatum, mineral oil	OTC; 14 g, 14.2 g, 28 g, 28.4 g, and 120 g tubes; 454 g jar	generic
Bacitracin zinc 500 units/g, polymyxin B sulfate 10,000 units/g (ointment)	OTC; 28.3g tube	*Polysporin*®
Bacitracin zinc 400 units/g, neomycin 3.5 mg/g, polymyxin B sulfate 5000 units/g (ointment)	OTC; 14.2g, 28.3g tube	*Neosporin*®; generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Benzoyl Peroxide, Topical

(ben-***zoyl*** per-***oks***-ide)

Antibacterial

Indications/Actions

Benzoyl peroxide products are available as gels, shampoos, or cream rinses for topical use. Shampoos are generally used for oily dermatoses (eg, seborrhea oleosa), superficial and deep pyodermas, and furunculosis. Benzoyl peroxide shampoos can also be used as adjunctive therapy for generalized demodicosis and schnauzer comedo syndrome. Gels may be useful for treating localized superficial and deep pyodermas, chin acne, fold pyodermas, and localized lesions caused by *Demodex* spp.

Benzoyl peroxide is an antibacterial agent with comedolytic ("follicular flushing"), keratolytic,[1] and antiseborrheic actions. It also has some mild antipruritic activity and wound healing effects. Benzoyl peroxide is lipophilic and penetrates into the stratum corneum, where it is converted to benzoic acid. The antibacterial activity of benzoyl peroxide is due to free radical formation that oxidizes bacterial proteins.[2]

Suggested Dosages/Uses

Gels are commonly administered once to twice daily; shampoos may be used anywhere from once daily to once weekly. Shampoos should remain in contact with the skin for 5 to 10 minutes before rinsing well. For veterinary products, refer to the product label for details on use.

Precautions/Adverse Effects

Avoid contact with the eyes or mucous membranes. Benzoyl peroxide will bleach colored fabrics, jewelry, clothing, or carpets and might bleach the animal's haircoat. Owners should be advised to keep treated animals away from fabrics during treatment. Do not let the animal lick or chew at treated areas for a few minutes after application.

In some animals, benzoyl peroxide can be drying or irritating. Adverse effects can include erythema, pruritus, or pain at the treated site, particularly when higher concentrations (greater than 5%) are used. Reducing the frequency of use, applying emollients after bathing, or using shampoos with moisturizing microvesicles may alleviate or prevent these effects. Benzoyl peroxide causes photosensitivity in humans.[3] Limit sun exposure for dogs with light or thin coats. **NOTE:** Benzoyl peroxide shampoos do not lather well.

Veterinary-Labeled Products

CREAM RINSE

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Benzoyl peroxide 2.5%	Labeled for dogs and cats	OTC; 12 oz	generic

GEL

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Benzoyl peroxide 5% Other ingredients: propylene glycol, docusate sodium, DMDM hydantoin	Labeled for dogs and cats	OTC; 30 g bottle	*Micro BP*

SHAMPOOS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Benzoyl peroxide 2.5% Other ingredients: *Novasome®* microvesicles	Labeled for dogs and cats	OTC; 12 oz, 1 gallon	*Micro-Pearls Advantage Benzoyl-Plus®*
Benzoyl peroxide 2.5%	Labeled for dogs and cats	OTC; 12 oz, 1 gallon	generic
Benzoyl peroxide 2.5% Other ingredients: glycolic acid 1%	Labeled for dogs and cats	OTC; 8 oz	*GlycoBenz®*
Benzoyl peroxide 2.5%, salicylic acid 1%, sulfur 1% (shampoo)	Labeled for dogs and cats	OTC; 12 oz	*Sulfur Benz®*
Benzoyl peroxide 2.5%, sulfur 2%	Labeled for dogs and cats	OTC; 16 oz	generic
Benzoyl peroxide 3%	Labeled for dogs and cats	OTC; 16 oz, 1 gallon	*BPO-3®*
Benzoyl peroxide 3%, salicylic acid 2%, sulfur 2%	Labeled for dogs and cats	OTC; 8 oz, 12 oz, and 16 oz	*Oxibenz-3®; Peroxiderm®;* generic

Human-Labeled Products

There are many human products available that contain benzoyl peroxide (2.5% to 10%); however, with the possible exception of the 5% gel products, the veterinary formulations would be more suitable for dogs or cats. Benzoyl peroxide 5% gel can be labeled as either Rx or OTC, depending on the product, and are available as generics or with the following trade names: *Benzac®*, *Desquam-X®*, or *PanOxyl®*.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Clindamycin, Topical

(klin-da-***mye***-sin) *Cleocin®*, *ClinzGard®*

For systemic use, see ***Clindamycin****.*

Indications/Actions

Topical clindamycin may be used for treating feline acne or other localized skin infections caused by bacteria susceptible to it. It has been recommended by some veterinary dermatologists that topical clindamycin only be used when other topical antibiotics, such as gentamicin, have failed. Ideally, treatment should be based on culture and susceptibility results. Clindamycin is a lincosamide antibiotic and inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. Its primary activity is against anaerobic and gram-positive aerobic bacteria. For more information on clindamycin pharmacology, see ***Clindamycin***.

Suggested Dosages/Uses

When used for feline acne, topical clindamycin is generally applied in a thin film once to twice daily. *ClinzGard*® is used as a single dose application in bite and puncture wounds, surgical incisions, and anal sacs after cleaning and debridement (when applicable).

Precautions/Adverse Effects

In patients with a history of hypersensitivity to clindamycin or lincomycin, topical clindamycin should not be used. Avoid contact with eyes. Clients should wash their hands after application or wear gloves when applying. Contact reactions (pain, burning, erythema, itching, drying, peeling) are possible. Clindamycin lotions and gels may cause less burning than topical solutions or foams. Because clindamycin can be absorbed through the skin, systemic adverse effects are possible. Although clindamycin-related diarrhea and colitis are rare in animals, severe diarrhea, bloody diarrhea, and severe colitis have been reported in humans with topical use.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Clindamycin hydrochloride 1% (gel)	Labeled for dogs and cats	Rx; sterile 1 or 2 mL single-dose syringes (box with 4 units)	*ClinzGard*® Indicated for anal sac disease, tissue abscesses, bites, puncture wounds, and surgical incisions. Single-dose with a sustained release over 7 to 10 days. Must be applied to clean and debrided (if necessary) surfaces

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Clindamycin phosphate 1% (lotion)	Rx; 60 mL	*Clindamax*®; *Cleocin T*®; generic
Clindamycin phosphate 1% (gel)	Rx; 30 g, 60 g, 40 mL, and 75 mL	*Clindagel*®; *Clindamax*®; *Cleocin T*®; generic
Clindamycin phosphate 1% (solution)	Rx; 30 mL, 60 mL	*Cleocin T*®; generic
Clindamycin phosphate 1% (aerosol foam) Other ingredients: cetyl alcohol, ethanol 58%, stearyl alcohol, propylene glycol	Rx; 50 g, 100 g	*Evoclin*® Dispense into cap or onto a cool surface, then use fingertips to apply small amounts. Dispensing directly onto hands or affected area(s) will cause the foam to melt on contact. If the foam seems runny or warm, run canister under cold water.
Clindamycin phosphate 1% (pledgets) Other ingredients: isopropyl alcohol, propylene glycol	Rx; 60 per box/jar	*Cleocin T*®; *Clindacin*®; *Clindacin*®*-P*; *Clindets*®; generic

Gentamicin, Topical

(jen-ta-***mye***-sin)

For systemic use, see ***Gentamicin.***

For otic use, see ***Corticosteroid/Antimicrobial Agents, Otic.***

Indications/Actions

Topical gentamicin can be useful for treating both primary and secondary superficial bacterial skin infections caused by susceptible bacteria. Some products indicate prophylactic use after lacerations, abrasions, or after minor surgery; however, dermatologists recommend using topical gentamicin only to treat confirmed bacterial skin infections, as resistance may occur. Therefore, the use of gentamicin solely for the treatment of yeast and/or inflammation without evidence of bacterial infection should also be avoided. In small animal medicine, topical gentamicin is frequently used in combination with a corticosteroid (typically betamethasone or mometasone) to treat superficial lesions, including "hot spots" (acute moist dermatitis and pruritic lesions).

Some products containing betamethasone or mometasone and clotrimazole are labeled for otic use but can also be used extra-label for bacterial and yeast skin infections that are sensitive to gentamicin and clotrimazole when an anti-inflammatory effect is also desired. Formulations containing a corticosteroid are best suited for focal (eg, pedal) or multifocal lesions for relatively short durations (ie, less than 2 months). Initially, topical gentamicin/corticosteroid products are used 1 to 4 times daily until control is achieved, and then the frequency is tapered. Sole ingredient gentamicin sulfate topical forms are available with human labeling.

Gentamicin is an aminoglycoside antibiotic. It binds to the bacterial 30S ribosomal unit and inhibits protein synthesis. It has minimal activity against *Streptococcus* spp, variable activity against *Staphylococcus* spp, and good activity against gram-negative bacteria, including

Klebsiella spp, *Escherichia coli*, and some strains of *Pseudomonas* spp. Gentamicin-resistant strains of *Pseudomonas* spp are an ongoing issue.

Suggested Dosages/Uses

Topical gentamicin creams, ointments, or suspensions are generally applied to affected areas up to 4 times daily. Topical gentamicin/corticosteroid sprays are labeled for use 1 to 4 times daily for up to 7 days. Creams are typically used for infections associated with greasy skin, whereas ointments are typically used for infections associated with dry skin.

Precautions/Adverse Effects

Topical gentamicin may be systemically absorbed if it is used on ulcerated, burned, or denuded skin. Creams are more likely to be absorbed than are ointments; however, systemic toxicity is unlikely to occur unless gentamicin is applied to a significant percentage of body surface area or is used for prolonged times. Do not let the animal lick or chew at treated areas for at least 20 to 30 minutes after application.

Products that contain corticosteroids can cause HPA axis suppression with long-term or frequent use, cutaneous atrophy that can be associated with skin fragility, superficial follicular cysts (milia), reduced or lack of hair regrowth, and comedones. Risk can be reduced by only treating for as long as necessary on the smallest area possible. Mometasone is considered a "soft steroid", delivering its potent anti-inflammatory effect close to its site of action while reducing the degree of systemic exposure and thus limiting the associated systemic and local adverse effects. Nevertheless, care should be taken to avoid excessive or prolonged use. Residual activity may affect intradermal or serum allergy tests, and it has been suggested to discontinue use of corticosteroid-containing products 2 weeks before allergy testing.[1]

Avoid contact with eyes. Clients should wash their hands after application or wear gloves when applying.

Veterinary-Labeled Products

Refer to the individual product labeling for specific dosing recommendations and inactive ingredients.

NOTE: There are no topical gentamicin-only labeled veterinary products in the United States.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Gentamicin 3 mg/g, clotrimazole 10 mg/g, mometasone 1 mg/g (suspension)	Approved for otic use in dogs	Rx; 7.5 g, 15 g, 30 g, and 215 g tubes	*Mometamax®; Mometavet®* Extra-label when used topically in dogs and cats
Gentamicin 3 mg/mL, betamethasone 1 mg/mL (solution)	Approved for otic use in dogs and cats	Rx; 7.5 mL, 15 mL, 240 mL	*Vet Beta-gen®* Extra-label when used topically
Gentamicin 0.57 mg/mL, betamethasone 0.284 mg/mL (spray)	Approved for use in dogs	Rx; 15 g, 30 g, 60 mL, 72 mL, 120 mL, and 240 mL	*Betagen®; Gentacalm®; GentaSoothe®; GentaSpray®; GentaVed®; Gentocin®*
Gentamicin 3 mg/g, clotrimazole 10 mg/g, betamethasone 1 mg/g (ointment)	Approved for otic use in dogs	Rx; 7.5 g, 10 g, 15 g, 20 g, 25 g, 30 g, and 210 g bottles or tubes	*MalOtic®; Otomax®; Tri-Otic®; Vetromax®* Extra-label when used topically in dogs and cats

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Gentamicin 0.1% (cream and ointment)	Rx; 15g, 30 g tubes	generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Honey, Topical

Topical Antibacterial Agent *HoneyCure®, Medihoney®*

Indications/Actions

Honey is a natural product that is generally made from flower nectar by honey bees. Honey is composed of over 200 biologically active compounds, including sugars, proteins, vitamins, minerals, phenols, and plant- and bee-derived enzymes. Medical-grade honey is honey that has been filtered to a higher level than food-grade honey and has been sterilized to kill, remove, or inactivate bacteria, fungi, protozoa, and spores. Medical-grade Manuka honey is commonly used for medicinal purposes and is produced by honey bees foraging only from flowers of the Manuka tree (*Leptospermum scoparium*) from Australia and New Zealand. Other monofloral (eg, honeydew, Scottish Heather) or regionally derived (eg, Saudi, Italian, Portugese) honeys have also been studied for potential medicinal purposes. Medical-grade honey can be used as an alternative or adjunctive therapy for many dermatologic conditions (eg, traumatic wounds, burn wounds) in animals.

Honey is considered to be an effective broad-spectrum topical antibacterial agent, and unlike traditional antibiotics, resistance to its killing effects has not been reported.[1] Honey's antimicrobial properties have been attributed to its osmotic action and low pH (3.4 to 6.1) as well as to the presence of hydrogen peroxide. Additionally, Manuka honey also has nonperoxide bacteriostatic properties due to the presence of methylglyoxal (MGO), which may inhibit bacterial protein synthesis.[1–3] In vitro studies have shown honey to have potent antimicrobial effects against important skin microbes, including *Staphylococcus aureus*, *Staphylococcus pseudintermedius*, *Acinetobacter baumannii, and Escherichia coli,* in addition to antibiotic-resistant strains of bacteria such as methicillin-resistant *Staphylococcus aureus* (MRSA) and *Pseudomonas aeruginosa.*[3–8] It has also been demonstrated to be effective against fungal organisms such as *Malassezia pachydermatis and Candida albicans* and viruses.[7,9]

Suggested Dosages/Uses

NOTE: Ensure that enough honey is in full contact with the wound bed and that honey dressings extend to cover any areas of inflammation surrounding the wound. If a nonadherent dressing is used between the honey dressing and the wound bed, it must be sufficiently porous to allow the honey to diffuse through.

Honey gels, ointments, or creams: Clean and dry wound. Apply product directly to the affected skin, beyond wound edges, as a thin layer once daily or more often according to the severity of the treated wounds or lesions. Cover with bandage if needed.

Honey dressings: Commercially available honey-impregnated dressings facilitate the application of honey to wounds, especially those that need to be maintained in place or protected longer. Alternatively, honey dressings may be prepared in the clinic by applying medical-grade honey to a clean, dry dressing (eg, sterile gauze pad, adhesive bandage); apply the dressing to the skin, then apply a suitable secondary dressing, such as a semi-occlusive dressing, to prevent leakage of honey. Replace the dressing when drainage from the wound saturates the dressing. Change the dressings frequently enough to prevent the honey from being washed away or excessively diluted by wound exudate. The frequency of dressing change will vary but the typical interval is every 2 to 3 days. As the wound heals, the dressing changes will likely be less frequent. When using honey to debride hard eschar, softening the eschar by soaking with saline will allow better penetration of the honey. Owners may be instructed on how to replace the dressings so that it can be done at home.

Precautions/Adverse Effects

Honey is safe when applied to healthy skin and wounds. Although it is safe if ingested, if applying honey without a bandage, do not let the animal lick or chew affected areas for at least 20 to 30 minutes to prevent removing the honey from the treated area(s). An Elizabethan collar may be helpful. Do not use honey to control heavy bleeding, in third-degree burns, and in patients with known sensitivity to honey. Honey may crystalize, making it difficult to use. If crystalized, gently warm the honey, mix it well, and allow it to cool before use. Honey applied directly to the skin (instead of the bandage) may be "messy" (eg, ooze or leak from or around application site).

Veterinary-Labeled Products

Many different brands of medical-grade honey or medical-grade Manuka honey exist. Listed below are representative examples of products available. Refer to specific product labels for further details.

Active ingredients	Label status/sizes	Trade names; other information
Manuka honey	OTC; labeled for use on dogs; 1 oz and 2 oz tubes; 2 oz, 4 oz, 16 oz, 32 oz jars	*HoneyCure®*
Manuka honey 10%, *MicroSilver BG®* 0.1% (spray gel)	OTC; labeled for use on all animals; 8 oz bottle	*Silver Honey® Hot Spot and Wound Repair*; *Silver Honey® Rapid Wound Repair*
Manuka honey 11%, *MicroSilver BG®* 0.2% (ointment)	OTC; labeled for use on all animals; 2 oz tube	*Silver Honey® Hot Spot and Wound Repair*; *Silver Honey® Rapid Wound Repair*
Manuka honey 40% (gel)	OTC; labeled for use on large and small animals; 1.75 oz tube	*L-Mesitran®*
Untreated raw honey (cream)	OTC; labeled for use on horses; 190 mL jar	*HoneyHeel®*

Human-Labeled Products

Many different brands of medical-grade honey or medical-grade Manuka honey exist. Listed below are representative examples of products available. Refer to specific product label for further details.

Active ingredients	Label status/sizes	Trade names; other information
Manuka honey 30% (cream)	OTC; 0.5 oz tube	*First Honey®*
Manuka honey 63% impregnated hydrogel colloidal sheet (nonadhesive dressing)	OTC; 2.4" x 2.4" and 8" x 12" sheets of 1/pack, 10/pack	*MediHoney® HCS*
Manuka honey 80% (gel)	OTC; 1.5 oz tube	*MediHoney®*
Manuka honey 100% (adhesive dressing)	OTC; 1" x 3.25" adhesive dressings of 12/box, 3" x 4" adhesive dressings of 6/box	*FirstHoney®*
Manuka honey 100% (gel)	OTC; 0.5 oz and 1.5 tubes. Single-use, 10/box or individual tube	*TheraHoney®*

Active ingredients	Label status/sizes	Trade names; other information
Manuka honey 100% (hydrocolloid paste)	OTC; 0.5 oz, 1.5 oz, and 3.5 oz tubes. 1/pack, 4/pack, 10/pack, 12/pack	*MediHoney®*
Manuka honey 100% (ointment)	OTC; 0.5 oz tube	*First Honey®*
Manuka honey 100% (impregnated in calcium alginate pads)	OTC; single-use sheets: 2" x 2", 4" x 5", and ¾" x 12" sheets of 1/pack, 5/pack, 10/pack, and 100/case	*MediHoney® Calcium Alginate Dressing* Calcium alginate provides wound fluid absorption capabilities.
Manuka honey 100% (impregnated in sheets [nonadhesive dressing])	OTC; single-use sheets; can be cut to fit within the wound edges. 2" x 2", 4" x 5" sheets of 10/box; rope: 1" x 12"; 4" x 4" flexible foams of 10/box	*TheraHoney®; TheraHoney® HD; TheraHoney® Foam Flex*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Mupirocin, Topical
Pseudomonic Acid A
(myoo-***pye***-roe-sin)

Indications/Actions

Mupirocin is FDA-approved for the treatment of bacterial skin infections in dogs caused by susceptible strains of *Staphylococcus aureus* or *Staphylococcus pseudintermedius*, including beta-lactamase producing and methicillin-resistant strains. Mupirocin can be used to treat superficial pyoderma, fold pyoderma, interdigital cysts, draining tracts, acne, and pressure point pyodermas. It may also be used in other species for conditions such as feline acne or equine pyoderma.

Mupirocin also has activity against other gram-positive pathogens, including strains of *Corynebacterium* spp, *Clostridium* spp, *Proteus* spp, and *Actinomyces* spp., Although it has activity against some gram-negative bacteria, it is not used clinically for gram-negative infections. *Pseudomonas* spp is particularly resistant to mupirocin. Mupirocin has anecdotally been reported to have some effect against Malassezia spp in dogs and cats.

Mupirocin is an antibiotic isolated from the bacterium *Pseudomonas fluorescens*. It is unique in that it belongs to the monocarboxylic acid class and is structurally related to other commercially available antibiotics. Pseudomonic acid A represents 90% to 95% of the mixture and is the main compound of mupirocin. It acts by inhibiting bacterial protein synthesis by binding and inhibiting bacterial isoleucyl transfer-RNA synthetase. It is bacteriostatic at low concentrations or bactericidal at high concentrations. While bacterial resistance is rare in animals, resistant strains of *Staphylococcus* spp have been identified. Resistance transference is thought to be plasmid-mediated and thought to occur more frequently when mupirocin is used on large areas of the skin for prolonged durations. Therefore, mupirocin may be best suited for short-term treatment on small, localized areas. Cross-resistance with other antimicrobials has not been identified. Because this antibiotic is a salient treatment used in humans with resistant infections, its use should be limited to select or resistant cases in veterinary patients.

Mupirocin is not significantly absorbed through the skin into the systemic circulation but does penetrate well into granulomatous and deep pyoderma lesions. It is not suitable for application to burns.

Suggested Dosages/Uses

After cleansing the lesion, mupirocin should be applied to completely cover the affected area. It is labeled for twice daily application on dogs. It requires 10 minutes of contact time to be active. The maximum duration of treatment is 30 days.[1]

In cats with chin acne, once- to twice-daily applications have been shown to be effective for treating secondary infections.[2]

Precautions/Adverse Effects

Mupirocin is contraindicated in patients with a history of hypersensitive reactions to it or other ointments containing polyethylene glycol. Use veterinary labeled products with caution on extensive deep lesions due to risk for nephrotoxicity from absorption of large quantities of polyethylene glycol.[1] Do not let animals lick or chew treated areas for at least 20 to 30 minutes after application. Avoid contact with eyes; not for ophthalmic use.

Mupirocin is typically well tolerated. Contact reactions such as pain, erythema, and itching are possible. Overgrowth of non-susceptible organisms (superinfection) is also possible with prolonged use.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Mupirocin 2% (ointment)	Labeled for use on dogs	Rx; 15 g, 22 g	*Muricin®*; generic

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Mupirocin 2% (ointment)	Rx; 1 g, 15 g, 22 g, and 30 g	*Bactroban®; Centany®*; generic
Mupirocin 2% (cream)	Rx; 15 g, 30 g	*Bactroban®*; generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Nitrofurazone, Topical

(nye-troe-***fur***-ah-zone) *Furazone®*

Indications/Actions

Nitrofurazone is FDA-approved for prevention or treatment of surface bacterial infections of wounds, burns, and cutaneous ulcers in dogs and horses; some products are also approved for cats.[1] It is a synthetic nitrofuran with a broad-antibacterial spectrum. Nitrofurazone is bactericidal against many bacteria involved in surface infections, including gram-positive (eg, *Staphylococcus aureus*, *Streptococcus* spp) and gram-negative (eg, *Escherichia coli*) bacteria; however, evidence supporting clinical efficacy is unavailable. Nitrofurazone has also anecdotally been used for the adjunctive treatment of myiasis.

The mechanism of action of nitrofurazone is thought to be associated with inhibiting bacterial enzymes that primarily degrade glucose and pyruvate.

Suggested Dosages/Uses

Nitrofurazone ointment should be applied directly to the affected area with a spatula or placed on gauze that is then applied to the lesion.[1] Dressing may be changed several times a day or left in place for longer periods of time, but the ointment should remain in contact with the lesion for at least 24 hours. Nitrofurazone powder should be applied to the affected areas daily to several times per day.[2] Refer to specific product labels for details on use.

Precautions/Adverse Effects

Nitrofurazone has been shown to cause mammary tumors when fed in high doses to rats and ovarian tumors in mice. **Federal law prohibits the use of nitrofurazone products in or on food animals, including horses intended to be used for food**. Wear gloves and wash hands after applying.

Ointments may contain polyethylene glycols. If used on large areas of denuded skin, significant amounts of polyethylene glycol could be absorbed, which can cause nephrotoxicity.[3] Avoid contact with eyes and mucous membranes. Do not allow animals to lick or chew treated areas for at least 30 minutes. Avoid exposing products to sunlight, strong fluorescent lighting, excessive heat, or alkaline materials.

Topical nitrofurazone is typically well tolerated. Hypersensitivity reactions and skin reactions, including pain, erythema, and itching, are possible but uncommon. Discontinue use of this product if redness, irritation, or swelling persists or increases. Overgrowth of nonsusceptible organisms (superinfection) is also possible with the prolonged use of nitrofurazone.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Nitrofurazone 0.2% (ointment)	Labeled for dogs, cats, and horses	OTC; 1 lb jars	*Fura-Septin®; NFZ® Wound Dressing;* generic
Nitrofurazone 0.2% (ointment)	Labeled for horses	OTC; 1 lb jars	*Fura-Zone®*
Nitrofurazone 0.2% (powder)	Labeled for dogs, cats, and horses	OTC; 45 g	*NFZ® Puffer;* generic Shake or rotate to loosen powder. Restricted drug in California

Human-Labeled Products

None. Nitrofurazone products are no longer available for humans in the United States.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Silver Sulfadiazine, Topical

(***sil***-ver sul-fa-***dye***-ah-zeen)

Indications/Actions

Silver sulfadiazine (SSD) is used in veterinary medicine to treat localized bacterial skin infections, particularly those caused by *Pseudomonas* spp. In humans, topical SSD is labeled for prophylaxis and treatment of second- and third-degree burns.

SSD has extensive antimicrobial activity. It is bactericidal against many gram-positive and gram-negative bacteria and is also effective against yeasts. SSD disrupts microbial cell membranes and cell walls; this differs from the antibacterial actions of silver nitrate or sodium sulfadiazine. It can enhance epithelization but can impair granulation.

Suggested Dosages/Uses

For burns, silver sulfadiazine (SSD) is applied once to twice daily at a thickness of ≈1/16th of an inch (1 to 2 mm). Dressings may be applied over the cream. For localized bacterial infections, once- to twice-daily application is recommended. Apply to affected area(s) and massage the cream into the skin.

Precautions/Adverse Effects

Use with caution in patients that are hypersensitive to sulfonamides, as they may also react to silver sulfadiazine (SSD). Patients with significant hepatic or renal dysfunction have increased risk for drug accumulation, particularly when used over large areas. Because SSD can impair granulation, avoid use in non-granulated wounds. Avoid contact with eyes. Wear gloves or wash hands after applying. Do not let animals lick or chew at affected areas for at least 20 to 30 minutes after application.

Adverse effects of systemic sulfonamides are possible, such as blood dyscrasias and keratoconjunctivitis sicca (KCS). Use over large areas or use for extended durations can increase the risk for systemic adverse effects. Refer to ***Sulfa-/Trimethoprim*** for more information. In humans, transient leukopenia has been reported with SSD treatment. However, leukocyte normalization occurs even with continuation of therapy. Other uncommon adverse effects in humans include skin necrosis, erythema multiforme, skin discoloration, rashes, and interstitial nephritis.

Veterinary-Labeled Products

There are no topical SSD products labeled for veterinary patients. An otic preparation (*Baytril® Otic*) contains silver sulfadiazine (SSD). See ***Antibacterials, Otic*** for more information.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Silver sulfadiazine 1%, enrofloxacin 0.5% (emulsion)	Labeled for otic use in dogs	Rx; 15 mL, 30 mL	*Baytril®* Extra-label when used topically

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Silver sulfadiazine 1% (cream)	Rx; 20 g, 25 g, 50 g, 85 g, 400 g, and 1000 g	*Silvadene®; SSD®; Thermazene®*

Antifungal Agents, Topical

Clotrimazole, Topical

(kloe-***trye***-ma-zole) *Lotrimin®*

For otic products containing clotrimazole, see ***Corticosteroid/Antimicrobial Agents, Otic.***

Indications/Actions

Topical clotrimazole has activity against dermatophytes and yeasts; it may be useful for localized lesions associated with *Malassezia* spp. There is limited information regarding its efficacy for the treatment of dermatophytosis in cats and dogs. Most veterinary clotrimazole products also contain gentamicin and a glucocorticoid and are labeled for otic use; these products may be used in an extra-label manner for bacterial and yeast skin infections with concurrent inflamed, pruritic skin.

Intranasal instillation of clotrimazole via sinus trephination or direct frontal sinuscopy may also be used for the treatment of sinonasal aspergillosis in dogs, with or without systemic antifungal therapy.[1–3]

Clotrimazole is an imidazole antifungal that inhibits the biosynthesis of ergosterol, a component of fungal cell membranes, leading to increased membrane permeability and probable disruption of membrane enzyme systems.

Suggested Dosages/Uses

If sprays are prescribed, twice-daily applications are usually recommended. Ointments are generally applied to affected areas up to 4 times daily. For veterinary products, refer to the product label for details on individual use.

Precautions/Adverse Effects

Avoid contact with eyes and mucous membranes. Clients should wash their hands after application or wear gloves when applying. Skin irritation is possible but unlikely to occur with products containing only clotrimazole. Avoid contact with eyes. Do not allow the animal to lick or chew at affected sites for at least 20 to 30 minutes after application.

Cribriform plate damage has historically been suggested as a contraindication for topical antifungal treatment for sinonasal aspergillosis; however, small retrospective reviews of dogs with cribriform plate lysis cases found no neurologic adverse effects following infusion of clotrimazole 1% into the frontal sinuses.[4,5]

Products containing clotrimazole in combination with a glucocorticoid (eg, betamethasone):

Several veterinary topical products list pregnancy as a contraindication. Systemic glucocorticoids can be teratogenic or induce parturition during the third trimester of pregnancy in animals. If considering use during pregnancy, weigh the respective risks with treating versus potential benefits. Clients should wash their hands after application or wear gloves when applying. Avoid contact with eyes. Do not allow the animal to lick or chew at affected sites for at least 20 to 30 minutes after application. Residual activity from glucocorticoids may affect intradermal or serum allergy tests; it has been suggested to stop use 2 weeks prior to allergy testing.

Use care when treating large areas or when using on smaller patients. Risks can be reduced by treating for only as long as necessary on as small an area as possible. When higher concentrations of glucocorticoids are used or with longer durations of therapy (ie, longer than 7 days), there is an increased risk for HPA axis suppression, systemic glucocorticoid effects (eg, polydipsia, polyuria, polyphagia), skin changes (eg, atrophy, skin fragility, alopecia, localized pyoderma, comedones), and delayed wound healing. Vomiting and diarrhea have been reported with the use of products containing betamethasone. Local skin reactions (burning, itching, redness) are possible but unlikely to occur.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Clotrimazole 1% (ointment) Other ingredients: bentonite, jojoba oil, magnesium oxide, peppermint oil, silver oxide, tea tree oil, white wax, zinc oxide	Not specified	OTC, not FDA approved; 100 g	*Aardora®*
Clotrimazole 1% (solution)	Labeled for dogs and cats	OTC, not FDA approved or Rx; 30 mL	*ClotrimaTop®*
Clotrimazole 1% (spray or dropper) Other ingredients: propylene glycol, SD alcohol 40, benzoyl alcohol, chloroxylenol	Labeled for dogs and cats	OTC, not FDA approved; 1 oz dropper, 2 oz spray	*Clotrimazole Antifungal®*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Clotrimazole 1% (solution)	OTC/Rx (status determined by labeling); depending on the product: 10 mL, 30 mL	generic
Clotrimazole 1% (cream)	OTC/Rx (status determined by labeling); depending on the product: 10 mL, 30 mL	*Lotrimin AF®; Desenex®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Enilconazole, Topical

(ee-nil-***kon***-a-zole) *Imaverol®*

Indications/Actions

Although no dosage forms are currently commercially available for topical use in the United States, compounded enilconazole is used topically for treating dermatophytosis in small animals and horses. In other countries, a commercially available topical rinse is available for canine, equine, and bovine species. The use of topical enilconazole on cats with dermatophytosis is somewhat controversial as no products are labeled for feline use; however, there are reports of successful use of enilconazole as a sole therapy or in combination with oral itraconazole for cats with dermatophytosis.

Intranasal instillation of enilconazole after plaque debridement has also been shown useful in treating nasal aspergillosis in small animal species.[1]

A poultry environmental disinfectant product (*Clinafarm EC®*) is available in the United States and has been used extra-label to treat dermatophytosis; however, as this product is regulated by the EPA, it is illegal to use it in a manner inconsistent with its labeling.

Enilconazole, like other azoles, inhibits the biosynthesis of ergosterol, an essential component of fungal cell membranes, leading to increased membrane permeability and allowing leakage of intracellular components.

Suggested Dosages/Uses

The commercial product should be diluted by adding 1 part of concentrated solution to 50 parts warm water, which yields a 2 mg/mL emulsion. Remove crusts with a hard brush that has been soaked in the diluted emulsion. It is highly recommended that the animal be sprayed entirely at the first treatment to reach subclinical lesions as well.

Dogs: Clip longhaired dogs prior to treatment, and wash with the diluted emulsion 4 times at 3 to 4 day intervals.[2] During application, the hair should be brushed thoroughly in the opposite direction of hair growth to make sure that the skin is thoroughly wet. Alternatively, dogs may be dipped in a bath of the diluted emulsion.[3]

Horses: The lesions and surrounding skin are to be washed with the diluted emulsion 4 times at 3 to 4 day intervals.[2]

Cattle: Wash with diluted emulsion or apply with a sprayer or high-pressure cleaning unit 3 to 4 times at 3-day intervals.[3]

Dispose of all unused diluted solutions. Refer to the product label for details on use.

Precautions/Adverse Effects

Avoid contact with eyes. Clients should wear gloves and eye protection when applying.

When enilconazole is used topically in cats, hypersalivation, vomiting, anorexia, weight loss, muscle weakness, and a slight increase in serum ALT levels have been reported.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Enilconazole 10% (concentrate)	Labeled for dogs and horses; also labeled for cattle in some countries	50 mL, 100 mL, 1 L	*Imaverol®; Austrazole®* Not available in the United States. Refer to specific product labels for withdrawal periods.

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Ketoconazole, Topical

(kee-toe-***kah***-na-zole) *Nizoral®, KetoChlor®*

For systemic use, see ***Ketoconazole.***

For otic use, see ***Corticosteroid/Antimicrobial Agents, Otic****.*

Indications/Actions

Topical ketoconazole has activity against dermatophytes and yeasts, and ketoconazole shampoos can be an effective treatment for *Malassezia* spp dermatitis and dermatophytosis; however, topical ketoconazole shampoos are generally ineffective or minimally effective when used alone for dermatophytosis. Patients with severe, generalized infections may require additional systemic therapy.

Ketoconazole, like other azoles, inhibits the biosynthesis of ergosterol, an essential component of fungal cell membranes, leading to increased membrane permeability and allowing leakage of intracellular components.

Suggested Dosages/Uses

Sprays and wipes should be applied 1 to 2 times daily. Shampoos and conditioners can be used daily to weekly. Shampoo should typically be left in contact with the skin for at least 10 minutes prior to rinsing; see specific product label for details.

Precautions/Adverse Effects

Avoid contact with eyes; skin irritation is possible. Clients should wash their hands after application or wear gloves when applying. Do not allow animals to chew or lick application site for at least 20 to 30 minutes after application.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Ketoconazole 1%, chlorhexidine 2%, TrisEDTA (spray)	Labeled for dogs and cats	OTC; 8 oz	*Mal-A-Ket® Plus*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine 0.02% (spray)	Labeled for dogs and cats	OTC; 8 oz	*Vetraseb®* CK
Ketoconazole 0.3%, chloroxylenol 0.3% (spray)	Labeled for dogs, cats, and horses	OTC; 4 oz, 8 oz	*Pharmaseb®*
Ketoconazole 0.15%, hydrocortisone 1%, acetic acid 1%, boric acid 2% (spray)	Labeled for dogs, cats, and horses	OTC; 8 oz	*MalAcetic® ULTRA*
Ketoconazole 0.15%, hydrocortisone 1%, acetic acid 1%, boric acid 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz	*MalAcetic® ULTRA*
Ketoconazole 1%, chlorhexidine 2%, acetic acid 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 1 gallon	*Mal-A-Ket®*
Ketoconazole 1%, chlorhexidine 2% (shampoo)	Labeled for dogs and cats; some products also labeled for horses	OTC; 8 oz, 16 oz, and 1 gallon	*KetoHex®; Ceraven CHX+KET; VetraSeb® CeraDerm® CK*
Ketoconazole 1%, chloroxylenol 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 1 gallon	*Pharmaseb®*
Ketoconazole 1%, chlorhexidine 2.3% (shampoo)	Labeled for dog and cats	OTC; 8 oz, 16 oz	*KetoChlor®*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine 0.05% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz	*Phyto CHX+KET®; Ketoseb +PS®*
Ketoconazole 1%, chlorhexidine 2%, glycolic acid 1% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz	*GlyChlorK®*
Ketoconazole 17 mg, chlorhexidine 20 mg, phytosphingosine 0.4 mg (wipes)	Labeled for dogs and cats	OTC; 50 count jar	*Ketoseb +PS*
Ketoconazole 1%, chlorhexidine 2% (wipes)	Labeled for dogs, cats, and horses	OTC; 50 count jar	*KetoHex®; VetraSeb®; CeraDerm® CK*
Ketoconazole 1%, chlorhexidine 2%, acetic acid 2% (wipes)	Labeled for dogs and cats	OTC; 50 count jar	*Mal-A-Ket®*
Ketoconazole 0.3%, chloroxylenol 0.3% (wipes)	Labeled for dogs, cats, and horses	OTC; 50 count	*Pharmaseb®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Ketoconazole 1%, chloroxylenol 2% (mousse)	Labeled for dogs, cats, and horses	OTC; 7 oz	*Pharmaseb®*
Ketoconazole 1%, chlorhexidine 2% (mousse)	Labeled for dogs and cats; some products also labeled for horses	OTC; 6.8 oz	Ceraven CHX+KET; *VetraSeb®*; *CeraDerm®* CK
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine 0.05% (mousse)	Labeled for dogs and cats	OTC; 6.8 oz	*Ketoseb+PS*
Ketoconazole 0.1%, TrisEDTA (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz, 12 oz	*T8 Keto®*
Ketoconazole 0.15%, chlorhexidine 0.15%, TrisEDTA (flush)	Labeled for dogs and cats	OTC; 4 oz, 12 oz	*Mal-A-Ket® Plus*
Ketoconazole 0.3%, chloroxylenol 0.3% (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz, 8 oz	*Pharmaseb®*
Ketoconazole 0.15%, TrisEDTA (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz, 12 oz	*TrizUltra® + Keto*
Ketoconazole 0.2%, chlorhexidine 0.2% (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz	*KetoHex®*
Ketoconazole 0.1%, phytosphingosine 0.01% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*VetraSeb® CeraDerm® Tris*
Ketoconazole 0.2%, chlorhexidine 0.2%, phytosphingosine 0.02% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*Ketoseb+PS®*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Ketoconazole 1% (shampoo)	OTC; 4 oz, 7 oz	*Nizoral A-D®*
Ketoconazole 2% (shampoo)	Rx; 4 oz	*Nizoral®*; generic
Ketoconazole 2% (cream)	Rx; 15 g, 30 g, 60 g	*Nizoral®*; generic
Ketoconazole 2% (foam)	Rx; 50 g, 100 g	*Extina®*; *Ketodan®*; generic
Ketoconazole 2% (gel)	Rx; 45 g	*Xolegel®*

Lime Sulfur, Topical

(***lyme sul***-fur)

Indications/Actions

Lime sulfur is used as a generalized topical treatment for dermatophytosis. It is considered one of the most effective agents against *Microsporum canis.* In addition, it is efficacious for treatment of demodicosis in cats.[1] Lime sulfur can also be useful as an adjunctive treatment of *Malassezia* spp dermatitis, cheyletiellosis, chiggers, sarcoptic and notoedric mange, fur mites, lice, canine demodicosis, and sarcoptic mange, in addition to chorioptic mange in horses.

Lime sulfur has antibacterial and antifungal properties secondary to the formation of pentathionic acid and hydrogen sulfide after application. Lime sulfur may also have keratolytic, keratoplastic, antiparasitic, and antipruritic effects.

Suggested Dosages/Uses

Concentrated dips should be shaken well prior to use and must be diluted. Typically, 4 oz of concentrate should be diluted in 1 gallon of water, mixed well, and carefully applied to affected areas.[2] For chronic or resistant cases, 8 oz of concentrate may be diluted per gallon of water. Treatment may be repeated every 5 to 7 days. See specific product label for detailed dilution and application instructions. Do not rinse or blow-dry animals after application. Do not allow animals to ingest lime sulfur; apply a protective collar until dry if necessary to prevent ingestion.

When used for dermatophytosis, twice-weekly treatments have been recommended if tolerated.[3] Continue treatment until 2 consecutive negative cultures at weekly intervals are obtained.

For confirmed cases of feline surface demodicosis, applications are recommended every 5 to 7 days for a total of 6 treatments. For surface demodicosis treatment trials, 3 applications should be performed; if a significant improvement in clinical signs is noted, 3 more applications should be performed to complete treatment. If no significant improvement is seen after 3 applications, demodicosis should be ruled out and other diagnoses should be considered.

Precautions/Adverse Effects

Use caution when handling lime sulfur as adverse effects may occur after accidental dermal exposure, oral ingestion, or inhalation. Wear gloves and wash hands after handling. Avoid contact with eyes and mucous membranes; safety glasses are recommended.[2] Lime sulfur can stain porous surfaces such as concrete and porcelain and can permanently discolor jewelry. Lime sulfur should only be used in a well-ventilated area, or a respirator should be worn during application.

Lime sulfur is safe to use on pregnant and nursing animals and kittens/puppies over 2 to 3 weeks of age.

Lime sulfur may cause skin irritation or drying. Lime sulfur can temporarily stain light-colored fur and can rarely cause hair loss on the pinnae in cats. The odor of lime sulfur may persist on treated animals after application but is generally tolerable after drying. Oral ingestion can rarely cause nausea and oral ulcers, particularly in cats; an Elizabethan collar can help prevent ingestion from licking and grooming. Systemic toxicosis secondary to dermal exposure of lime sulfur has been reported after improper dilution of the concentrate.[4] Therefore, detailed dilution instructions are important for owners when recommending this therapy.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Lime sulfur 97.8% (dip)	Labeled for dogs, puppies, kittens, and cats; some products also labeled for horses	OTC; 4 oz, 16 oz, 1 gallon	*LimePlus*®; generic Shake well and dilute before use.

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Miconazole, Topical

(mye-***kah***-nah-zole)

For ophthalmic use, see ***Miconazole Ophthalmic***

For otic use, see ***Antifungal Agents, Otic*** *and* ***Corticosteroid/Antimicrobial Agents, Otic***

Indications/Actions

Topical miconazole is used for the treatment of *Malassezia* spp dermatitis and infections caused by *Microsporum canis, Microsporum gypseum, and Trichophyton mentagrophytes*.[1] Lotions, sprays, and creams are generally used for localized lesions; patients with severe, generalized infections may require systemic antifungal therapy. Topical miconazole products are typically only minimally effective for dermatophytosis when used as a single agent; adjunctive systemic treatment is commonly required.

Miconazole is an imidazole antifungal with activity against dermatophytes and yeast. Its actions are a result of altering the permeability of fungal cellular membranes and interfering with peroxisomal and mitochondrial enzymes, leading to intracellular necrosis. Miconazole products are fungicidal with repeated application. Miconazole has been shown to be active against a series of gram-positive bacteria and to help repair the skin barrier function and mitigate some inflammatory cell reactions.[2]

Suggested Dosages/Uses

Sprays, creams, lotions, or wipes should be applied 1 to 2 times daily. Shampoos and conditioners may be used daily to weekly. Typically, shampoo should be left in contact with the skin for at least 10 minutes prior to rinsing. For veterinary products, refer to the specific product label for details on use.

Precautions/Adverse Effects

Avoid contact with eyes. Wash hands thoroughly after application to prevent the spread of fungal infections.[1] Do not allow treated animals to lick or chew at affected areas for at least 20 to 30 minutes after application.

Skin irritation is uncommon but possible. Wipes and sprays that contain alcohol may be severely irritating to inflamed, eroded, or ulcerated skin.

Veterinary-Labeled Products

NOTE: Miconazole nitrate is the salt generally used in pharmaceutical products. Some products express concentration as base, others express concentrations of the salt.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Miconazole 1% (spray)	Labeled for dogs and cats	Rx; 60 mL, 120 mL, 240 mL	*MicaVed®; Micazole®; Priconazole®; Conzol®*
Miconazole nitrate 2% (spray)	Labeled for dogs, cats, and horses	OTC; 4 oz, 16 oz	generic
Miconazole nitrate 2%, chlorhexidine 2% (spray)	Labeled for dogs, cats, and horses	Rx; 8 oz	*Malaseb®*
Miconazole nitrate 2%, chlorhexidine 2%, TrisEDTA (spray)	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, 32 oz	*MiconaHex+Triz®*
Miconazole nitrate 0.2%, chlorhexidine 0.2% (flush)	Labeled for dogs, cats, and horses	Rx; 4 oz, 12 oz	*Malaseb®*
Miconazole 0.1% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*BioTris®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Miconazole 1% (lotion)	Labeled for dogs and cats	Rx; 30 mL, 60 mL	*Conzol®; Priconazole®; MicaVed®; Miconosol®; Micazole®*
Miconazole nitrate 2%, chlorhexidine 2%, TrisEDTA (wipes)	Labeled for dogs, cats, and horses	OTC; 50 count jar	*MiconaHex+Triz®*
Miconazole nitrate 15.1 mg/mL, gentamicin sulfate 1.5 mg/mL, hydrocortisone aceponate 1.11 mg/mL (suspension)	Labeled for otic use in dogs	Rx; 10 mL	*Easotic®* Topical use is extra-label.
Miconazole nitrate 23 mg/mL, polymyxin B sulfate 0.5293 mg/mL, prednisolone acetate 5 mg/mL (suspension)	Labeled for otic use in dogs	Rx; 15 mL, 30 mL	*Surolan®* Topical use is extra-label.
Miconazole nitrate 2%, chlorhexidine 2% (shampoo)	Labeled for dogs, cats, and horses	Rx; 8 oz, 16 oz	*Malaseb®*
Miconazole nitrate 2%, chlorhexidine 2%, TrisEDTA (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, 1 gallon	*MiconaHex+Triz®*
Miconazole nitrate 2%, chloroxylenol 1%, salicylic acid 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, 1 gallon	*Sebozole®*
Miconazole 2% (shampoo)	Labeled for dogs and cats	OTC; 12 oz	generic
Miconazole nitrate 2%, colloidal oatmeal 2% (shampoo)	Labeled for dogs, cats, puppies, and kittens	OTC; 12 oz, 1 gallon	generic
Miconazole nitrate 2%, chlorhexidine 2% (shampoo)	Labeled for dogs and cats	OTC; 8 oz, 12 oz, 16 oz, 1 gallon	*BioHex®; Micoseb®*
Miconazole nitrate 2%, chlorhexidine 2%, TrisEDTA (mousse)	Labeled for dogs, cats, and horses	OTC; 7.1 oz	*MiconaHex+Triz®*

Human-Labeled Products

In addition to the products listed below, there are 2% topical vaginal creams, vaginal suppositories, 2% powders, and spray powders available. Most human-labeled products are OTC.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Miconazole 2% (ointment)	OTC; depending on the product: 4 g, 30 g, 57 g, 56.7 g, 113 g, and 142 g packages	generic
Miconazole 2% (aerosol spray or powder)	OTC; 30 g, 43 g, 85 g, 90 g, 113 g, and 133 g cans	generic
Miconazole 2% (cream)	OTC; many sizes available	generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Nystatin/Nystatin Combinations, Topical

(nye-***sta***-tin)

For systemic use, see ***Nystatin.***

Indications/Actions

Nystatin is typically used in combination with antibiotics and glucocorticoids in small animal medicine; it is generally not used as a single agent due to limited dosage forms and alternative antifungal agents. Combination products containing neomycin, thiostrepton, and triamcinolone are FDA-approved in dogs and cats for treatment of inflammatory dermatologic disorders, particularly associated with bacterial or *Candida albicans* infections.[1] Use for topical yeast infections without a bacterial component should be avoided to prevent antibiotic resistance.

Nystatin is active against many yeasts and yeast-like fungi, such as *Malassezia spp* and *Candida* spp. Nystatin binds to sterols in fungal cell membranes, thereby increasing membrane permeability and allowing leakage of intracellular components. Nystatin does not have activity against bacteria and is ineffective against other fungi.

Suggested Dosages/Uses

Although nystatin is rarely used as a single topical agent, human products can be used if needed. In humans, creams are preferred to ointments in intertriginous areas; topical powders are used for moist lesions.

After cleaning affected areas, nystatin combination products should be applied sparingly in a thin film. Frequency of application is based on severity of infection and may range from once daily to once weekly for mild infections and up to 2 to 3 times daily for severe infections.[1] Frequency can be decreased as infection improves. Products that contain triamcinolone are best suited for focal (eg, pedal) or multifocal lesions that require relatively short treatment durations (eg, less than 2 months).

Precautions/Adverse Effects

Avoid empiric use of combination products or use in the absence of concurrent bacterial infection to help prevent bacterial resistance.

Avoid contact with eyes. Wear gloves or wash hands after application. Do not let the animal lick or chew at the treated area(s) for at least 20 to 30 minutes after application.

Topical nystatin is considered nontoxic and is generally well-tolerated, although hypersensitivity reactions are possible.[2] Potential adverse effects reported in humans include burning, itching, rash, and eczema.[3] The combination veterinary products are usually well-tolerated when used on the skin; however, neomycin can cause localized sensitivity, and glucocorticoids can potentially cause cutaneous atrophy or HPA-axis suppression.[1] At least a 2-week withdrawal period is recommended prior to intradermal or allergy serum testing for products containing corticosteroids, such as triamcinolone.[4]

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Nystatin 100,000 units/g, neomycin sulfate 2.5 mg/g, thiostrepton 2500 units/g, triamcinolone acetate 1 mg/g (cream)	Labeled for dogs and cats	Rx; 7.5 g, 15 g	*Animax®*
Nystatin 100,000 units/g, neomycin sulfate 2.5 mg/g, thiostrepton 2500 units/g, triamcinolone acetonide 1 mg/g (ointment)	Labeled for dogs and cats	Rx; 7.5 mL, 15 mL, 30 mL, and 240 mL	*Animax®; Quadritop®; Derma-Vet®; Dermalog®; Dermalone®; EnteDerm®*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Nystatin 100,000 units/g (powder)	Rx; 15 g, 30 g, and 60 g	*Nyamyc®; Nystop®*; generic
Nystatin 100,000 units/g (ointment)	Rx; 15 g, 30 g	generic
Nystatin 100,000 units/g (cream)	Rx; 15 g, 30 g	generic
Nystatin 100,000 units/g, triamcinolone acetate 1mg/g (ointment)	Rx; 15 g, 30 g, and 60 g	generic
Nystatin 100,000 units/g, triamcinolone acetate 1 mg/g (cream)	Rx; 15 g, 30 g, and 60 g	generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Selenium Sulfide, Topical

(si-***leen***-ee-um ***sul***-fide)

Indications/Actions

Selenium sulfide may be useful in dogs for treating seborrheic disorders (mainly for seborrhea oleosa) and for the adjunctive treatment of *Malassezia* dermatitis, particularly in dogs exhibiting signs of waxy, greasy, or scaly (seborrheic) dermatitis.

Selenium sulfide possesses antifungal and sporicidal activity, keratolytic, keratoplastic, and degreasing properties. It alters epidermal turnover in cells of the epidermis and follicular epithelium and interferes with hydrogen bond formation of keratin, thereby reducing corneocyte production. The antifungal mechanism of action of selenium sulfide is not well understood.

Suggested Dosages/Uses

Selenium sulfide shampooing frequency is patient dependent. In general, medicated shampoos should remain in contact with the skin for 10 minutes. Alternatively, allow 3 to 5 minutes of contact time, rinse thoroughly, and then repeat the procedure. Thoroughly rinse off the shampoo to prevent skin irritation and excessive drying.

Precautions/Adverse Effects

Selenium sulfide products should not be used on cats. Avoid contact with eyes and mucous membranes. Selenium sulfide can damage or discolor jewelry. Clients should wear gloves when using these products.

Selenium sulfide can be irritating, especially to the scrotal area. It can also cause excessive drying and hair-coat staining or discoloration. Do not use these products on broken or inflamed skin. After discontinuation, rebound seborrhea can occur and can be more severe than pretreatment presentation.

GI effects such as nausea, vomiting, or diarrhea may occur if selenium sulfide is ingested orally. Neurologic signs are possible if large quantities are ingested. In case of substantial oral ingestion, consultation with a 24-hour poison consultation center specializing in providing veterinary-specific information is recommended. For general information related to overdose and toxin exposures, as well as contact information for poison control centers, refer to ***Appendix***.

Veterinary-Labeled Products

There are no veterinary labeled products containing selenium sulfide currently available in the United States; products may be available in other countries.

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Selenium sulfide 1% (shampoo)	OTC; 11 oz	*Selsun Blue*®, generic
Selenium sulfide 2.25% (shampoo)	Rx; 6 oz	generic
Selenium sulfide 2.3% (shampoo)	Rx; 6 oz	generic
Selenium sulfide 2.5% (lotion)	Rx; 4 oz	generic
Selenium sulfide 2.25% (foam)	Rx; 2.5 oz	*Tersi*® Marketing status uncertain

Terbinafine, Topical

(ter-***bin***-a-feen)

For systemic use, see ***Terbinafine.***

For otic use, see ***Corticosteroid/Antimicrobial Agents, Otic.***

Indications/Actions

Terbinafine is an allylamine antifungal agent that may be useful for localized lesions associated with *Malassezia* infections. In an in vitro study, topical 1% terbinafine was shown to have good efficacy as adjuvant focal therapy against *Microsporum canis* with good residual activity.[1]

Terbinafine inhibits the biosynthesis of ergosterol by inhibiting the fungal squalene epoxidase enzyme. The resultant depletion of ergosterol within the fungal cell membrane and the intracellular accumulation of squalene are believed to be responsible for the fungicidal effect of terbinafine. It is fungicidal against dermatophytes but may only be fungistatic against yeast organisms.

Suggested Dosages/Uses

Topical terbinafine may be applied to skin areas affected with *Malassezia* once to twice a day. Based on in vitro data, less frequent application (ie, once daily to every other day) may be sufficient for adjuvant therapy of *M canis* skin infections due to the demonstrated residual activity of terbinafine.[1]

Precautions/Adverse Effects

Avoid contact with eyes and mucous membranes. Wash hands after application. Do not allow animals to lick or chew at the treated area(s) for at least 30 minutes after application.

Local irritation is possible, including burning, pruritis, or rash.

Veterinary-Labeled Products

None labeled for use on skin.

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Terbinafine hydrochloride 1% (cream)	OTC; 12 g, 15 g, 24 g, 30 g, 42 g	*Lamisil AT*®; generic
Terbinafine hydrochloride 1% (spray)	OTC; 125 mL	*Lamisil AT*®

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Antiseptic Agents, Topical

Acetic Acid/Boric Acid, Topical

(ah-***see***-tik ***ass***-id; ***bor***-ik ***assi***-id)

For otic products, refer to ***Antiseptic Agents, Otic***

Indications/Actions

Acetic and boric acids are indicated for the topical treatment of skin infections caused by bacteria, including *Staphylococcus* spp,[1] *Pseudomonas* spp,[2,3] and yeast, such as *Malassezia* spp. They are also indicated for fold dermatitis, acute moist dermatitis, pododermatitis, and seborrhea oleosa.

Acetic and boric acids are potent antibacterial and antifungal agents with a rapid killing effect. They also possess ceruminolytic, keratolytic, keratoplastic, and astringent properties.

Some products also contain other antimicrobials, such as chlorhexidine and ketoconazole, or anti-inflammatory agents, such as hydrocortisone for anti-pruritic effects.

Suggested Uses/Dosages

Sprays and wipes may be applied up to 2 to 3 times a day. Shampoos and conditioners can be used daily to weekly, during or after routine bathing. Leave shampoos in contact with the skin for at least 10 minutes prior to rinsing. Refer to the product label for specific instructions.

Acetic acid and boric acid topical products have concentrations ranging from 1% to 2% for each acid. One study found that acetic acid and boric acid concentrations as low as 0.5% have bactericidal effects against *Staphylococcus pseudintermedius* in vitro.[1]

Precautions/Adverse Effects

Skin redness and irritation may occur. Do not let the animal lick or chew at affected areas for at least 20 to 30 minutes after application. For products that also contain hydrocortisone, caution should be used to avoid excessive or frequent use due to its potential cutaneous and systemic adverse effects.

Veterinary-Labeled Products

SHAMPOOS

Acetic acid 2%, boric acid 2%	Labeled for dogs, cats, and horses	OTC; 12 oz, 16 oz, and 1 gallon	*MalAcetic®*
Ketoconazole 0.15%, hydrocortisone 1%, acetic acid 1%, boric acid 2% Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 8 oz	*MalAcetic® ULTRA*
Ketoconazole 1%, acetic acid 2%, chlorhexidine 2%	Labeled for dogs, cats, and horses	OTC; 8 oz and 1 gallon	*Mal-A-Ket®*
Ketoconazole 1%, acetic acid 2%, chlorhexidine 2%		OTC; 12 oz	*HexaChlor-K®*

SPRAYS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Acetic acid 2%, boric acid 2%	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz	*MalAcetic®*
Ketoconazole 0.15%, hydrocortisone 1%, acetic acid 1%, boric acid 2% Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 8 oz	*MalAcetic® ULTRA*
Ketoconazole 1% acetic acid 2%, chlorhexidine 2%	Labeled for dogs and cats	OTC; 8 oz	*HexaChlor-K®*

WIPES

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Acetic acid 2%, boric acid 2%	Labeled for dogs and cats	OTC; 25 count, 100 count jar	*MalAcetic®*
Ketoconazole 1%, acetic acid 2%, chlorhexidine 2%	Labeled for dogs and cats	OTC; 50 count jar	*HexaChlor-K®; Mal-A-Ket®*
Hydrocortisone 1%, acetic acid 1%, boric acid 1%; other ingredients: ceramide complex	Labeled for dogs and cats	OTC; 25 count jar	*MalAcetic® HC*

Human-Labeled Products

There are several OTC human products available containing acetic acid or boric acid (alone or containing other ingredients). For more information on human-labeled acetic acid or boric acid products, refer to a comprehensive human drug reference or contact a pharmacist.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Chlorhexidine, Topical

(klor-***heks***-ih-deen) *Nolvasan®*

Indications/Actions

Chlorhexidine is a synthetic phenol-related bisbiguanide that exists as acetate, gluconate, and hydrochloride salts. It is a topical antiseptic that has activity against many bacteria and fungi. It appears to have greater efficacy for bacterial infections, particularly *Staphylococcus* spp, than for yeast or dermatophyte infections. Chlorhexidine prevents the outgrowth of bacterial spores but does not prevent germination and is not sporicidal. It can be bacteriostatic or bactericidal depending on concentration.[1] Chlorhexidine is a membrane-active agent that acts by damaging bacterial cytoplasmic membranes. Antifungal activity can be obtained with chlorhexidine concentrations of 2% or higher.

Because chlorhexidine causes less drying and is usually less irritating than benzoyl peroxide, it is sometimes used in patients that cannot tolerate benzoyl peroxide. However, it does not have the keratolytic, comedolytic, or degreasing effects of benzoyl peroxide. Chlorhexidine possesses some residual effects and can remain active on the skin after rinsing.

In one study comparing in vitro efficacy of shampoos, chlorhexidine was shown to be the most effective biocide against *Pseudomonas* spp, methicillin-susceptible and methicillin-resistant *Staphylococcus* spp, and *Malassezia* spp.[2] Another controlled clinical study showed that bathing twice weekly with a 4% chlorhexidine-based shampoo for 4 weeks was as effective as systemic therapy with amoxicillin/clavulanic acid for the treatment of canine superficial pyoderma associated with susceptible and methicillin-resistant *Staphylococcus pseudintermedius*.

Another study found chlorhexidine 2% solution with alcohol had similar efficacy as povidone-iodine 7.5% in reducing total skin bacteria load and preventing surgical site infections in dogs.[3]

Chlorhexidine products may also contain other ingredients, such as antifungals (eg, ketoconazole, miconazole, climbazole), which provide superior antifungal activity as compared with chlorhexidine alone, salicylic acid, phytosphingosine, and hydrocortisone.

Suggested Dosages/Uses

For wound irrigation or foot soaking, diluting the chlorhexidine solution with water to 0.05% to 0.1% is recommended. Sprays, mousses, and wipes/pads are typically applied 1 to 2 times daily. Shampoos and conditioners are typically used daily to weekly. Shampoos should be left in contact with the skin for at least 10 minutes prior to rinsing. For veterinary products, refer to the product label for details and specific instructions.

A nosocomial outbreak of chlorhexidine-resistant *Serratia marcescens* has been linked to improper disinfectant procedures in one veterinary hospital.[4]

Precautions/Adverse Effects

Keep chlorhexidine products away from the eyes. Chlorhexidine is safely used on cats, although irritation and corneal ulcers have been reported. Chlorhexidine may impair wound healing; it is not recommended for long-term use, particularly on granulating lesions.

Hypersensitivity and local skin irritant reactions are possible. Clients should wash their hands after application or wear gloves when applying. The likelihood of irritation increases with increased concentrations. Do not let the animal lick or chew treated areas for at least 20 to 30 minutes after application. Advise clients that blue chlorhexidine solutions can stain furniture, carpets, and the animal.

Veterinary-Labeled Products

Several teat dip and udder wash products, a lubricant, and oral rinses are also available. Several trade names are used for chlorhexidine products, including *Nolvasan®*, *Chlorhexiderm®*, *Dermachlor®*, *Chlorasan®*, *Chloradine®*, *Privasan®*, and *Chlorhex®*.

OINTMENTS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 1%		OTC; 7 oz, 16 oz	*Chlorhex®; Nolvasan®*
Chlorhexidine 2%	Labeled for dogs, cats, and horses	OTC; 7 oz, 16 oz	
Chlorhexidine 4%	Labeled for dogs, cats, and horses	OTC; 1 lb	*Vetasan®*

SPRAYS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine (various concentrations)	Labeled for dogs, cats, and horses	OTC; various sizes	Some products contain aloe.
Chlorhexidine 4%, TrizEDTA	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz	*TrizChlor 4®*
Hydrocortisone 1%, chlorhexidine 4%, TrizEDTA	Labeled for dogs, cats, and horses	OTC; 8 oz	*TrizChlor 4HC®* Avoid excessive or prolonged use due to product's steroid content.
Ketoconazole 1%, chlorhexidine 2%, TrizEDTA	Labeled for dogs and cats	OTC; 8 oz	*Mal-A-Ket®Plus*
Miconazole 2%, chlorhexidine 2%, TrizEDTA Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 32 oz	*MiconaHex®*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine 0.02% Other ingredients: aloe barbadenis, lactic acid	Labeled for dogs and cats	OTC; 8 oz	*VetraSeb CK*
Chlorhexidine 3%, phytosphingosine 0.05%	Labeled for dogs and cats	OTC; 6.8 oz	*DOUXO®*
Chlorhexidine 4% Other ingredients: isopropyl alcohol 4%	Labeled for dogs and cats	OTC; 8 oz	*Chlorhex®*

SHAMPOOS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 4%	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, 16 oz, 1 gallon	*ChlorhexiDerm®; Chlorhex®; Max Chlorhexidine®*
Chlorhexidine 4%, TrizEDTA	Labeled for dogs, cats, and horses	OTC; 8 oz, 1 gallon	*TrizCHLOR®*
Hydrocortisone 1%, chlorhexidine 4%, TrizEDTA	Labeled for dogs, cats, and horses	OTC; 8 oz	*TrizCHLOR®4HC* Avoid excessive or prolonged use due to its steroid content.
Chlorhexidine 4% Other ingredients: ceramide III, microsilver	Labeled for dogs and cats	OTC; 8 oz, 16 oz, and 1 gallon	*Hexaderm®* Soap-free
Chlorhexidine 2%		OTC; 8 oz, 16 oz, and 1 gallon	
Chlorhexidine 3%, ophytrium 0.5%	Labeled for dogs and cats	OTC; 200 mL, 500 mL	*DOUXO S3® PYO*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine salicyloyl 0.05%	Labeled for dogs and cats	OTC; 6.8 oz	*Vetraseb CK®*
Ketoconazole 1%, hydrocortisone 1%, chlorhexidine 2% Other ingredients: vitamin-rich emollients	Labeled for dogs, cats, and horses	OTC; 16 oz	*EquiShield®*
Ketoconazole 1%, acetic acid 2%, chlorhexidine 2%	Labeled for dogs, cats, and horses	OTC; 8 oz, 1 gallon	*Mal-A-Ket®*
Miconazole 2%, chlorhexidine 2%, TrizEDTA Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 1 gallon	*MiconaHex®*
Ketoconazole 1%, chlorhexidine 2% Other ingredients: gylcerin, isopropyl alcohol	Labeled for dogs and cats	OTC; 8 oz, 16 oz, and 1 gallon	*KetoHex®*
Ketoconazole 1%, chlorhexidine 2.3% Other ingredients: alkyl polyglucoside (polysaccharide); chitosanide; D-mannose, D-galactose, L-rhamnose (monosaccharides); *Spherulite®* microcapsules	Labeled for dogs and cats	OTC; 8 oz, 16 oz, and 1 gallon	*KetoChlor®*
Miconazole nitrate 2%, chlorhexidine 2%	Labeled for dogs, cats, and horses	OTC; 8 oz, 12 oz, 16 oz, and 0.5 gallon	*Malaseb®*
Miconazole nitrate 2%, chlorhexidine 2% Other ingredients: ceramide III, microsilver	Labeled for dogs and cats	OTC; 8 oz, 16 oz, and 1 gallon	*BioHex®* Soap-free

SOLUTIONS, FLUSHES, AND SCRUBS

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 2% (solution)	May be labeled for use on dogs, horses, cattle, and swine	OTC; 16 oz, 1 gallon	
Chlorhexidine 20% (solution)		OTC	*Chlorhexidine®* For 1%: Dilute 6 oz into 1 gallon of water or shampoo. For 2%: Dilute 12 oz into 1 gallon of water or shampoo.
Chlorhexidine (flush)		OTC; 4 oz, 12 oz	
Chlorhexidine 0.2% (flush) Other ingredients: glycerin, isopropyl alcohol	Labeled for dogs and cats	OTC; 4 oz	*Chlorhexis®*
Chlorhexidine 0.15%, TrizEDTA (flush)	Labeled for dogs and cats	OTC; 4 oz	*TrizChlor®*
Ketoconazole 0.15%, chlorhexidine 0.15%, TrizEDTA (flush)	Labeled for dogs and cats	OTC; 4 oz, 12 oz	*Mal-A-Ket®* A multi-cleanse flush to aid in the treatment of bacterial and fungal (dermatophytosis and *Malassezia* spp) infections
Chlorhexidine 0.2%, lidocaine 0.5% (flush) Other ingredients: benzoic acid, glycerin, malic acid, propylene glycol, salicylic acid		OTC; 4 oz	*Dermachlor®*
Miconazole 2%, chlorhexidine 2% (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz, 12 oz	*Malaseb®*
Chlorhexidine 2% (flush)		OTC; 4 oz, 8 oz, and 16 oz	
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine 0.02% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*Ketoseb®*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine salicyloyl 0.05% (flush)	Labeled for dogs and cats	OTC; 6.8 oz	*Vetraseb CK®*
Chlorhexidine 2%, chlorhexidine 4% (scrub)		OTC; 1 gallon	

WIPES

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 3%, ophytrium 0.5%	Labeled for dogs and cats	OTC; 30 count jar	*DOUXO S3® PYO* Alcohol-free
Ketoconazole 17 mg, chlorhexidine 20 mg, phytosphingosine 0.4 mg	Labeled for dogs and cats (*Ketoseb+PS®*) and horses (*PhytoVet CK®*)	OTC; 50 count jar	*Ketoseb+PS®; PhytoVet CK®*
Ketoconazole 1%, chlorhexidine 2%, phytosphingosine salicyloyl 0.04% Other ingredients: glycerin, lactic acid, propylene glycol	Labeled for dogs and cats	OTC; 60 count jar	*Vetraseb CK®*
Ketoconazole 17 mg, chlorhexidine 20 mg	Labeled for dogs, cats, and horses	OTC; 50 count jar	*KetoHex®*

Ketoconazole 1%, acetic acid 2%, chlorhexidine 2%	Labeled for dogs and cats	OTC; 50 count jar	*Mal-A-Ket®*
Miconazole 2%, chlorhexidine 2%, TrizEDTA Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 50 count jar	*MiconaHex®*
Chlorhexidine 4%, TrizEDTA	Labeled for dogs and cats	OTC; 50 count jar	*TrizChlor®*
Climbazole, chlorhexidine 2% Other ingredients: ceramide III, microsilver	Labeled for dogs and cats	OTC; 50 count jar	*BioHex®*

MOUSSES

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 3%, ophytrium 0.5%	Labeled for dogs and cats	OTC; 5 oz	*DOUXO S3® PYO* Water-based. Leave-on. To be used between bathing or when bathing is not practical
Miconazole 2%, chlorhexidine 2%	Labeled for dogs and cats	OTC; 6.8 oz	*Ketoseb®* Water-based. Leave-on. To be used between bathing or when bathing is not practical
Chlorhexidine 4%, TrizEDTA	Labeled for dogs, cats, and horses	OTC; 7.1 oz	*TrizChlor 4®*
Miconazole 2%, chlorhexidine 2%, TrizEDTA Other ingredients: ceramide complex	Labeled for dogs, cats, and horses	OTC; 7.1 oz	*Miconahex+Triz®*
Climbazole 0.5%, ceramides III 0.05%, chlorhexidine 3%, microsilver 1%	Labeled for dogs and cats	OTC; 6.8 oz	*BioHex®* Leave-on. To be used between bathing or when bathing is not practical

ORAL RINSES

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chlorhexidine 0.12%	Labeled for dogs and cats	OTC; 8 oz	*Dentahex®* For daily use after each meal. Avoid touching tip of bottle to gums.
Chlorhexidine 0.13%	Labeled for dogs and cats	OTC; 8 oz	*Clenz-a-dent®* Orange flavored

Human-Labeled Products

There are several topical skin cleansers available in the 2% to 4% range. Trade names include *Hibiclens®*, *Hibistat®*, *Betasept®*, *Exidine®*, *Dyna-Hex®*, and *BactoShield®*.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Chloroxylenol, Topical

PCMX

(kloro-***zye***-len-ol)

For otic use, see ***Cleaning/Drying Agents, Otic*** *and* ***Antiseptic Agents, Otic.***

Indications/Actions

Chloroxylenol is an antiseptic with demonstrated efficacy against gram-negative and gram-positive bacteria, in addition to a wide variety of fungal organisms, and against RNA and DNA viruses. It is slower acting than other antiseptic agents[1] and has minimal residual effects.[2] It can be used in presurgical preparation of the skin; in cleaning wounds; and in the treatment of bacterial, fungal, and yeast skin infections. An in vitro study comparing the efficacy of antimicrobial shampoos showed that a shampoo containing chloroxylenol 2% was not effective against *Pseudomonas* spp and methicillin-susceptible and methicillin-resistant *Staphylococcus* spp; however, it was effective against *Malassezia* spp.[3] Another in vitro study found that chloroxylenol was inferior to chlorhexidine against MRSA biofilms.[4]

Chloroxylenol, also known as PCMX, is a chlorinated phenolic antiseptic. Its antibacterial action is thought to be due to the disruption of bacterial membranes.

Suggested Uses/Dosages

Shampoos should be left in contact with the skin for at least 10 minutes prior to rinsing. Refer to product label for details on individual use. Sprays and wipes/pads can be applied 1 to 3 times daily according to the patient's needs. Do not let the animal lick or chew at treated areas for at least 20 to 30 minutes after application.

Precautions/Adverse Effects

Chloroxylenol may cause skin irritation.

Veterinary-Labeled Products

This list represents only products where chloroxylenol is one of the main active ingredients.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Chloroxylenol 2% (scrub) Other ingredients: deionized water, hydroxyethylcellulose, polysorbate-20, polyquaternium-7, propylene glycol, sodium laureth sulfate, sodium lauryl sulfate		OTC; 16 oz, 1 gallon	Used for surgical scrub and preoperative skin preparation. Do not dilute.
Chloroxylenol 2%, salicylic acid 2%, sodium thiosulfate 2% (shampoo) Other ingredients: citric acid, propylene glycol	Labeled for dogs and cats	OTC; 16 oz, 1 gallon	Has antiseborrheic effect
Miconazole nitrate 2%, chloroxylenol 1% (shampoo) Other ingredients: propylene glycol, salicylic acid	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 1 gallon	*Sebozole®* Has antibacterial, antifungal, and antiseborrheic activity. Shake well.
Ketoconazole 1%, chloroxylenol 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 8 oz, 16 oz, and 1 gallon	*Pharmaseb®*
Ketoconazole 0.3%, chloroxylenol 0.3% (wipes)	Labeled for dogs, cats, and horses	OTC; 60 count	*Pharmaseb®* Water-based
Ketoconazole 0.3%, chloroxylenol 0.3% (flush) Other ingredients: benzyl alcohol, propylene glycol	Labeled for dogs, cats, and horses	OTC; 4 oz, 8 oz	*Pharmaseb®* Water-based. Can be used as a cleansing agent for the skin and ears
Ketoconazole 0.33%, chloroxylenol 0.33% (flush) Other ingredients: benzyl alcohol, propylene glycol, sodium borate decahydrate	Labeled for dogs, cats, and horses	OTC; 8 oz	*Ketomax®* Contains sodium borate decahydrate to obtain pH of 8.5
Ketoconazole 0.3%, chloroxylenol 0.3% (spray) Other ingredients: benzyl alcohol, propylene glycol	Labeled for dogs, cats, and horses	OTC; 4 oz, 8 oz	*Pharmaseb®* Water-based

Human-Labeled Products

There are several topical products available containing chloroxylenol that are usually combined with other ingredients, such as hydrocortisone, menthol, pramoxine, and benzocaine. They are presented in different forms (creams, ointments, lotions, and shampoos) but are not commonly used in dogs and cats. Trade names include *Aurinol®*, *Calamycin®*, *Cortamox®*, *Cortane-B®*, *Dermacoat®*, and *Foille®*.

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Enzymes, Topical (Lactoperoxidase, Lysozyme, Lactoferrin)

For otic use, see ***Cleaning/Drying Agents, Otic.***

Indications/Actions

Lactoperoxidase, lysozyme, and lactoferrin are topical enzymes that work synergistically when combined and are reported to be effective against bacterial, fungal, and viral microorganisms. They are listed by the manufacturer to be effective against *Staphylococcus* spp, *Pseudomonas* spp, *Malassezia* spp, *Candida albicans*, and *Microsporum* spp. Lactoperoxidase combined with hydrogen peroxide, thiocyanate, and/or iodide produces hypothiocyanite or hypoiodite ions that are bactericidal by oxidizing components of bacterial cell walls. Hypoiodite also is fungicidal. Lactoferrin acts as a bacteriostatic agent against many bacteria by depriving them of iron.

They can be used alone but are often used in combination with hydrocortisone for mild itching or as an adjunctive treatment for more severe itching, such as atopic dermatitis.

Suggested Uses/Dosages

Sprays and creams are labeled for use 1 to 2 times a day. Shampoos and conditioners should be used daily to weekly. Shampoo should be in contact with the skin for 3 to 5 minutes prior to rinsing well. Products are labeled as worry-free if licked, but animals should be prevented from licking or chewing treated areas for at least 20 to 30 minutes after application if possible. Refer to product label for details on individual product use.

Precautions/Adverse Effects

Overall, these products appear to be safe. There are no reported adverse effects, but skin irritation is possible. Caution should be used to avoid excessive or long-term use of products containing glucocorticoids due to potential adverse effects such as cutaneous atrophy and systemic absorption.

Veterinary-Labeled Products

Refer to each product label for details on use, inactive ingredients, and any precautions.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Lactoferrin, lactoperoxidase, lysozyme (spray)	Labeled for all pets of all ages	OTC; 2 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme, hydrocortisone 0.5% (spray)	Labeled for all pets of all ages	OTC; 2 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme, hydrocortisone 1% (spray)	Labeled for all pets of all ages	OTC; 2 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme (cream)	Labeled for companion animals	OTC; 1 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme, hydrocortisone 0.5% (cream)	Labeled for companion animals	OTC; 1 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme, hydrocortisone 1% (cream)	Labeled for companion animals	OTC; 1 oz	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme (shampoo)	Labeled for all pets of all ages	OTC; 12 oz, 1 gallon	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme (leave-on conditioner)	Labeled for all pets of all ages	OTC; 12 oz, 1 gallon	*Zymox®*
Lactoferrin, lactoperoxidase, lysozyme (shampoo)	Labeled for horses	OTC; 12 oz, 1 gallon	*Zymox®Equine Defense*
Lactoferrin, lactoperoxidase, lysozyme (conditioner)	Labeled for horses	OTC; 12 oz, 1 gallon	*Zymox® Equine Defense*
Lactoferrin, lactoperoxidase, lysozyme (spray)	Labeled for horses	OTC; 8 oz	*Zymox® Equine Defense*
Lactoferrin, lactoperoxidase, lysozyme (cream)	Labeled for horses	OTC; 2.5 oz	*Zymox® Equine Defense*

Human-Labeled Products

None

Hypochlorous Acid, Topical

(hye-poe-***klor***-us ***ass***-id)

For ophthalmic use, see ***Irrigating Solutions, Ophthalmic****.*

For otic use, see ***Antiseptic Agents, Otic*** *and* ***Cleaning/Drying Agents, Otic****.*

Indications/Actions

Topical hypochlorous acid is used for the management of wounds, abscesses, cuts, abrasions, skin irritations, ulcers, postsurgical incision sites, burns, wound odors, and to accelerate healing. It may be used for the prevention or treatment of bacterial skin infections, including methicillin-resistant *Staphylococcus* spp and *Pseudomonas* spp; however, one small, controlled study investigating the efficacy of hypochlorous acid 0.011% applied to the skin twice daily for 4 weeks for the treatment of canine superficial pyoderma showed no statistically significant difference as compared with placebo. Hypochlorous acid also has antifungal and antiviral properties and is reported to reduce inflammation, pain, and itching.

Hypochlorous acid products act rapidly against gram-positive and gram-negative bacteria and fungal/yeast organisms. Some products contain oxychlorine (bleach-like) compounds; many products have a neutral pH.

The mechanism of action of hypochlorous acid is to disrupt the cellular membrane of single-cell organisms. Because the mechanism of action is primarily chemical in nature, resistance appears unlikely.

Suggested Dosages/Uses

Clip hair if necessary and apply to affected areas up to 3 times daily; may be used with dressing applications on wounds or in combination with a cleanser. These products are generally safe to use around the eyes, nose, and mouth, and no rinsing is typically required; refer to specific product labels for details on use.

Precautions/Adverse Effects

No precautions or adverse effects have been reported. Hypochlorous acid products are generally non-toxic and safe if licked or ingested. Although they do not sting or irritate the skin, the treated area may redden secondary to increased blood flow. Clients should be warned about possible bleaching/staining of carpets, clothing, and jewelry.

Veterinary-Labeled Products

Below is a representative sample of the products available.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Hypochlorous acid 0.009% (liquid)	Labeled for cats	OTC; 2 oz	*Vetericyn Plus® Feline Facial Therapy*
Hypochlorous acid 0.012% (spray)	Labeled for all animals	OTC; 3 oz, 4 oz, 8 oz, and 16 oz	*Vetericyn Plus® Reptile Wound and Skin Care; Vetericyn Plus® Poultry Care; Vetericyn Plus® All Animal Wound and Skin Care; Vetericyn Plus® Equine Wound and Skin Care*
Hypochlorous acid 0.012% (spray gel)	Labeled for all animals	OTC; 4 oz, 8 oz, and 16 oz	*Vetericyn Plus® Hydrogel; Vetericyn Plus® Feline Hydrogel*
Hypochlorous acid 0.015% (spray)	Labeled for all animals	OTC; 16 oz	*Vetericyn® Pink Eye Spray*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Hypochlorous acid 0.008%, sodium hypochlorite 0.002% (spray/gel/solution)	Depending on the labeling: OTC/Rx; 1.5 oz to 990 mL	*Microcyn®*
Hypochlorous acid 0.00025%, sodium hypochlorite 0.0036% (spray)	OTC; 2 oz	*Myclyns®* pH: 6.2 to 7.8.

Povidone Iodine, Topical

(***poe***-vi-done ***eye***-oh-dine) *Betadine®*

Indications/Actions

Povidone iodine is used as a topical first-aid agent and a presurgical skin antiseptic. It may be used for superficial pyoderma and *Malassezia* spp dermatitis; however, it is infrequently used in small animal dermatology due to its drying, irritating, and staining effects.

Povidone iodine, an iodophor antiseptic, is rapidly bactericidal at low concentrations against both gram-positive and gram-negative bacteria. It is also fungicidal and sporicidal as a 1% aqueous solution. Povidone acts by slowly releasing iodine to tissues. It has prolonged activity (4-6 hours) but is not as long acting as chlorhexidine. Povidone iodine also has mild degreasing and debriding activity.

Suggested Dosages/Uses
Thoroughly wet affected area(s) to ensure complete coverage. For veterinary products, refer to the specific product label for details on use.

Precautions/Adverse Effects
Povidone may be drying and irritating to skin and can be extremely irritating to the scrotal skin and external ears. Use with emollients may alleviate the drying effects. It may also stain skin, hair, and fabrics. Avoid contact with eyes.

Wear gloves or wash hands after application. Use with caution on deep or puncture wounds or on serious burns. Systemic absorption can result in renal and thyroid dysfunction.

Veterinary-Labeled Products
NOTES:
1. The following products are only representative and not meant to be all-inclusive. There are many available povidone iodine products, including hoof dressings, teat dips, boluses (to prepare a solution), and udder washes. Povidone iodine products are also available in combination with lidocaine and as a combination of povidone and potassium iodine.
2. 10% povidone iodine yields 1% titratable iodine; labels may be confusing. Povidone iodine may also be listed as polyvinylpyrrolidone iodine. Products below are listed as povidone iodine.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Povidone iodine 5% (shampoo)	OTC; Depending on the manufacturer: 8 oz, 16 oz, 32 oz, 1 gallon	*Aloedine*®, generic
Povidone iodine 5% (spray)	OTC; 16 oz	generic
Povidone iodine 7.5% (scrub)	OTC; 32 oz, 1 gallon	*Vetadine*®; *Pivodine*®; *Povidine*®; *Poviderm*®
Povidone iodine 10% (gel)	OTC; 20 oz	*Biozide*®
Povidone iodine 10% (solution)	OTC; 32 oz, 1 gallon	*Poviderm*®; *Prodine*®; *Vetadine*®

Human-Labeled Products
NOTE: There are a large number of povidone iodine products available. In addition to those listed, vaginal gels, swabs, sponges, and foaming skin cleansers are also available.

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Povidone iodine 5% (powder spray)	OTC; 2 oz	*Betadine*®
Povidone iodine 5% (spray)	OTC; 3 oz	*Betadine*®
Povidone iodine 7.5% (scrub)	OTC; 4 oz, 16 oz, 32 oz, 1 gallon	*Betadine*®; generic
Povidone iodine 10% (ointment)	OTC; 1 g (packet), 28.4 g	generic
Povidone iodine 10% (solution)	OTC; 0.5 oz, 4 oz, 8 oz, 16 oz, 32 oz, 1 gallon	*Betadine*®; generic

Antiparasitic Agents, Topical
For topical agents that are administered topically but absorbed systemically through the skin to treat ecto- and endoparasites (eg, ***Eprinomectin, Fluralaner, Ivermectin, Moxidectin, Selamectin),*** refer to the respective monographs in the systemic section.

Amitraz Collar, Topical
(a-mi-***traz***) *Preventic*®

Indications/Actions
A water-resistant amitraz collar is available for dogs 12 weeks of age and older that is labeled to kill and detach ticks for up to 3 months[1]; the collar does not control fleas. It starts working within 24 hours of placement. **Do NOT apply the amitraz collar to cats or rabbits as it will cause toxicity**.

Topical amitraz dip was FDA-approved for the treatment of generalized demodicosis in dogs; however, this product is no longer marketed. It was also used to treat canine sarcoptic mange, cheyletiellosis, and feline surface (*Demodex gatoi)* or follicular demodicosis (*Demodex cati)*, and as a general insecticidal/miticidal agent in several other species.

Amitraz is thought to act through agonism of octopamine receptors in ectoparasites. Octopamine receptors are considered the invertebrate equivalent to adrenergic receptors in vertebrates. The overstimulation of octopamine receptors leads to overexcitation, paralysis, and death.[2,3]

Suggested Dosages/Uses
DOGS:

Kill and detach ticks (label dosage; EPA-approved): Apply collar based on dog's body weight. Dogs up to 27.3 kg (60 lb): 18-inch adjustable collar; dogs over 27.3 kg (60 lb): 25-inch adjustable collar. The collar must be worn tightly enough to make contact with the skin. Replace collar every 90 days.

Precautions/Adverse Effects/Drug Interactions

Do not reuse the collar or container; wrap it in a newspaper and throw it in the trash. Do not use the collar on puppies under 12 weeks of age.

Amitraz can be toxic to cats and rabbits and should not be used in these species.

Amitraz collars may cause toxicity if swallowed (by either animals or humans). Treatment should consist of emesis, retrieval of the collar using endoscopy if possible, and administration of activated charcoal and a cathartic to remove any remaining collar fragments. Yohimbine 0.11 – 0.2 mg/kg IV (start with low dosage) may be of benefit to treat adverse effects associated with the overdose. Because yohimbine has a short half-life, it may need to be repeated, particularly if the animal has ingested an amitraz-containing collar that could has not been retrieved from the GI tract. See ***Yohimbine***. Atipamezole has also been used to treat amitraz toxicity; see ***Atipamezole.*** For patients that have experienced or are suspected to have experienced an overdose, consultation with a 24-hour poison consultation center specializing in providing veterinary-specific information is recommended. For general information related to overdose and toxin exposures, as well as contact information for poison control centers, refer to ***Appendix.***

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Amitraz 9% (collar)	Labeled only for dogs 12 weeks of age and older	OTC (EPA); 18 inches (for dogs less than 27.2 kg [60 lb]), 25 inches (for dogs more than 27.2 kg [60 lb])	*Preventic®* Adjustable (cut off excess). Effective for 3 months

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Crotamiton, Topical

(***kroe***-ta-mye-ton)

Indications/Actions

Crotamiton is a topical miticide/scabicide used primarily for adjunctive treatment (with ivermectin) of mite infections (eg, *Knemidocoptes* spp) in birds. Crotamiton has both miticidal and antipruritic actions. Its miticidal mechanism of action is unknown; however, its antipruritic effect is reported to be due to its inhibition of the transient receptor potential vanilloid 4 (TRPV4) channel that is expressed on the skin peripheral neurons and keratinocytes.[1]

Suggested Dosages/Uses

Once- to twice-daily application is usually recommended. Shake well before use.

Precautions/Adverse Effects

Do not apply around the eyes or mouth. Little is known of the compound's safety profile. Irritation or hypersensitivity reactions are possible; do not use on patients with a known hypersensitivity to it.[2] Discontinue if irritation or sensitization develops. Do not apply to acutely inflamed skin or raw, weeping skin.

Prevent grooming or preening after application, as oral ingestion can cause irritation of the buccal, esophageal, and gastric mucosa; nausea; vomiting; and abdominal pain.

Adverse effects in humans include dermatitis, pruritis, and rash.[2]

For patients that have experienced or are suspected to have experienced an overdose, consultation with a 24-hour poison consultation center specializing in providing veterinary-specific information is recommended. For general information related to overdose and toxin exposures, as well as contact information for poison control centers. See ***Appendix***.

Veterinary-Labeled Products

None

Human-Labeled Topical Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Crotamiton 10% (lotion) Other ingredients: cetyl alcohol, emollient base	Rx; 60 g, 227 g, 454 g	*Crotan®*, *Eurax®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Deltamethrin/Deltamethrin Combination Products, Topical

(del-ta-***meeth***-rin)

Indications/Actions

In the United States, deltamethrin-impregnated collars are labeled for killing fleas (ie, *Ctenocephalides* spp) and ticks[1] (ie, *Amblyomma americanum, Rhipicephalus sanguineus, Dermacentor variabilis, Ixodes scapularis, I pacificus*) and repelling mosquitoes on dogs for up to 6 months.[2] Treatment with deltamethrin has been shown to be an effective strategy to reduce the incidence of leishmaniasis by repelling and

killing mosquitos and phlebotomine sandflies, which are important vectors.[3] The mosquito repellent activity may also help reduce the incidence of heartworm infection[4]; further research is needed.
Deltamethrin is a synthetic pyrethroid insecticide and acts by disrupting the sodium channel conduction in arthropod nerve cell membranes, resulting in paralysis and death. Some collars are combined with pyriproxyfen, an insect growth regulator.

Suggested Dosages/Uses

Collars should be worn continuously and be replaced every 6 months.[1,2] Maximum effect may not occur until 2 to 3 weeks after collar placement; therefore, it is recommended to apply the collar before the start of flea/tick season.[5] Wipe the collar with a damp paper towel prior to application to remove dust. The collar is placed so that 2 to 3 fingers can easily fit between the collar and neck. The collar does not need to be removed for bathing or swimming. Follow specific product label for use instructions.

Precautions/Adverse Effects

Do not use deltamethrin on cats. Collars containing deltamethrin should not be used on dogs younger than 12 weeks.[1,2] Collars may cause irritation if placed too tightly. Skin reactions (erythema, localized dermatitis, pruritis) and hair loss at the application site are possible.[6] In extremely rare cases, neurologic signs such as tremor or lethargy have been reported.[6] Use with caution in debilitated, pregnant, nursing, geriatric, or medicated animals.[1,2]

If a collar is ingested, uncoordinated movements, tremor, drooling, vomiting, or rigidity are possible. Treatment is supportive, and clinical signs usually subside within 48 hours.[6] For patients that have experienced or are suspected to have experienced an overdose, consultation with a 24-hour poison consultation center specializing in providing veterinary-specific information is recommended. For general information related to overdose and toxin exposures, as well as contact information for poison control centers, refer to ***Appendix***.

Avoid contact with eyes, skin, or clothing. Wash hands after handling. Mammalian exposure to deltamethrin is classified as safe; however, deltamethrin is highly toxic to aquatic animals (especially fish) and should be used very carefully around water sources.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Deltamethrin 4% (collar)	Labeled for dogs 12 weeks of age or older **Do NOT use on cats.**	OTC (EPA)	*Activyl®; Adams®; PetArmor®; Proact®; Sentry®; Zodiac®* Adjustable (one size fits all). Water-resistant
Deltamethrin 4% (collar)	Labeled for dogs 12 weeks of age or older **Do NOT use on cats.**	OTC (EPA)	*Protect®; Salvo®; Sentry®* Available for small, medium, and large dogs. Water-resistant
Deltamethrin 4%, pyriproxyfen 1% (collar)	Labeled for dogs 12 weeks of age or older **Do NOT use on cats.**	OTC (EPA)	*PetArmor®* One size fits all. Waterproof

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Dinotefuran/Pyriproxyfen Combinations, Topical

(die-noh-te-***fyoor***-an, pye-ri-***proks***-i-fen)

Indications/Actions

Dinotefuran/pyriproxyfen combinations are used in dogs and cats for the treatment and control of external parasites. The combination of dinotefuran and pyriproxyfen controls adult fleas and all immature flea stages, including eggs, larvae, and pupae. Some canine combination products also contain permethrin, which broadens the spectrum to also repel and kill ticks, mosquitos, lice, mites (excluding mange), biting flies, and sand flies. Some feline combination products contain fipronil, providing rapid speed of kill and added protection against ticks and chewing lice.

Dinotefuran is a nitroguanidine, neonicotinoid insecticide with a structure similar to acetylcholine. It permanently binds to the same insect receptor sites as acetylcholine and activates the nerve impulse at the synapse, which cause stimulation that results in tremors, incoordination, and insect death. Dinotefuran does not bind to mammalian acetylcholine receptor sites.

Pyriproxyfen is a second-generation insect growth regulator that mimics insect juvenile growth hormone, halting development during metamorphosis and larval development. It also concentrates in female flea ovaries, which causes nonviable eggs to be produced. When dinotefuran is combined with an adulticide (eg, permethrin, fipronil), all stages of the parasite are killed, and re-infestation is less likely. It is more resistant to UV light than is methoprene.

Suggested Dosages/Uses

Dinotefuran/pyriproxyfen combination products are recommended once monthly for dogs and cats. Refer to the package labeling for specific dosages and application instructions.

Apparently, bathing or swimming does not interfere with efficacy of canine products; however, the authors do not recommend bathing the pet until 2 days after application. Cats should be completely dry before the *Catego®* product is applied and they should remain dry for 24 hours after application.[1]

Precautions/Adverse Effects

The dog product containing permethrin (*Vectra 3D®*) **must not be used on cats or on dogs that cohabitate with cats**. The manufacturer recommends not using *Vectra 3D®* or *Vectra®* on debilitated, aged, medicated, pregnant, or nursing animals or on animals known to be sensitive to pesticide products. Mild transitory skin erythema may occur at the application site. Avoid eye and oral contact since dinotefuran can cause substantial but temporary eye irritation. *Vectra 3D®* was reported to be associated with the development of pemphigus foliaceus-like disease occurring at the site of application and at distant sites in 3 dogs.[1]

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Dinotefuran 22%, pyriproxyfen 3% (spot-on solution)	Labeled for dogs and puppies 8 weeks of age or older	OTC (EPA) 1.1 - 4.5 kg (2.5-10 lb): 1.3 mL/dose 5 - 9 kg (11-20 lb): 2 mL/dose 9.5 - 25 kg (21-55 lb): 4 mL/dose 25 - 45.4 kg (56-100 lb): 6 mL/dose	*Vectra®*
Dinotefuran 22%, pyriproxyfen 3% (spot-on solution)	Labeled for cats and kittens 8 weeks of age or older	OTC (EPA) 0.9 - 4.1 kg (2-9 lb): 0.8 mL/dose 4.1 kg (9 lb) or over: 1.2 mL/dose	*Vectra®*
Dinotefuran 22%, pyriproxyfen 3%, fipronil 8.92% (spot-on solution)	Labeled for cats and kittens 8 weeks of age or older	OTC (EPA) All cats weighing over 0.7 kg (1.5 lb): 0.5 mL/dose	*Catego®*
Dinotefuran 4.95%, pyriproxyfen 0.44%, permethrin 36.08% (spot-on solution)	Labeled ***only*** for dogs and puppies 8 weeks of age or older	OTC (EPA) 2.3 - 4.5 kg (5-10 lb): 0.8 mL/dose 5 - 9 kg (11-20 lb): 1.6 mL/dose 9.5 - 25 kg (21-55 lb): 3.6 mL/dose 25.4 - 43.1 kg (56-95 lb): 4.7 mL/dose Over 43.1 kg (95 lb): 8 mL/dose	*Vectra 3D®* **Must NOT be used on cats. In households where a dog cohabitates with cats, keep cat separated from treated dog for 24 hours.**

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Fipronil/Fipronil Combinations, Topical

(***fip***-roe-nil) *Frontline®*

See also ***Permethrin/Permethrin Combinations, Topical; Pyriproxyfen/Pyriproxyfen Combinations, Topical***

Indications/Actions

In the United States, fipronil is indicated for the treatment of fleas, ticks, and chewing lice infestations in dogs and cats. It has also been used successfully for *Trombicula autumnalis* (chigger) infestation, sarcoptic mange, cheyletiellosis, and otoacariasis.

Fipronil is a phenylpyrazole antiparasitic agent that interferes with the passage of chloride ions in GABA-regulated chloride channels in invertebrates, thereby disrupting CNS activity and causing death of the parasite. Fipronil collects in the oils of the skin and hair follicles and continues to be released over a period of time, which results in a long residual activity. When topically applied, the drug spreads over the body in ≈24 hours via translocation.

When fipronil is combined with the insect growth regulator (S)-methoprene, flea eggs and flea larvae are also killed. (S)-methoprene mimics flea juvenile growth hormone, which halts development during metamorphosis and larval development. It also concentrates in female flea ovaries, causing nonviable eggs to be produced. Additional combination products with permethrin, etofenprox, cyphenothrin, and pyriproxyfen (synthetic pyrethroids) or amitraz are available.

Suggested Dosages/Uses

Spot-on solutions are generally applied in one or more spots on the back of the neck or between the shoulder blades. The hair should be parted so the solution can be applied directly to the skin.

When applying spray, use one gloved hand to ruffle the coat while spraying sides, back, legs, abdomen, shoulders, and neck. To apply this product to the head and eye area, spray a gloved hand and gently rub into the coat. Alternatively, a towel soaked with fipronil may be used to treat the face. Apply this product until coat is thoroughly wet.[1]

Fleas, ticks, or chewing lice: Monthly treatments are generally recommended when fipronil is used for treatment and prevention of fleas, ticks, or chewing lice. See product labels for specific directions on administration and recommendations on bathing and swimming after administration. In some cases, for better efficacy, the interval between applications needs to be shortened to every 14 days.

Trombicula autumnalis **infestation:** Monthly application of fipronil *spray* (0.25%) has been successful in dogs. Apply the *spray* to the entire body, thoroughly wetting the coat, with particular emphasis on the areas typically affected (eg, feet, ears, face, perineum, and tail). Monthly applications should be administered throughout the trombiculid season at the dose of 3 – 6 mL/kg. In some cases, application every 14 days may be required. Fipronil seemed to have a shorter duration of effect in cats, with reinfestation seen after 7 to 10 days.[2]

Sarcoptic mange: Fipronil *spray* (0.25%) is recommended for confirmed cases using 2 protocols:

a) Apply once weekly for 4 weeks[3]
b) Apply every 2 to 3 weeks for 3 treatments[4]

Dosages ranging from 3 mL/kg to 39 mL/kg were used in these reports. Thoroughly wet the coat, paying special attention to the areas typically affected by the disease (eg, ears, face, ventrum, and extremities). A decrease in pruritus may be noticed as early as 7 days after the first application. Fipronil use is not recommended in treatment trials (ie, high clinical suspicion of mite infestation but mites not found on skin scrapings) because of anecdotal reports of treatment failures with this product.

Cheyletiellosis: Fipronil *spray* (0.25%) has been successfully used at a dose of 3 mL/kg to ensure complete and thorough body coverage, along with environmental permethrin treatment. Repeat fipronil treatment after 30 days.[5] Fipronil use is only recommended for confirmed cases and is not recommended in treatment trials (ie, high clinical suspicion of mite infestation but mites not found on skin scrapings) because of anecdotal reports of treatment failures.

Otoacariasis: Instill 1 drop of fipronil solution inside each ear canal and apply the rest of the dosing tube between the shoulder blades.[6] Fipronil needs to be applied in the ear canals to be effective. Resolution of clinical signs can be seen as early as 7 days posttreatment. Additional applications may be needed.

Precautions/Adverse Effects

Do not use fipronil on puppies or kittens less than 8 weeks of age. Fipronil compounds are reported to be contraindicated in rabbits, as toxicity deaths have occurred with the spray. Do not apply fipronil to or spray in or near the eyes. Temporary irritation may occur at the site of administration. Hypersensitivity has rarely been reported; animals that have demonstrated hypersensitivity reactions to fipronil should not be retreated. Use fipronil with caution on debilitated, elderly, or medicated patients.

Dispose of containers properly to avoid contamination of food or water sources. Avoid human contact with skin, eyes, or clothing. Wash well with soap and water if skin contact occurs. Wear gloves when applying fipronil and wash hands thoroughly after handling. Use sprays only in a well-ventilated area. Avoid contact with treated animals until coat is dry.

Most products are labeled as remaining effective after bathing, water immersion, or exposure to sunlight; however, avoid shampooing within 48 hours of application. Spotted areas may appear wet or oily for up to 24 hours after application.

Veterinary-Labeled Products

NOTE: This is not an all-inclusive list but is representative of the types of products available.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Fipronil 0.29% (spray)	Labeled for use on dogs, cats, puppies, and kittens 8 weeks of age or older	OTC (EPA); 8.5 oz, 17 oz	*Frontline®*
Fipronil 0.29%, (S)-methoprene 0.27% (spray)	Labeled for use on dogs, cats, puppies, or kittens 8 weeks of age or older	OTC (EPA); 8 oz, 16 oz	*Martin's FLEE® Plus*
Fipronil 9.7% (spot-on solution)	Labeled for use on dogs, cats, puppies, and kittens 8 weeks of age or older	OTC (EPA); various formulations and packages depending on each product	*Hartz® First Defense®; Pronyl OTC®; Sentry Fiproguard®; Spectra Sure®* The indication may differ among these products.
Fipronil 9.7% (spot-on solution)	Labeled for use on cats and kittens 8 weeks of age or older and over 0.7 kg (1.5 lb)	OTC (EPA); single-dose applicators 0.5 mL in packages of 3 and 6	*Adams Fortress®; Hartz® First Defense®; Sentry Fiproguard®*
Fipronil 9.8%, (S)-methoprene 11.8% (spot-on solution)	Labeled for use on cats or kittens 8 weeks of age or older and over 0.7 kg (1.5 lb)	OTC (EPA); single-dose applicators 0.5 mL in packages of 3 and 6	*Frontline®* Plus*; Martin's FLEE® Plus;* PetAction® *Plus*; *PetArmor® Plus; Pronyl OTC Plus®; Sentry Fibroguard®; Spectra Sure® Plus*
Fipronil 9.8%, (S)-methoprene 8.8% (spot-on solution)	Labeled for dogs and puppies 8 weeks of age or older **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3 and 6: up to 10 kg (22 lb): 0.67 mL; 10.4 - 20 kg (23-44 lb): 1.34 mL; 20.4 - 39.9 kg (45-88 lb): 2.68 mL; 40.4 - 59.9 kg (89-132 lb): 4.02 mL	*EctoAdvance® Plus*; *Frontline®Plus; PetAction® Plus*, *PetArmor® Plus; Pronyl OTC®Plus; Spectra Sure® Plus*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Fipronil 9.8%, etofenprox 15% (spot-on solution)	Labeled for all cats over 12 weeks of age	OTC (EPA); single-dose applicators 0.5 mL in packages of 3	*Martin's FLEE® Prefurred*
Fipronil 9.8%, cyphenothren 5.2%, (S)-methoprene 8.8% (spot-on solution)	Labeled for use on dogs 12 weeks of age or older and at least 1.8 kg (4 lb)	OTC (EPA); single-dose applicators in packages of 3	*Frontline® Trita®*
Fipronil 9.8%, etofenprox 15%, (S)-methoprene 11.8% (spot-on solution)	Labeled for use on cats or kittens 12 weeks of age or older	OTC (EPA); single-dose applicators 0.5 mL in packages of 3	*Frontline® Tritak*
Fipronil 9.8%, pyriproxyfen 0.25%, (S)-methoprene 11.8% (spot-on solution)	Labeled for cats and kittens 8 weeks of age or older and less than or equal to 0.7 kg (1.5 lb)	OTC (EPA); single-dose applicators 0.5 mL in packages of 3 and 6	*Frontline® Gold*
Fipronil 9.58%, pyriproxyfen 11.65% (spot-on solution)	Labeled for cats and kittens 8 weeks of age or older and less than or equal to 0.7 kg (1.5 lb)	OTC (EPA); single-dose applicators 0.5 mL in packages of 3	*Effipro®*
Fipronil 9.78%, pyriproxyfen 2.94% (spot-on solution)	Labeled for dogs and puppies 8 weeks of age or older **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3: 2.3 - 10.4 kg (5-22.9 lb): 0.67 mL; 10.4 - 20 kg (23-44.9 lb): 1.34 mL; 20.4 - 40.3 kg (45-88.9 lb): 2.68 mL; 40.4 - 59.9 kg (89-132 lb): 4.02 mL	*Effipro®*
Fipronil 9.8%, cyphenothrin 5.2% (spot-on solution)	Labeled for dogs and puppies 12 weeks of age or older **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3: up to 10 kg (22 lb): 0.67 mL; 10.4 - 20 kg (23-44 lb): 1.34 mL; 20.4 - 39.9 kg (45-88 lb): 2.68 mL; 40.4 - 59.9 kg (89-132 lb): 4.02 mL	*Martin's Prefurred® Plus; Parastar® Plus*
Fipronil 9.8%, cyphenothrin 5.2%, (S)-methoprene 8.8% (spot-on solution)	Labeled for use on dogs 12 weeks of age or older and at least 1.8 kg (4 lb) **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3: up to 10 kg (22 lb): 0.67 mL; 10.4 - 20 kg (23-44 lb): 1.34 mL; 20.4 - 40.3 kg (45-88 lb): 2.68 mL; 40.4 - 59.9 kg (89-132 lb): 4.02 mL	*Frontline® Tritak*
Fipronil 6.01%, permethrin 44.88% (spot-on solution)	Labeled for dogs and puppies at least 8 weeks of age **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3: 2.3 - 5 kg (5-10.9 lb): 0.5 mL; 5 - 10.4 kg (11-22.9 lb): 1 mL; 20.4 - 40.3 kg (45-88.9 lb): 4 mL; 40.4 - 59.9 kg (89-132 lb): 6 mL	*Effitix®*
Fipronil 6%, permethrin 44.8%, pyriproxyfen 1.8% (spot-on solution)	Labeled for dogs and puppies at least 8 weeks of age **Do NOT use on cats.**	OTC (EPA); single-dose applicators in packages of 3: 2.3 - 5 kg (5-10.9 lb): 0.5 mL; 5 - 10.4 kg (11-22.9 lb): 1 mL; 20.4 - 40.3 kg (45-88.9 lb): 4 mL; 40.4 - 59.9 kg (89-132 lb): 6 mL	*Effitix® Plus; Frontline® Shield*
Fipronil 8.92%, dinotefuran 22%, pyriproxyfen 3% (spot-on solution)	Labeled for cats and kittens over 0.7 kg (1.5 lb) and over 8 weeks of age	OTC (EPA); packages of 3 and 6	*Catego®*
Fipronil 9.8%, (S)-methoprene 8.8%; Amitraz 22.1% (spot-on solution)	Labeled for dogs and puppies 8 weeks of age or older and at least 2.3 kg (5 lb) body weight **Do NOT use on cats.**	OTC (EPA); 1.7 mL, 2.14 mL, 4.28 mL, and 6.42 mL per applicator	*Certifect®*

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Imidacloprid/Imidacloprid Combinations, Topical

(eye-mi-da-***kloe***-prid) *Advantage*®, Seresto®

For systemic use, see ***Imidacloprid, Systemic and Moxidectin/Moxidectin Combination Products***

Indications/Actions

Imidacloprid topical solution is indicated for the prevention and treatment of flea infestations in cats and flea and lice infestations in dogs. Imidacloprid is effective against both adult fleas and flea larvae.[1] The combination of imidacloprid and pyriproxyfen is indicated for the prevention and treatment of all flea stages in dogs and cats and lice infestations in dogs. The combination of imidacloprid, pyriproxyfen, and permethrin is indicated in dogs only for the prevention and treatment of fleas, ticks, mosquitoes, biting flies, and lice.

The combination of imidacloprid and flumethrin impregnated into a slow-release collar is indicated for the prevention and treatment of ticks, fleas, and lice in dogs and prevention and treatment of ticks and fleas in cats. It has also been shown to prevent babesiosis[2] and leishmaniosis[3] in dogs and is indicated for the prevention of leishmaniosis in some countries.

Imidacloprid's mechanism of action as an insecticide is to act on nicotinic acetylcholine receptors on the postsynaptic membrane, causing CNS impairment and death of the ectoparasite. Certain insect species are more sensitive to these agents than are mammalian receptors. This is a different mechanism of action than other insecticidal agents (organophosphates, pyrethrins, carbamates, insect growth regulators [IGRs], and insect development inhibitors [IDIs]). Imidacloprid is reported to be nonteratogenic, nonhypersensitizing, nonmutagenic, nonallergenic, noncarcinogenic, and nonphotosensitizing. When applied topically, imidocloprid is not absorbed into the bloodstream or internal organs. Permethrin and flumethrin, found in some imidacloprid products, are neurotoxins that slow the movement of sodium ions through sodium channels in neuron membranes of the ectoparasite. Pyriproxyfen may also be found in some imidacloprid products; it mimics insect juvenile growth hormone, halting development during metamorphosis and larval development. It also concentrates in female flea ovaries, causing non-viable eggs to be produced.

Suggested Dosages/Uses

Refer to the package information for product-specific application instructions. Imidacloprid products are generally administered once monthly; however, extra-label dosages, such as once weekly or every other week, have been used. Although swimming, bathing, and rain do not significantly affect the duration of action, animals that have been shampooed repeatedly may require additional treatment(s) before the next monthly dosing interval, not to exceed once weekly administration.

Precautions/Adverse Effects

When used as directed, imidacloprid topical solutions are usually well tolerated and adverse effects are minimal. Dermal irritation has been reported at the site of application.

Imidacloprid and pyriproxyfen combinations are contraindicated in puppies younger than 7 weeks, dogs weighing less than 3 lbs, and kittens younger than 7 or 8 weeks (depending on product). Use with caution in debilitated, older, pregnant, or nursing animals.

Combination products that contain permethrin must not be used on cats. They should be used with caution in households with cats, particularly if cats are in close contact with or will groom treated dogs.

Avoid contact with the eyes and mouth; oral contact may cause excessive salivation. If eye contact occurs in humans or animals, flush well with an ophthalmic irrigation solution or water. Wearing gloves is not required during administration but is recommended to avoid contact with human skin. Wash hands with soap and water after handling. Keep out of reach of children. Store products away from food sources to avoid the possibility of contamination. Carefully dispose of products in the trash.

Most toxicities are seen following oral dosing of topical products. In cats, clinical signs may include hypersalivation, vomiting, lethargy, and hiding; oral ulcers have also been rarely reported in cats. In dogs, reported clinical signs include vomiting, hypersalivation, and trembling.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Imidacloprid 9.1% (spot-on solution)	Dogs and puppies 7 weeks of age and older; cats and kittens 8 weeks of age and older	OTC (EPA); packages of 4 or 6 applicators: Cats 4.1 kg (9 lb) and under: 0.4 mL Cats over 4.1 kg (9 lb): 0.8 mL Dogs under 4.5 kg (10 lb): 0.4 mL Dogs 5 - 9.1 kg (11-20 lb): 1 mL Dogs 9.5 - 25 kg (21-55 lb): 2.5 mL Dogs over 25 kg (55 lb): 4 mL	*Advantage*® (product no longer available in the United States)
Imidacloprid 10%, flumethrin 4.5%, (collar)	Dogs and puppies 7 weeks of age and older; cats and kittens 10 weeks of age and older	OTC (EPA); available in 3 collar sizes: Cat: 15 inches Small dog (up to 8.2 kg [18 lb]): 15 inches Large dog (over 8.2 kg [18 lb]): 27.5 inches	*Seresto*®

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Imidacloprid 9.1%, pyriproxyfen 0.46% (spot-on solution)	Dogs and puppies 7 weeks of age and older; cats and kittens 8 weeks of age and older	OTC (EPA); packages of 4 or 6 applicators: Cats 0.9 - 2.3 kg (2-5 lb): 0.23 mL Cats 2.3 - 4.1 kg (5-9 lb): 0.4 mL Cats over 4.1 kg (9 lb): 0.8 mL Dogs 1.4 - 4.5 kg (3-10 lb): 0.4 mL Dogs 5 - 9.1 kg (11-20 lb): 1 mL Dogs 9.5 - 25 kg (21-55 lb): 2.5 mL Dogs over 25 kg (55 lb): 4 mL	*Advantage® II; Adventure Plus®*
Imidacloprid 9.1%, pyriproxyfen 0.46% (spot-on solution)	Cats and kittens at least 7 weeks of age	OTC (EPA); packages of 4 applicators: 2.3 - 4.1 kg (5-9 lb): 0.4mL (red) Over 4.1 kg (9 lb): 0.8 mL (purple)	*Provecta® II*
Imidacloprid 7.12%, permethrin 35.6%, pyriproxyfen 0.36% (spot-on solution)	Dogs and puppies over 7 weeks of age **Do NOT use on cats.**	OTC (EPA); packages of 4 applicators for use on dogs ONLY: 1.8 - 4.5 kg (4-10 lb): 0.48 mL 5 - 9.1 kg (11-20 lb): 1.2 mL 9.5 - 25 kg (21-55 lb): 3 mL Over 25 kg (55 lb): 4.8 mL	*Activate® II*
Imidacloprid 8.8%, permethrin 44%, pyriproxyfen 0.44% (spot-on solution)	Dogs and puppies 7 weeks of age **Do NOT use on cats.**	OTC (EPA); packages of 4 or 6 applicators for use on dogs ONLY: 1.8 - 4.5 kg (4-10 lb): 0.4 mL 5 - 9.1 kg (11-20 lb): 1 mL 9.5 - 25 kg (21-55 lb): 2.5 mL Over 25 kg (55 lb): 4 mL	*K9 Advantix® II; Provecta® Advanced*

Human-Labeled Products
None

References
For the complete list of references, see **wiley.com/go/budde/plumb**

Isopropyl Myristate, Topical
(eye-so-***proe***-pil ***meer***-is-tate) *Resultix®*

Indications/Actions
Isopropyl myristate is a noninsecticidal product labeled for the removal and killing of attached and crawling ticks on dogs and cats. It acts by dissolving the outer wax layer covering the hard shell (cuticle) of the tick, resulting in uncontrollable water loss and death. Ticks generally die and fall off within 3 hours.

Suggested Dosages/Uses
Spray tick until it is covered with solution, ≈2 sprays. Tick should be dead within 3 hours and will either fall off the animal or will be immobile when it is removed. If the tick has not fallen off after 3 hours, carefully remove it with gloves or tweezers. Dispose of the tick by placing it in a sealable plastic bag. Wash hands after handling product, and after any accidental exposure to ticks.

Isopropyl myristate can be used as often as needed.[1]

Precautions/Adverse Effects
Do not use near the eyes or on irritated skin. Discontinue use if skin irritation or skin infection develops.

Isopropyl myristate is toxic to invertebrates and fish.[1] Never dispose of product down any drain; do not discharge into lakes, streams, ponds, estuaries, oceans, or other waters where aquatic invertebrates or fish may be found.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Isopropyl myristate 50% (spray)	Labeled for dogs and cats	OTC (EPA); 20 mL	*Resultix®*

Human-Labeled Products
None

References
For the complete list of references, see **wiley.com/go/budde/plumb**

Methoprene/Methoprene Combinations, Topical

(meth-oh-***preen***)

Indications/Actions

Methoprene is an insect growth regulator that prevents the maturation of flea and mosquito eggs and larva. Methoprene mimics insect juvenile growth hormone, halting development during metamorphosis and larval development. It also concentrates in female flea ovaries, resulting in the production of nonviable eggs.

Methoprene is typically combined with an adulticide to kill all stages of the parasites and prevent reinfestation. Fipronil and pyrethroids (eg, cyphenothrin, etofenprox, permethrin) are commonly used in combination with methoprene. These agents also provide coverage against additional parasites; see specific product labels for details on parasite activity.

Suggested Dosages/Uses

For use and dosage recommendations, refer to the specific product insert.

Precautions/Adverse Effects

Methoprene is commonly found in products containing **pyrethroids that are toxic to cats**, especially small kittens. **Only use products on cats that are specifically labeled for use on cats**. Use pyrethroid-containing products cautiously on dogs that cohabitate with cats.

Hypersensitivity reactions and skin irritation are possible with methoprene-containing products. Avoid eyes and mucous membranes during application. When used as a single agent, methoprene has low toxicity in mammals.

Additional precautions and adverse effects vary based on other active ingredients; refer to specific product insert.

Methoprene products should be protected from light as methoprene is broken down by UV light.

Veterinary-Labeled Products

NOTES:

1. Methoprene is available in a variety of combinations. Use caution selecting products labeled for different species as different combinations of active ingredients may be marketed under the same trade name.
2. There is an extensive number of products marketed, including collars, topical sprays, foggers, and spot-on solutions; the following products are only representative.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Methoprene 2.9% (spot-on solution)	Labeled for cats and kittens 12 weeks of age and older	OTC (EPA); single-dose 1 mL applicators in packages of 3	*Hartz® UltraGuard® OneSpot®*
Methoprene 3%, permethrin 45% (spot-on solution)	Labeled for dogs 6 months of age and older **Do NOT use on cats.**	OTC (EPA); single-dose applicators per weight in packages of 4	*Zodiac® Spot On® for Dogs*
Methoprene 3.6%, etofenprox 40% (spot-on solution)	Labeled for kittens 12 weeks of age and older	OTC (EPA); single-dose 1.8 mL applicators in packages of 3 for cats 2.3 kg (5 lb) and over	*Hartz® UltraGuard Plus® for Cats; Hartz® UltraGuard Pro® for Cats*
Methoprene 4.05%, etofenprox 46%	Labeled for cats and kittens 12 weeks of age and older	OTC (EPA); single-dose applicators per weight in packages of 3	*Adams® Plus for Cats*
Methoprene 8.8%, fipronil 9.8% (spot-on solution)	Labeled for dogs or puppies 8 weeks of age and older	OTC (EPA); single-dose applicators per weight in packages of 3 and 6	*Frontline® Plus for Dogs; Onguard® Plus for Dogs; PetArmor® Plus for Dogs; Sentry® Fiproguard Plus for Dogs*
Methoprene 8.8%, fipronil 9.8%, cyphenothrin 5.2% (spot-on solution)	Labeled for use on dogs 12 weeks of age and older **Do NOT use on cats.**	OTC (EPA); single-dose applicators per weight in packages of 3	*Frontline® Tritak® for Dogs*
Methoprene 11.8%, fipronil 9.8% (spot-on solution)	Labeled for use on cats or kittens 8 weeks of age and older	OTC (EPA); single-dose applicators of 0.5 mL in packages of 3 and 6	*Frontline® Plus for Cats; Onguard® Plus for Cats; PetArmor® Plus for Cats; Sentry® Fiproguard Plus for Cats*
Methoprene 11.8%, fipronil 9.8%, etofenprox 15% (spot-on solution)	Labeled for use on cats or kittens 12 weeks of age or older	OTC (EPA); single-dose applicators of 0.5 mL in packages of 3	*Frontline® Tritak® for Cats*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Methoprene 8.8%, fipronil 9.8%, pyriproxyfen 0.25% (spot-on solution)	Labeled for dogs and puppies 8 weeks of age and older	OTC (EPA); single-dose applicators of 0.5 mL in packages of 3 and 6 per weight	*Frontline® Gold for Dogs*
Methoprene 11.8%, fipronil 9.8%, pyriproxyfen 0.25% (spot-on solution)	Labeled for cats and kittens 8 weeks of age and older	OTC (EPA); single-dose applicators of 0.5 mL in packages of 3 and 6 for cats 0.7 kg (1.5 lb) or greater	*Frontline® Gold for Cats*
Methoprene 0.27%, piperonyl butoxide 0.37%, pyrethrins 0.2% (spray)	Labeled for use on dogs, cats, puppies, kittens, horses, and ponies **NOT for puppies or kittens less than 12 weeks old**	OTC (EPA); 16 oz, 24 oz, 1 gallon	*Vet-Kem Ovitrol Plus®; Bio-Spot®*
Methoprene 0.27%, fipronil 0.29% (spray)	Labeled for use on cats or kittens 8 weeks of age or older	OTC (EPA); 6.5 oz (aerosol spray), 8 oz (trigger spray)	*Martin's FLEE®*
Methoprene 1.1%, piperonyl butoxide 1.05%, pyrethrins 0.15% (shampoo)	**NOT for puppies or kittens less than 12 weeks old**	OTC (EPA); 12 oz	*Vet-Kem®*
Methoprene 1.02%, tetrachlorvinphos 14.55% (collar)	Labeled for dogs and cats	OTC (EPA); sizing based on neck circumference and varies based on manufacturer	*Hartz® UltraGuard Plus®; Hartz® UltraGuard Pro®; PetArmor® for Cats; Adams® Plus*
Methoprene 1.02%, deltamethrin 4% (collar)	Labeled for dogs 12 weeks and older	OTC (EPA); one size, fits dogs with necks up to 26"	*Hartz® InControl®; Hartz® UltraGuard ProMAX*

Human-Labeled Products
None

Permethrin/Permethrin Combinations, Topical
(per-***meth***-rin)

Indications/Actions
Permethrin is a synthetic pyrethroid insecticide that acts as an adulticide and miticide. It has knockdown activity against fleas, ticks, lice, and some mites, including *Cheyletiella* spp and *Sarcoptes scabiei*. Permethrin also has repellant activity against flies, gnats, and mosquitoes.

In dogs, permethrin is primarily used to prevent fleas and ticks. In large animal and food animal species, it is primarily used for flies, lice, mites, mosquitoes, ticks, and keds.

Permethrin acts by disrupting the sodium channel current in arthropod nerve cell membranes, resulting in paralysis and death.

Suggested Dosages/Uses
Permethrin products can typically be applied every 2 weeks or every 4 weeks; refer to specific product label for details on use.

Precautions/Adverse Effects
Permethrin and other synthetic pyrethroids are toxic to cats, especially small kittens. Do not use permethrin-containing products on cats and avoid use on dogs that cohabitate with cats.

Permethrin may cause hypersensitivity reactions. Do not use permethrin near the eyes or mucous membranes. Adverse effects are uncommon but can include pruritus or mild skin irritation at the application site.

Wear gloves and wash hands after application. In case of skin contact, wash off with water.

Permethrin is extremely toxic to aquatic organisms and bees. Apply and dispose of products with caution as not to contaminate the environment.

Veterinary-Labeled Products
NOTES:
1. This is not an inclusive list; there are an extensive number of products marketed, including shampoos, pour-ons, sprays, foggers, and dusts. Use caution selecting products, as different combinations of active ingredients may be marketed under the same trade name.
2. **Permethrin products must NOT be used on cats** and should not be used on dogs that cohabitate with cats.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Permethrin 45% (spot-on solution)	Labeled for horses over 3 months of age	OTC (EPA); single-dose applicators in packages of 3 or 6	*Equi-Spot®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Permethrin 50% (spot-on solution)	Labeled for horses over 3 months of age	OTC (EPA); single-dose applicators in packages of 3 or 6	*Pro-Force®50*
Permethrin 45%, pyriproxyfen 1.9% (spot-on solution)	Labeled for dogs over 3 months of age	OTC (EPA); single-dose applicators in packages of 3: 6.8 kg (15 lb) and under: 1 mL 6.8 - 15 kg (15-33 lb): 1.5 mL 15 -30 kg (33-66 lb): 3 mL 30 kg (66 lb) and over: 4.5 mL	*Sentry®*
Permethrin 45%, pyriproxyfen 5% (spot-on solution)	Labeled for dogs older than 3 months of age	OTC (EPA); single-dose applicators in packages of 4: 2.3 - 6.8 kg (5-15 lb): 1 mL 7.3 - 15 kg (16-33 lb): 1.5 mL 15.4 - 30 kg (34-66 lb): 3 mL Over 30 kg (66 lb): 4.5 mL	*ShieldTec Plus®*
Permethrin 44%, imidacloprid 8.8%, pyriproxyfen 0.44% (spot-on solution)	Labeled for dogs 7 weeks of age and older	OTC (EPA); in packages of 2, 4, or 6: 1.8 - 4.5 kg (4-10 lb): 0.4 mL 5 - 9.1 kg (11-20 lb): 1 mL 9.5 - 25 kg (21-55 lb): 2.5 mL Over 25 kg (55 lb): 4 mL	*K9 Advantix® II*
Permethrin 35.6%, imidacloprid 7.12%, pyriproxyfen 0.36% (spot-on solution)	Labeled for dogs 7 weeks of age and older	OTC (EPA); in packages of 4: 1.8 - 4.5 kg (4-10 lb): 0.48 mL 5 - 9.1 kg (11-20 lb): 1.2 mL 9.5 - 25 kg (21-55 lb): 3 mL Over 25 kg (55 lb): 4.8 mL	*Activate® II*
Permethrin 36.08%, dinotefuran 4.95%, pyriproxyfen 0.44% (spot-on solution)	Labeled for dogs 7 weeks of age and older	OTC (EPA); in packages of 3 or 6: 2.3 - 4.5 kg (5-10 lb): 0.8 mL 5 - 9.1 kg (11-20 lb): 1.6 mL 9.5 - 25 kg (21-55 lb): 3.6 mL 25.4 -4 3.1 kg (56-95 lb): 4.7 mL Over 43.1 kg (95 lb): 8 mL	*Vectra 3D®*
Permethrin 45%, methoprene 3% (spot-on solution)	Labeled for dogs 6 months of age and older	OTC (EPA); in packages of 4: 3.2 6.8 kg (7-15 lb): 0.65 mL 7.3 - 13.6 kg (16-30 lb): 1 mL 14.1 - 27.2 kg (31-60 lb): 2 mL Over 27.2 kg (60 lb): 3 mL	*Zodiac®*
Permethrin 0.1%, 2-[1 methyl-2-(4-phenoxyphenoxy) ethoxy] pyridine 0.125%, N-octyl bicycloheptene dicarboximide 0.479%, pyrethrins 0.112% (spray)	Labeled for dogs over 6 months of age	OTC (EPA); 8 oz bottle	*Veterinary Formula Clinical Care®*
Permethrin 0.1%, piperonyl butoxide 0.5%, pyrethrins 0.05% (spray)	Labeled for dogs over 3 months of age	OTC (EPA); 32 oz bottle	*Flea Halt®*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Permethrin 1% (creme rinse)	OTC; 59 mL	*Nix®*; generic Labeled for treatment of head lice in humans
Permethrin 5% (cream)	Rx; 60 g	*Acticin®; Elimite®*; generic Labeled for treatment of scabies in humans

Pyrethrin/Pyrethrin Combinations, Topical

(pye-***ree***-thrin)

For otic use, see ***Antiparasitic Agents, Otic.***

Indications/Actions

Pyrethrins are natural insecticides derived from the pyrethrum flower that have knockdown activity against fleas, lice, ticks, and *Cheyletiella* spp. In small animal medicine, pyrethrins are used primarily to treat fleas and ticks on dogs and cats. In large animal and food animal medicine, there are many products available for pour-on, dusting, and spray use.

The natural pyrethrins are made up of 6 compounds. They act by disrupting the sodium channel current in arthropod nerve cell membranes, resulting in paralysis and death. Pyrethrins are often found in combination with insect growth regulators, such as methoprene or pyriproxyfen, or with synergistic agents such as piperonyl butoxide. Insect growth regulators add activity against egg and larval growth stages. Piperonyl butoxide inhibits insect metabolic enzymes (P450 system), which prevents the breakdown of primary insecticides and allows for a lower dose to be used.

Suggested Dosages/Uses

For dosage recommendations, refer to the specific product label.

Precautions/Adverse Effects

Pyrethrins are generally considered to be some of the safest insecticidal products. Avoid contact with the eyes and mucous membranes. Do not apply over wounds or irritated skin. Wear gloves when applying or wash hands after handling.

Cats are more susceptible to pyrethrin toxicity than other species. **Only use products on cats that are specifically labeled for cats.** Use pyrethrin products with caution in dogs that cohabitate with cats, and do not allow cats to groom application areas before dry. Avoid applying to hypothermic patients or applying when ambient temperatures are low. Pyrethrin products are toxic to fish and aquatic animals; do not allow products to enter drains or other water sources during application or disposal.

Adverse effects are uncommon but may include pruritus or mild skin irritation at the application site. Hypersensitivity reactions can also occur.

Veterinary-Labeled Products

There are a wide variety of available products, including shampoos, pour-ons, sprays, dusts, and ointments. The following is not an all-inclusive list.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pyrethrins 0.2%, piperonyl butoxide 0.37%, (S)-methoprene 0.27% (spray) Other ingredients: N-octyl bicycloheptene dicarboximide 0.62%	Labeled for use on dogs, cats over 3 pounds and 12 weeks of age; horses; and ponies.	OTC (EPA); 16 oz, 1 gallon	*Vet-Kem Ovitrol Plus®*
Pyrethrins 0.05%, permethrin 0.1%, piperonyl butoxide 0.5% (spray)	Labeled for dogs older than 3 months of age. **Do NOT use on cats or on dogs that cohabit with a cat.**	OTC (EPA); 8 oz, 32 oz, 40 oz	*Flea Halt®*
Pyrethrins 0.18%, 2-[1-Methyl-2-(4-phenoxyphenoxy) ethoxy] pyridine 0.125%, N-octyl bicycloheptene dicarboximide 0.957% (spray)	Labeled for dogs over 6 months of age or cats over 7 months of age	OTC (EPA); 8 oz, 15 oz	*Advantage®*
Pyrethrins 0.18%, N-octyl bicycloheptene dicarboximide 0.479%, piperonyl butoxide 0.3% (spray)	Labeled for dogs and puppies over 6 months of age	OTC (EPA); 8 oz, 15 oz	*Advantage®*
Pyrethrins 0.05%, piperonyl butoxide 0.5% (shampoo)	Labeled for dogs and cats over 12 weeks of age	OTC (EPA); 12 oz	*Zodiac®*
Pyrethrins 0.15%, piperonyl butoxide 1.88%, (S)-methoprene 0.1% (shampoo)	Labeled for dogs and cats over 12 weeks of age	OTC (EPA); 12 oz	*Adams® Plus*
Pyrethrins 0.15%, piperonyl butoxide 1.88%, (S)-methoprene 0.1% (shampoo)	Labeled for dogs and cats over 12 weeks of age	OTC (EPA); 12 oz	*Vet-Kem® Ovitrol Plus*
Pyrethrins 0.15%, N-oxyl bicycloheptene dicarboximide 0.5%, piperonyl butoxide 0.3% (shampoo)	Labeled for dogs and cats over 12 weeks of age	OTC; 12 oz, 1 gallon	*Davis®*
Pyrethrins 0.15%, N-octyl bicycloheptene dicarboxamide 0.479%, piperonyl butoxide 0.3% (shampoo)	Labeled for dogs, cats, ferrets, and horses over 12 weeks of age	OTC (EPA); 16 oz, 1 gallon	*Veterinary Formula®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pyrethrins 0.97%, di-n-propyl isocinchomerate 1.94%, N-octyl bicycloheptene dicarboxamide 5.7%, piperonyl butoxide 3.74% (dip)	Labeled for dogs and cats over 12 weeks of age	OTC (EPA); 4 oz	*Adams Plus®*
Pyrethrins 0.3%, piperonyl butoxide 3% (dip)	Labeled for dogs and cats over 12 weeks of age	OTC (EPA); 8 oz, 1 gallon	*Bio-Groom®*

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Pyrethrins 0.33%, piperonyl butoxide 4% (shampoo)	OTC; 2 oz, 4 oz, 8 oz	*Rid®*; generic Indicated for treatment of lice in humans

Pyriproxyfen/Pyriproxyfen Combinations, Topical

(pye-ri-***proks***-i-fen)

See also ***Deltamethrin, Dinotefuran/Pyriproxyfen Combinations, Fipronil/Fipronil Combinations, Imidacloprid/Imidacloprid Combinations, Methoprene/Methoprene Combinations, Permethrin/Permethrin Combinations****, and* ***Pyrethrin/Pyrethrin Combinations***

Indications/Actions

Pyriproxyfen is a second-generation insect growth regulator and is added to topical and environmental products to extend the insecticidal spectrum to include eggs and larva.

Pyriproxyfen mimics insect juvenile growth hormone, halting development during metamorphosis and larval development. It also concentrates in female flea ovaries, causing non-viable eggs to be produced. Pyriproxyfen is combined with adulticides such as permethrin or fipronil to kill all stages of targeted parasites, making re-infestation less likely.

Suggested Dosages/Uses

For dosage recommendations, refer to the specific product label.

Precautions/Adverse Effects

Pyriproxyfen can be found in products that contain **permethrin, which is toxic to cats**, particularly small kittens. **Only use products containing pyrethroids that are specifically labeled for use on cats**. Pyriproxyfen has low toxicity in mammals, but skin irritation and hypersensitivity reactions are possible.

Wear gloves when applying products containing insecticides and wash hands after handling. Pyriproxyfen is more resistant to UV light than is methoprene.

Veterinary-Labeled Products

NOTE: This is not an all-inclusive list but is representative of the types of products available.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pyriproxyfen 0.46%, imidacloprid 9.1% (spot-on solution)	Labeled for dogs and puppies 7 weeks of age and older; cats and kittens 8 weeks of age and older	OTC (EPA); packages of 3, 4 or 6 applicators: Cats 0.9 - 2.3 kg (2-5 lb): 0.23 mL Cats 2.3 - 4.1 kg (5-9 lb): 0.4 mL Cats over 4.1 kg (9 lb): 0.8 mL Dogs 1.4 - 4.5 kg (3-10 lb): 0.4 mL Dogs 5 - 9.1 kg (11-20 lb): 1 mL Dogs 9.5 - 25 kg (21-55 lb): 2.5 mL Dogs over 25 kg (55 lb): 4 mL	*Advantage®II; Adventure Plus®; Cross Block® II; Pet Armor Advanced® II*
Pyriproxyfen 0.46%, imidacloprid 9.1% (spot-on solution)	Labeled for cats and kittens 7 weeks of age and older	OTC (EPA); packages of 4 applicators: 2.3 - 4.1 kg (5-9 lb): 0.4 mL (red) Over 4.1 kg (9 lb): 0.8 mL (purple)	*Provecta® II*
Pyriproxyfen 0.44%, imidacloprid 8.8%, permethrin 44% (spot-on solution)	Labeled for dogs and puppies 7 weeks of age and older **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); packages of 4 or 6 applicators for use on dogs ONLY: 1.8 - 4.5 kg (4-10 lb): 0.4 mL 5 - 9.1 kg (11-20 lb): 1 mL 9.5 -25 kg (21-55 lb): 2.5 mL Over 25 kg (55 lb): 4 mL	*K9 Advantix® II; Provecta® Advanced*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Pyriproxyfen 0.36%, imidacloprid 7.12%, permethrin 35.6% (spot-on solution)	Dogs and puppies over 7 weeks of age **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); packages of 4 applicators for use on dogs ONLY: 1.8 - 4.5 kg (4-10 lb): 0.48 mL 5 - 9.1 kg (11-20 lb): 1.2 mL 9.5 - 25 kg (21-55 lb): 3 mL Over 25 kg (55 lb): 4.8 mL	*Activate® II*
Pyriproxyfen 0.44%, dinotefuran 4.95%, permethrin 36.08% (spot-on solution)	Labeled for dogs over 7 weeks of age **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); in packages of 3 or 6: 2.3 – 4.5 kg (5-10 lb): 0.8 mL 5 - 9.1 kg (11-20 lb): 1.6 mL 9.5 -25 kg (21-55 lb): 3.6 mL 56 to 95 lb: 4.7 mL Over 95 lb: 8 mL	*Vectra 3D®*
Pyriproxyfen 1.9%, permethrin 45%, (spot-on solution)	Labeled for dogs over 3 months of age **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); single-dose applicators in packages of 3: 6.8 kg (15 lb) and under: 1 mL 6.8 - 15 kg (15-33 lb): 1.5 mL 15 - 29.9 kg (33-66 lb): 3 mL 29.9 kg (66 lb) and over: 4.5 mL	*Sentry®*
Pyriproxyfen 5%, permethrin 45% (spot-on solution)	Labeled for dogs over 3 months of age **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); single-dose applicators in packages of 4: 2.3 - 6.8 kg (5-15 lb): 1 mL 7.3 - 15 kg (16-33 lb): 1.5 mL 15.4 - 29.9 kg (34-66 lb): 3 mL Over (29.9 kg) 66 lb: 4.5 mL	*ShieldTec Plus®*
Pyriproxyfen 3%, dinotefuran 22%, fipronil 8.92% (spot-on solution)	Labeled for cats and kittens over 8 weeks of age	OTC (EPA); available in packages of 3 or 6 applicators for cats over 0.7 kg (1.5 lb)	*Catego®*
Pyriproxyfen 2%, cyphenothrin 40%, (spot-on solution)	Labeled for use on dogs more than 12 weeks of age **Do NOT use on cats or on dogs that cohabitate with a cat.**	OTC (EPA); available in packages of 3 applicators: 4.1 - 9.1 kg (9 - 20 lb), 9.5 - 17.7 kg (21 - 39 lb), 18.1 - 27.2 kg (40 - 60 lb), and at least 27.7 kg (61 lb)	*TriForce®*

Human-Labeled Products
None

Spinetoram, Topical
(***spin***-et-oh-ram)

Indications/Actions
Spinetoram is an insecticide used for the prevention and treatment of flea infestations in cats.

Spinetoram is a semi-synthetic, second-generation derivative of the natural insecticide spinosad. It acts on both GABA and nicotinic acetylcholine receptors in the CNS of insects, leading to CNS hyperexcitation, followed by paralysis and death. Spinetoram begins to act within 30 minutes after application.[1]

Suggested Dosages/Uses
To treat and prevent flea infestations, spinetoram should be applied once monthly to the skin at the base of the head. For detailed application instructions, refer to the specific product label.

Precautions/Adverse Effects
Spinetoram should not be used on kittens less than 8 weeks of age. Do not allow cats to ingest product. Only a single dose should be applied at one time, regardless of the weight of the cat (ie, large cats do not require additional doses). Use spinetoram with caution on debilitated, aged, pregnant, or nursing animals, or any animal known to be sensitive to pesticides. Avoid contact with the eyes or mouth. Spinetoram should not be used in dogs, as they are more sensitive to the toxic effects of spinetoram than mice, rats, or cats.[2]

The primary adverse effects are application site reactions, including hair loss, hair changes (eg, greasing, clumping, matting), redness, inflammation, and itching. Other reported adverse effects include inactivity, vomiting, and inappetence.[3] Do not allow cats to groom after application in order to prevent ingestion. Application at the base of the head, rather than the shoulder blades, decreases the risk for direct ingestion.

Spinetoram has been shown to produce reproductive adverse effects in female rats including treatment-related depletion of primordial ovarian follicles, growing ovarian follicles, dystocia and other parturition abnormalities, late resorptions or retained fetuses, and increased postimplantation loss. However, no adverse effects were observed on the offspring at dose levels that produced parental toxicity. The developmental toxicity and reproduction studies indicated no evidence of increased susceptibility of the offspring with pre- and/or postnatal exposures.[2]

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Spinetoram 11.2% (spot-on solution)	Labeled for cats and kittens 8 weeks of age and older	OTC (EPA); in packs of 1, 3, or 6: Cats and kittens over 0.82 kg (1.8 lb): 0.77 mL	*Cheristin®*

Human-Labeled Products

None

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Antiseborrheic Agents, Topical

Refer to:
Benzoyl Peroxide, Topical (Antimicrobial Agents, Topical; Antibacterial Agents)
Coal Tar/Coal Tar Combinations, Topical (Keratolytic Agents)
Fatty Acids (Omega), Topical (Antipruritic/Anti-Inflammatory Agents, Topical; Non-Corticosteroid Agents)
Phytosphingosine, Topical (Antipruritic/Anti-Inflammatory Agents, Topical; Non-Corticosteroid Agents)
Salicylic Acid, Topical (Keratolytic Agents)
Selenium Sulfide, Topical (Antimicrobial Agents, Topical; Antifungal Agents)
Sulfur, Precipitated, Topical (Keratolytic Agents)

Immunomodulator Agents, Topical

Imiquimod, Topical

(ih-***mi***-kwih-mod) *Aldara®, Zyclara®*

Indications/Actions

Imiquimod is an immune response modifier that may be useful in the treatment of a variety of topical conditions in animals. In dogs and cats, it may be beneficial in treating actinic keratosis, squamous cell carcinoma, pigmented epidermal plaques, cutaneous viral papillomatosis, localized solar dermatitis, and solar carcinoma in situ. It may also be useful for feline herpesvirus dermatitis, feline Bowen's disease, and canine melanocytomas. Many of these indications are based on anecdotal evidence, as there are very few studies published for veterinary use. In horses, imiquimod has been anecdotally used for treating equine sarcoids, aural plaques, and epitrichial sweat gland ductal carcinoma. In humans, it is labeled as a treatment for genital or perianal warts, superficial basal cell carcinomas, and actinic keratoses of the face and scalp.

Imiquimod is an imidazoquinoline compound that binds to toll-like receptor 7 and stimulates the immune system to release various cytokines, including interferon-alpha and interleukin-12. These locally generated cytokines induce a Th1 immune response with the generation of cytotoxic effectors. Imiquimod itself does not have in vitro activity against wart viruses but stimulates monocytes and macrophages to release cytokines that induce regression in viral protein production.

Suggested Dosages/Uses

Use of imiquimod in animals is limited and further studies are needed. Doses and treatment regimens vary depending on the disease treated and tolerance to the drug. Doses range from applying a thin film once daily to applying every other week. Exudate and crusts should be gently cleaned prior to application of imiquimod. Treatment duration and frequency may be adjusted depending on patient response and adverse reactions.

Precautions/Adverse Effects

Wear gloves when applying imiquimod. Avoid exposing the eyes and mucous membranes to imiquimod; however, dogs with oral mucosal papilloma and horses with corneal squamous cell carcinomas have been treated without significant problems. Although systemic absorption is unlikely, do not allow treated animals to lick or groom the application site.

Local skin reactions are common and expected with imiquimod therapy, including erythema, burning, tenderness, pain, irritation, oozing/exudate, and erosion. Treatment duration and frequency may require adjustment depending on irritant reactions. Depigmentation and hair loss may occur at application sites as posttreatment sequelae.

Increased liver enzymes, neutropenia, partial anorexia, and vomiting were reported in cats in one small study, possibly associated with ingestion postcutaneous application.[1] Consider a baseline hepatic profile with periodic monitoring. Do not cover application site with occlusive dressings. It is thought that imiquimod may increase risk for sunburn; avoid exposing application site to sunlight.

Veterinary-Labeled Products

None

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Imiquimod 2.5% (cream)	Rx; 7.5 g pump bottle, single-use 250 mg packets in boxes of 28	*Zyclara*®
Imiquimod 3.75% (cream)	Rx; 7.5 g pump bottle	*Zyclara*®; generic
Imiquimod 5% (cream)	Rx; single-use 250 mg packets in boxes of 12 or 24	*Aldara*®; generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Pimecrolimus, Topical

(pim-e-***kroe***-li-mus)

Indications/Actions

Topical pimecrolimus may be beneficial for the adjunctive treatment of atopic dermatitis, discoid lupus erythematosus, pemphigus erythematosus or foliaceous, pinnal vascular disease or other cutaneous vasculopathies, alopecia areata, vitiligo, perianal fistulas (concurrent therapy or as maintenance treatment after cyclosporine therapy), and for feline proliferative and necrotizing otitis externa. Unlike topical corticosteroids, pimecrolimus does not have atrophogenic or metabolic effects associated with long-term use or large treatment areas.

Pimecrolimus is a calcineurin inhibitor that acts similarly to cyclosporine and tacrolimus. It inhibits T-lymphocyte activation primarily by inhibiting the phosphatase activity of calcineurin. It also inhibits the release of inflammatory cytokines and mediators from mast cells and basophils. The mechanism of pimecrolimus may not be identical to tacrolimus, as pimecrolimus did not impair the primary immune response (as did tacrolimus) in mice after a contact sensitizer was applied. Both drugs did impair the secondary response, however. Any clinical significance associated with this difference is not yet clear.

Suggested Dosages/Uses

There is limited experience with pimecrolimus in veterinary patients. It is typically recommended to use twice daily until signs are controlled, then decrease to the least frequent application that controls clinical signs.

Precautions/Adverse Effects

Although a causal relationship has not been established, skin malignancy and lymphoma have rarely been reported in humans using topical pimecrolimus. Therefore, long-term use is not recommended in humans, and pimecrolimus application should be limited to only the affected area(s). The long-term adverse effects of topical pimecrolimus in veterinary patients are currently unknown. Close monitoring is recommended for patients using this medication on a continuous maintenance basis. When long-term treatment is required, the ultimate goal should be to achieve the lowest possible dose that controls clinical signs. Pimecrolimus should not be used in patients that are receiving systemic immunosuppressant therapy or are otherwise immunocompromised.

Wear gloves or use an applicator (eg, cotton swab) when applying the cream and wash hands after handling. Do not allow animals to lick or chew at the treated area(s) for at least 20 to 30 minutes after application to avoid ingestion and allow the medication to work.

Although topical pimecrolimus appears well tolerated in dogs, localized irritation and pruritus have been reported in both humans and dogs. Pimecrolimus has been anecdotally reported to be less irritating than tacrolimus in dogs; however, it may also be less effective. Pimecrolimus may be cost prohibitive.

Veterinary-Labeled Products

None

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Pimecrolimus 1% (cream)	Rx; 30 g, 60 g, and 100 g	*Elidel*®; generic

Tacrolimus, Topical

(ta-***kroe***-li-mus)

Indications/Actions

Tacrolimus ointment may be beneficial in veterinary patients as a sole or adjunctive treatment of atopic dermatitis, discoid lupus erythematosus, pemphigus erythematosus or foliaceous, pinnal vascular disease, alopecia areata, vitiligo, dermal arteritis of the nasal philtrum, perianal fistulas (adjunctive or maintenance treatment after cyclosporin therapy), sterile canine panniculitis, and feline proliferative and necrotizing otitis externa. Unlike topical corticosteroids, tacrolimus does not have atrophogenic or metabolic effects associated with long-term use or large treatment areas.

Tacrolimus is a calcineurin inhibitor that inhibits T-lymphocyte activation. It also inhibits the release of inflammatory cytokines and mediators from mast cells and basophils.

Suggested Dosages/Uses

There is limited experience with topical tacrolimus in veterinary patients. Twice-daily application is typically recommended until signs are controlled. Application can then be reduced to the minimum frequency needed to control clinical signs.

Precautions/Adverse Effects

Although a causal relationship has not been established, skin malignancy and lymphoma have rarely been reported in humans using topical tacrolimus. Therefore, long-term use is not recommended in humans, and tacrolimus application should be limited to only the affected area(s). The long-term adverse effects of topical tacrolimus in veterinary patients are currently unknown, and thus close monitoring is recommended for patients using this medication on a continuous maintenance basis. When long-term treatment is required, the ultimate goal should be to achieve the lowest possible dose that controls clinical signs. Tacrolimus should not be used in patients that are receiving systemic immunosuppressant therapy or are otherwise immunocompromised. Do not use this ointment on premalignant or malignant skin conditions.

Wear gloves or use an applicator (eg, cotton swab) when applying the ointment and wash hands after handling. Do not allow animals to lick or chew at the treated area(s) for at least 20 to 30 minutes after application to avoid ingestion and allow the medication to work.

The commercially available ointment should not be used as (or compounded into) an ophthalmic preparation for treating KCS in dogs, as it contains propylene carbonate, a known ocular toxin.

Although topical tacrolimus appears well tolerated in dogs, localized irritation and pruritus have been reported in both humans and dogs. Pimecrolimus has been anecdotally reported as less irritating than tacrolimus in dogs; however, it may also be less effective. The cost of topical tacrolimus may be prohibitive.

The National Institute for Occupational Safety and Health (NIOSH) classifies tacrolimus as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Veterinary-Labeled Products

None

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Tacrolimus 0.03% (ointment)	Rx; 30 g, 60 g, and 100 g	*Protopic®*; generic
Tacrolimus 0.1% (ointment)	Rx; 30 g, 60 g, and 100 g	*Protopic®*; generic

Keratolytic Agents, Topical

See also:
Benzoyl Peroxide, Topical (Antimicrobial Agents, Topical; Antibacterial Agents)
Phytosphingosine, Topical (Antipruritic/Anti-Inflammatory Agents, Topical; Non-Corticosteroid Agents)
Selenium Sulfide, Topical (Antimicrobial Agents, Topical; Antifungal Agents)

Coal Tar/Coal Tar Combinations, Topical

(kole tar)

Indications/Actions

The use of coal tar containing shampoos in veterinary medicine is somewhat controversial, particularly since almost all veterinary-labeled products have been withdrawn from the market. However, coal tar shampoos have been used for many years to treat greasy dermatoses (seborrhea oleosa) in dogs.

Coal tar possesses keratoplastic, keratolytic, vasoconstrictive, antipruritic, and degreasing actions. Coal tar's mechanism of keratoplastic (keratoregulating) action is probably secondary to decreasing mitosis and DNA synthesis of basal epidermal cells.[1]

Suggested Dosages/Uses

The frequency of shampooing will vary according to the patient's needs. When using a medicated shampoo, allow 10 minutes of contact time. Another acceptable regimen is to allow the shampoo to act for 3 to 5 minutes, rinse it thoroughly, then repeat the procedure. Refer to specific product labels for directions, as contact time recommendations may vary. Completely rinse to prevent skin irritation and/or excessive drying.

Precautions/Adverse Effects

The carcinogenic risks associated with coal tar products are debated. At present, most (including the FDA) believe that coal tar products with concentrations of 5% or less are safe for human use. However, should they be used on animals, clients should wear gloves when applying and wash off any product that contacts their skin. Carcinogenic risk assessment for dogs using coal tar products was not located. **Coal tar products should not be used on cats**, patients that have prior sensitivity reactions to tar products, or patients that have dry scaling dermatoses.

Be careful comparing coal tar concentrations on labels. Coal tar solution contains ≈20% coal tar extract or refined tar. For example, a 10% coal tar solution contains ≈2% coal tar (refined).

Photosensitization, skin drying, and skin irritation are possible with coal tar therapy. Adverse effects are more likely with coal tar concentrations greater than 3%. Residual odor is often bothersome to clients. Tar may stain fabrics or haircoats and discolor jewelry.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Coal tar solution 3%, chloroxylenol 0.6%, salicylic acid 2%, sulfur 2% (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 8 oz	*Imrex®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Coal tar solution 2%, sulfur 2% (shampoo) Other ingredients: allantoin, aloe vera extract, citric acid, sodium laureth sulfate	Labeled for dogs **Do NOT use on cats.**	OTC; 4 oz, 8 oz, and 12 oz	*Sulfodene®*
Coal tar solution 5% (equivalent to 1% refined coal tar) (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 12 oz, 1 gallon	*Nova Pearls®* Slow-release power moisturizers encapsulated in *Novasome®* microcarriers
Coal tar solution 2%, menthol 1%, salicylic acid 3% (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 16 oz, 1 gallon	*Pharmasal-T®*
Coal tar solution 1%, salicylic acid 0.22%, sulfur 2% (shampoo) Other ingredients: aloe vera, menthol, zinc oxide	Labeled for dogs **Do NOT use on cats.**	OTC; 12 oz, 1 gallon	

Human-Labeled Products

Several products labeled for human use and containing coal tar—including ointments, lotions, and creams—are available in concentrations ranging from 0.5% to 5% coal tar (not coal tar extract).

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Coal tar 0.5% (shampoo)	OTC; 120 mL, 177 mL, 240 mL, 355 mL, and 480 mL	*DHS Tar®; Polytar®; Tera-Gel®*
Coal tar 1% (shampoo)	OTC; 180 mL, 473 mL	*Ionil T®; PC-Tar®; Zetar®*
Coal tar 2% (shampoo)	OTC; 132 mL, 255 mL, and 480 mL	*Neutrogena® T/Gel Original®*
Coal tar 3% (shampoo)	OTC; 120 mL, 240 mL	*MG 217 Medicated Tar®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Salicylic Acid, Topical

(sal-i-***sil***-ic ***ass***-id)

Indications/Actions

Salicylic acid is used to treat seborrheic disorders with mild to moderate scaling and mild waxy or keratinous debris, including seborrhea sicca and oleosa. Salicylic acid is often combined with sulfur or benzoyl peroxide. In higher concentrations, salicylic acid can be used to remove localized excessive tissues associated with hyperkeratotic disorders, such as calluses and idiopathic thickening of the planum nasale and footpads. Salicylic acid is also used in some wound cleansers.

Salicylic acid has mild antipruritic, antibacterial (bacteriostatic), keratoplastic, and keratolytic actions. Lower concentrations are primarily keratoplastic, and higher concentrations are keratolytic. Salicylic acid lowers skin pH, increases corneocyte hydration, and dissolves the intercellular binder between corneocytes. Salicylic acid and sulfur are thought to be synergistic in their keratolytic actions.

Suggested Dosages/Uses

Salicylic acid shampoos are typically labeled for use 2 to 3 times weekly; however, the frequency will vary according to the patient's needs. Allow medicated shampoos to remain in contact with the skin for 5 to 10 minutes before rinsing well. Alternatively, allow for 3 to 5 minutes of contact time, rinse thoroughly, then repeat the procedure. Ensure shampoo is rinsed completely to prevent skin irritation and excessive drying. Refer to specific product labels for complete directions.

Precautions/Adverse Effects

Salicylic acid products that are combined with coal tar must not be used on cats.

Avoid contact with eyes and mucous membranes. Do not allow animals to lick during bathing in order to avoid ingestion. Avoid using on broken skin unless product is specifically labeled for wounds. Wash hands after application, as skin irritation is possible. Salicylic acid products may bleach colored fabrics.

Possible adverse effects include burning, itching, pain, erythema, and swelling at the application site, particularly when used in concentrations greater than 2%. A rebound seborrheic effect can occur.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Salicylic acid 6.6% (gel)	Labeled for dogs, cats, and horses	OTC; 1 oz	*Solva-Ker®*
Salicylic acid 1.6%, benzocaine 1.5% (ointment)	Labeled for dogs	OTC; 2 oz	*Sulfodene® 3-Way Ointment*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Salicylic acid, benzoic acid, malic acid (cream)	Labeled for dogs and cats	OTC; 1 oz, 14 oz	*Derma-Clens®*
Salicylic acid 2% (shampoo)	Labeled for dogs and cats; some products also labeled for horses	OTC; 8 oz, 16 oz	*BioSeb®; CeraSoothe® SA; KeraSeb®, Ceraven SA®*
Salicylic acid 1%, benzoyl peroxide 2.5% (shampoo)	Labeled for dogs, cats, and horses	OTC; 12 oz, 1 gallon	*DermaBenSs®*
Salicylic acid 2%, benzoyl peroxide 3% (shampoo)	Labeled for dogs, cats, and horses	OTC; 12 oz, 1 gallon	*EZ-Derm®*
Salicylic acid 2%, chloroxylenol 2%, sodium thiosulfate 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 16 oz, 1 gallon	*Universal Medicated Shampoo*
Salicylic acid 2%, sulfur 2% (shampoo)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, 16 oz	*Exfolux®*
Salicylic acid 2%, selenium sulfide 1%, sulfur 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 16 oz	*Siccaderm®*
Salicylic acid 2%, benzoyl peroxide 3%, sulfur 2% (shampoo)	Labeled for dogs and cats	OTC; 12 oz	*Pet MD®*
Salicylic acid 0.22%, sulfur 2%, coal tar solution 1%, zinc oxide 0.22% (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 12 oz, 1 gallon	*Sulfur & Tar®*
Salicylic acid 1%, benzoyl peroxide 2.5% (shampoo)	Labeled for dogs and cats	OTC; 12 oz	*Peroxiderm®*
Salicylic acid 1% (mousse)	Labeled for dogs and cats	OTC; 6.8 oz	*CeraSoothe® SA*
Salicylic acid 2% (mousse)	Labeled for dogs, cats, and horses	OTC; 6.8 oz	*BioSeb®*

Human-Labeled Products

NOTE: In addition to the prescription products listed, there are a wide variety of human-labeled topical salicylic acid products in a range of concentrations available OTC, including creams, ointments, transdermal patches, liquids, and gels. These products are typically labeled for wart removal or treatment of acne. There are also a wide variety of OTC skin cleansers and shampoos containing salicylic acid, often in combination with sulfur, coal tar, or menthol. For more information on these products, refer to a comprehensive human drug reference or contact a pharmacist.

Active Ingredients (formulation)	Label status; size(s)	Trade names/additional information
Salicylic acid 3% (ointment)	Rx; 30 g	*Bensal HP®; Keralyt®*; generic
Salicylic acid 6% (cream)	Rx; 16 oz	*Salex®*
Salicylic acid 6% (lotion)	Rx; 8 oz	*Salex®*
Salicylic acid 6% (foam)	Rx; 7.1 oz	*Salvax®*
Salicylic acid 6% (gel)	Rx; 40 g, 100 g	*Keralyt®*; generic

Sulfur, Precipitated, Topical

(***sul***-fer)

Indications/Actions

Sulfur-containing shampoos are often used to treat patients with seborrheic disorders exhibiting mild to moderate scaling, with mild waxiness and keratinous debris. Sulfur is often found in combination with salicylic acid.

Sulfur has keratoplastic and keratolytic actions. Lower concentrations of sulfur are primarily keratoplastic secondary to assisting the conversion of cysteine to cystine, which is thought to be an important factor in the maturation of corneocytes. Like salicylic acid, the keratolytic effects of sulfur increase with concentration. Salicylic acid and sulfur are believed to be synergistic in their keratolytic actions. Sulfur also has mild degreasing effects and can be mildly antipruritic. Sulfur has antibacterial, antifungal, and antiparasitic effects secondary to sulfur conversion to hydrogen sulfide and pentathionic acid by bacteria and keratocytes.

Suggested Dosages/Uses

Sulfur shampoo label directions range from once daily to once weekly; the frequency of use will vary depending on the patient. Allow medicated shampoos to remain in contact with the skin for 5 to 10 minutes before rinsing well. Alternatively, allow for 3 to 5 minutes of contact time, rinse thoroughly, then repeat the procedure. Ensure shampoo is rinsed completely to prevent skin irritation and excessive drying. Refer to the specific product label for complete instructions.

Precautions/Adverse Effects

Avoid contact with eyes, mucous membranes, and open cuts or sores. Wear gloves when applying and wash hands after application. Residual odor can be bothersome, and sulfur may stain fabrics and hair.

Sulfur can be irritating or drying and can cause pruritus. A rebound seborrheic effect can occur with sulfur-containing products.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Sulfur 2% (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 12 oz	*Sulfodene®*
Sodium thiosulfate 2%, chloroxylenol 2%, salicylic acid 2% (shampoo)	Labeled for dogs, cats, and horses	OTC; 16 oz, 1 gallon	*Universal Medicated Shampoo®* Sodium thiosulfate serves as source of soluble sulfur.
Sulfur 2%, salicylic acid 2%, selenium sulfide 1% (shampoo)	Labeled for dogs, cats, and horses	OTC; 16 oz	*Siccaderm®*
Sulfur 1%, benzoyl peroxide 2.5%, salicylic acid 1% (shampoo)	Labeled for dogs and cats	OTC; 12 oz, 1 gallon	*Sulfur Benz®*
Sulfur 2%, coal tar solution 1%, salicylic acid 0.22%, zinc oxide 0.22% (shampoo)	Labeled for dogs **Do NOT use on cats.**	OTC; 12 oz, 1 gallon	*Sulfur & Tar®*
Sulfur 2%, salicylic acid 2% (shampoo)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, 16 oz	*Exfolux®*

Human-Labeled Products

NOTE: There are several topical products containing sulfur labeled for human use, including creams, lotions, shampoos, and soaps that are typically labeled for acne or dandruff. For more information on these products, refer to a comprehensive human drug reference or contact a pharmacist.

Retinoid Agents, Topical

Tretinoin, Topical

All-Trans Retinoic Acid; Vitamin A Acid

(***tret***-in-oyn)

Indications/Actions

Topical tretinoin has been used anecdotally for treating localized follicular or hyperkeratotic disorders, such as idiopathic nasal and footpad hyperkeratosis, Schnauzer comedo syndrome, callous pyodermas, and chin acne.

The exact mechanism of action of tretinoin is not well understood, but it stimulates cellular mitotic activity, increases cell turnover, and decreases the cohesiveness of follicular epithelial cells. Similar to other retinoids, tretinoin is derived from vitamin A. It **may have anti-inflammatory and immunomodulatory effects by stimulating cytotoxic T-cells and natural killer cells, inhibiting polymorphonuclear cells and suppressing lymphocyte proliferation. Retinoids also have antineoplastic effects by maintaining epithelial cell differentiation and inhibiting tumor cell proliferation.**

Suggested Dosages/Uses

In small animal species, topical tretinoin is often used initially at a concentration of 0.05% and is applied once daily. Treatment is continued until clinical signs are controlled or until treatment is no longer tolerated. Once clinical signs are controlled, tretinoin frequency is then reduced to only as needed. In animals that are unable to tolerate therapy, a lower concentration of 0.025% to 0.01% may be tried to minimize adverse effects.

Precautions/Adverse Effects

Avoid excessive sun exposure during treatment with tretinoin. Avoid contact with eyes, nostrils, inner ears, and mouth. Wear gloves when applying and wash hands after handling. **Do not use this product in pregnant or nursing animals. Do not let animals lick or chew at the treated area(s) for at least 20 to 30 minutes after application. Some gel products contain soluble fish proteins and should be used with caution in patients with fish allergies.**

Adverse effects can include hypersensitivity reactions and local irritation (eg, erythema, dryness, peeling, pruritus).

The National Institute for Occupational Safety and Health (NIOSH) classifies tretinoin as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Veterinary-Labeled Products

None

Human-Labeled Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
Tretinoin 0.02% (cream)	Rx; 20 g, 40 g, 44 g, 60 g	*Renova®*
Tretinoin 0.025% (cream)	Rx; 20 g, 45 g	*Avita®; Retin-A®*; generic
Tretinoin 0.05% (cream)	Rx; 20 g, 40 g	*Refissa®; Retin-A®*; generic
Tretinoin 0.1% (cream)	Rx; 20 g	*Retin-A®*; generic
Tretinoin 0.01% (gel)	Rx; 15 g, 45 g	*Retin-A®*; generic
Tretinoin 0.025% (gel)	Rx; 15 g, 20 g, 45 g	*Avita®; Retin-A®*; generic
Tretinoin 0.05% (gel)	Rx; 45 g	*Atralin®*; generic
Tretinoin 0.04% (gel microspheres)	Rx; 20 g, 45 g, 50 g	*Retin-A Micro®*; generic
Tretinoin 0.06% (gel microspheres)	Rx; 50 g	*Retin-A Micro®*
Tretinoin 0.08% (gel microspheres)	Rx; 50 g	*Retin-A Micro®*
Tretinoin 0.1% (gel microspheres)	Rx; 20 g, 45 g, 50 g	*Retin-A Micro®*; generic
Tretinoin 0.05% (lotion)	Rx; 20 g, 45 g, 50 g	*Altreno®*

Otic Agents, Topical

Although this is not a complete list, the following examples are representative of the types of topical otic preparations that are available to veterinarians. Included are both veterinary-labeled and human-labeled products that may be of use in veterinary medicine. Portions of this section are adapted from Koch SN, Torres SMF, Plumb, DC. *Canine and Feline Dermatology Drug Handbook.* John Wiley & Sons, 2012. This reference provides additional detail on these and other compounds and products.

Antibacterial Agents, Otic

See also ***Corticosteroid/Antimicrobial Agents, Otic***

Indications/Actions

Antibacterial ear preparations are commonly used to treat infections caused by various bacteria, such as *Staphylococcus* spp or *Pseudomonas* spp. Ear cytology should be performed to confirm a bacterial infection prior to prescription of topical antibiotics, mainly to prevent antibiotic overuse and bacterial resistance. In long-term cases and when rods are seen on cytology, culture and susceptibility may be recommended to identify multidrug-resistant organisms, such as *Pseudomonas* spp. Very few otic products that contain solely an antibiotic are commercially available to treat bacterial otitis; therefore, ophthalmic products, compounded products, or injectable products applied topically into the ear canals are sometimes used to treat these infections, mainly when steroids and antifungals are not needed. Listed below are the commercial products containing an antibiotic agent exclusively.

Suggested Dosages/Uses

These products should be used 2 to 3 times daily in adequate amounts to coat the entire ear canal. When using acidifying ear cleansers with an aminoglycoside- or fluoroquinolone-containing agent, it is recommended to use these products about 1 hour apart since low pH can decrease the activity of aminoglycosides and fluoroquinolones. Advise the client to discontinue the medication and contact the veterinarian if the animal's ears become redder or more inflamed at any time during treatment.

Patients should be rechecked before discontinuing treatment, and treatment must be continued until 1 week after negative cytology is obtained.

Precautions/Adverse Effects

There is little information available regarding ototoxicity from topical antibiotics use in dogs and cats; therefore, caution should be used when prescribing various topical otic medications in patients with ruptured tympanic membranes due to potential ototoxicity. Ototoxicity has been reported with aminoglycosides in dogs and cats, and cats may be more sensitive to their potential ototoxicity; however, this concern may be overstated. One study showed no vestibulotoxic or ototoxic effects from 21 days of otic gentamicin applied every 12 hours to ears with ruptured tympanic membranes. Another study showed that instillation of aqueous solutions of tris-EDTA combined with marbofloxacin or gentamicin into the middle ear of dogs with otitis media either did not change or improved the brainstem auditory-evoked responses (BAER); however, tobramycin solution impaired the BAER. Based on this study, the concern for ototoxicity is higher with tobramycin compared with marbofloxacin and gentamicin.[1]

Veterinary- and Human-Labeled Products

Refer to each product's label for inactive ingredients and precautions or contraindications.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Enrofloxacin 0.5%, silver sulfadiazine (SSD) 1%	Labeled for dogs	Rx; 15 mL, 30 mL	*Baytril®*
Gentamicin sulfate 0.3% (solution, ointment)	Labeled for dogs and cats	Rx; 5 mL (solution), 3.5 g (ointment)	*Gentocin®* Extra-label use in the ears. Many human ophthalmic products are also available.
Oxytetracycline hydrochloride 5 mg/g, polymixin B 10,000 units/g (ointment)	Labeled for dogs, cats, and horses	3.5 g tube	*Terramycin®* Extra-label use in the ears
Ciprofloxacin hydrochloride 0.3% (solution, ointment)	Human-labeled product	Rx; 5 mL, 10 mL (solution), 3.5 g (ointment)	*Ciloxan®;* generic Extra-label use in the ears
Ciprofloxacin hydrochloride 0.2% (solution)	Human-labeled product	Rx; 0.25 mL single use	*Cetraxal®*; generic
Ofloxacin 0.3% (solution)	Human-labeled product	Rx; 5 mL, 10 mL	*Floxin®*; generic
Tobramycin 0.3% (solution, ointment)	Human-labeled product	Rx; 5 mL	*Tobrex®;* generic Extra-label use in the ears

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Antibiotic-Potentiating Agents, Otic

Tromethamine-ethylenediaminetetraacetic acid (tris-EDTA) has antimicrobial and antibiotic potentiating activity. Cell surfaces of gram-negative bacteria are damaged when exposed to EDTA. Tromethamine (tris) enhances the effects of EDTA, and the tris-EDTA combination is bacteriostatic.[1] Exposure to tris-EDTA results in increased permeability to extracellular solutes and leakage of intracellular solutes. Cell surfaces of gram-positive bacteria are more resistant to tris-EDTA than gram-negative bacteria.[2] Tris-EDTA has also shown synergistic effects with various antibiotics, such as aminoglycosides and fluoroquinolones in vitro.[1–4] Tris-EDTA also has antibiofilm activity through inhibition of bacterial adhesion.[5]

Tris-EDTA is non-ototoxic and safe to use in the middle ear. Tris-EDTA–containing products are most effective when applied 15 to 30 minutes before the topical antibiotic. Products containing tris-EDTA are available as a sole ingredient or combined with chlorhexidine or ketoconazole. When recommending ear flushing as part of the treatment regimen, it is important to demonstrate the cleaning technique in the examination room and advise the client to discontinue application and contact the veterinarian if the ears become redder or more inflamed at any time during the course of cleaning. The frequency of cleaning varies according to the needs of the individual patient. ***Refer to each product's label for inactive ingredients and precautions or contraindications.***

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Tris-EDTA (flush)	Labeled for dogs, cats, and horses	4 oz, 16 oz	*TrizEDTA®*
Tris-EDTA (crystals flush)	Labeled for dogs and cats	4 oz, 16 oz	*TrizEDTA crystals®*
Tris-EDTA, chlorhexidine 0.15% (flush)	Labeled for dogs and cats	4 oz	*TrizCHLOR®*
Tris-EDTA, ketoconazole 0.15% (flush)	Labeled for dogs, cats, and horses	4 oz, 12 oz	*TrizULTRA + Keto®*
Tris-EDTA, chlorhexidine 0.15%, ketoconazole 0.15% (flush)	Labeled for dogs and cats	4 oz, 12 oz	*Mal-A-Ket Plus TrizEDTA®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Antifungal Agents, Otic

See also ***Corticosteroid/Antimicrobial Agents, Otic***

Indications/Actions

Antifungal ear preparations are usually used to treat otitis caused by *Malassezia* spp and rarely other otic fungal diseases, such as aspergillosis and candidiasis. Most commercially available products also contain an antibiotic and/or a glucocorticoid. Therefore, ear cytology should be performed to confirm a fungal/yeast infection prior to prescription of topical antifungals, particularly if an antimicrobial and antibiotic combination is being considered.

Topical azole antifungals, such as miconazole and clotrimazole, have been shown to possess in vitro activity against *Staphylococcus* spp. Listed in the ***Veterinary-Labeled Products*** section are exclusively the products without antibiotic or glucocorticoid agents. These products are not labeled as otic medications but have commonly been used in veterinary patients.

Suggested Dosages/Uses

Most products are labeled to be used 1 to 2 times daily in adequate amounts to coat the patient's entire ear canal.

Precautions/Adverse Effects

Advise owners to discontinue the medication and contact their veterinarian if their animal's ears become redder or more inflamed at any time during treatment. There is little information available regarding ototoxicity from topical antifungal use in dogs and cats; therefore, caution should be used when prescribing various topical otic medications in animals with ruptured tympanic membranes. In one study in dogs, administration of clotrimazole (15 g tube [2% cream] mixed 50:50 with sterile water) did not change the BAER score, and vestibular signs were not reported.[1] No other safety studies in dogs or cats with otic antifungals are available. Patients should be rechecked before discontinuing treatment, and treatment should be continued until 1 week after negative cytology is obtained.

Refer to each product's label for inactive ingredients and precautions or contraindications. Some inactive ingredients, including alcohol or alcohol-based solvents such as propylene glycol, can be irritating and cause inflammation, particularly in sensitive, eroded, or ulcerated ears, and these ingredients should be avoided in ears with ruptured tympanum.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Clotrimazole 1% (solution)	Labeled for dogs and cats	Rx; 10 mL, 30 mL, and 60 mL	*ClotrimaTop®; Clotrimazole®*
Miconazole nitrate 1.15% (equivalent to 1% miconazole base by weight) (lotion)	Labeled for dogs and cats	Rx; 60 mL, 120 mL	*Conzol®; MicaVed®; Miconosol®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Antiparasitic Agents, Otic

NOTE: This monograph includes only preparations labeled to be applied directly into the ear canals for the treatment of otoacariosis; however, parasiticides with systemic or more generalized effect are preferred because *Otodectes cynotis* mites are known to also live outside ear canals and can re-infest the ears. See also ***Afoxolaner; Fipronil/Fipronil Combinations, Topical; Fluralaner; Sarolaner; Selamectin***. Refer to specific product labels for complete instructions regarding use, inactive ingredients, precautions, and contraindications.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Piperonyl butoxide 0.5%, pyrethrins 0.05%	Labeled for dogs and cats 12 weeks of age or older	OTC; 3 mL, 9 mL, 0.5 oz, 1 oz	*Adams®; Happy Jack Mitex®; Hartz UltraGuard®* Clean ears. Apply daily for 7 to 10 days. Repeat in 2 weeks if necessary
Piperonyl butoxide 0.6%, pyrethrins 0.06%	For use on dogs and cats 12 weeks of age or older	OTC; 1 oz, 3 oz, 4 oz	*No-Bite®; PetArmor; Sentry HC EARMITE free®; Sergeant's Vetscription MITEaway®* Clean ears. Apply twice daily until ticks/mites are eliminated
Piperonyl butoxide 1%, pyrethrins 0.15%	Labeled for dogs; some products labeled for cats and horses NOT for use on puppies or kittens under 12 weeks old OR for use on horses/foals intended for slaughter	OTC; 4 oz, 6 oz	*Ear-Rite®* (dogs)*; Ear Mite®* (dogs, cats, horses) Clean ears. Apply daily for 7 to 10 days. Do not repeat treatment for 3 to 5 days if used as a repellent. Horses are treated daily for 2 to 3 days, then every 3 to 5 days as needed for controlling ectoparasites (eg, ticks, gnats).
Piperonyl butoxide 1.05%, pyrethrins 0.1%	For use on dogs 12 weeks of age or older	OTC; 0.75 oz	*Ear Mite Remedy* Clean ears. Apply once. If needed, repeat every other day until the infestation is resolved.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Piperonyl butoxide 1.5%, pyrethrins 0.15%	Species depends on product No products available for use on dogs, cats, or rabbits under 12 weeks of age	OTC; 0.5 oz, 1 oz	*Ear-Rite*® (cats and kittens); *Eradimite*® (dogs and cats); *Otomite*® *Plus* (dogs, cats, puppies, kittens) Clean ear. Apply as a single dose. May repeat every 2 days until the condition has cleared up or repeat in 7 days depending on product directions. As a preventative treatment, apply every 15 days
Piperonyl butoxide 1%, pyrethrins 0.15% (lotion)	For use on dogs and cats at least 12 weeks old	OTC; 6 oz	*Performer*® Clean ears. Apply for 7 to 10 days
Piperonyl butoxide 1.5%, pyrethrins 0.15% (lotion)	For use on dogs and cats over 12 weeks of age	OTC; 12 mL, 22 mL	*Mita-Clear*® Clean ear; instill enough to wet ear canal and massage. Do not reapply for 7 days
Milbemycin oxime 0.1% (solution)	For use on cats or kittens 4 weeks of age or older	Rx; box of 10 pouches of 2 tubes of 0.25 mL each	*MilbeMite*® Clean ear. Apply entire contents of tube in the external ear canal; one tube per ear. Repeat in 30 days, if recommended by the veterinarian Safe use in cats used for breeding purposes, pregnant, or lactating has not been evaluated.
Ivermectin 0.01% (suspension)	For use in cats and kittens 4 weeks of age or older	Rx; box of 12 foil pouches of 2 ampules of 0.5mL each	*Acarexx*® Apply 0.5 mL in each ear. Repeat 1 time if necessary based on ear mite life cycle and response to treatment
Neomycin sulfate 0.25%, thiabendazole 4%, dexamethasone 0.1%	Labeled for otoacariasis and bacterial and fungal otitis in dogs and cats for a maximum of 1 week duration	OTC; 7.5 mL, 15 mL	*Tresaderm*® Clean ears. Apply twice daily until the infestation is resolved

Antiseptic Cleaning Agents, Otic

See also ***Cleaning/Drying Agents, Otic***

Indications/Actions

Topical antiseptic ear flushes include acetic acid, salicylic acid, boric acid, chlorhexidine, ketoconazole, miconazole, hypochlorous acid, and microsilver-containing products. These products are typically used as adjunctive therapy for ear infections (eg, bacterial and/or yeast) but can also be used as sole treatment in mild, first-time infections and for maintenance in some cases. Acetic and boric acids work as an acidifying, drying, and antimicrobial agent. Chlorhexidine has activity against gram-positive and gram-negative bacteria and fungi. Chloroxylenol is mostly active against bacteria. Ketoconazole and miconazole are fungistatic antifungals, although miconazole has been shown to be fungicidal against *Candida* spp. Hypochlorous acid is a broad-spectrum antimicrobial with rapid activity against gram-positive and gram-negative bacteria and fungal/yeast organisms. Its mode of action against micro-organisms mimics that of neutrophils in the body. The main route of attack against microorganisms is disruption of the cellular membrane. Microsilver has both antibiofilm and antibacterial properties, generating silver ions that kill gram-positive and gram-negative bacteria, such as *Pseudomonas aeruginosa*, *Staphylococcus* spp, *Streptococcus* spp, including multidrug-resistant organisms, in addition to *Malassezia* spp.

Suggested Dosages/Uses

When recommending ear flushing as part of the treatment regimen, it is important to demonstrate the cleaning technique in the examination room and advise the client to discontinue application and contact the veterinarian if the ears become redder or more inflamed at any time during the course of cleaning.

The frequency of cleaning varies according to the needs of the individual patient. Some of these products may also be used as antiseptics for the skin. Refer to each product label for inactive ingredients, details on use, and any precautions.

Precautions/Adverse Effects

There is little information regarding the ototoxicity of ear cleaners in dogs and cats. Agents that are believed to be safe in dogs and cats for ear flushing in the presence of ruptured tympanic membrane include sterile water, sterile saline, tris-EDTA, and possibly chlorhexidine at a concentration of less than 0.05%. More concentrated chlorhexidine-containing products should be used cautiously in ear canals with ruptured tympanic membranes because of the potential for ototoxicity, particularly in cats.

Alcohol-containing products should be avoided if the status of the tympanic membrane cannot be determined or if it is ruptured. Avoid products containing alcohol for eroded, ulcerative, and/or painful and sensitive ears as that alcohol can be irritant.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Acetic acid 2%, boric acid 2% (cleanser)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, and 16 oz	*MalAcetic®* Other ingredients: water, glycerin, polysorbate 20, fragrance, sodium hydroxide
Acetic acid 2% (solution)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*OtoCetic®* Other ingredients: water, witch hazel, benzyl alcohol, polysorbate 80, DMDM hydantoin (as preservative), fragrance
Acetic acid 1%, ketoconazole 0.15%, hydrocortisone 1% (cleanser)	Labeled for dogs, cats, and horses	OTC; 2 oz, 8 oz	*MalAcetic® Ultra* Other ingredients: ceramide complex
Acetic acid 1%, boric acid 2%, hydrocortisone 1%, ketoconazole 0.15% (flush)	Labeled for dogs and cats	OTC; 8 oz	*Oticetic®* Other ingredients: propylene glycol, ceramides, glycerin, polysorbate, triethanolamine, fragrance
Chlorhexidine 0.15%, ketoconazole 0.15%, tris-EDTA (flush)	Labeled for dogs and cats	OTC; 4 oz, 12 oz	*Mal-A-Ket Plus TrizEDTA®* Other ingredients: decolorized aloe vera, glycerine, glycolic acid, methylisothiazolinone, nonoxynol-9, tris-EDTA, water
Chloroxylenol 0.2%, ketoconazole 0.2% (flush)	Labeled for dogs, cats, and horses	OTC; 4 oz, 8oz	*Pharmaseb®*
Chloroxylenol 0.33%, ketoconazole 0.33% (flush)	Labeled for dogs, cats, and horses	OTC; 8 oz	*Ketomax®* Other ingredients: benzyl alcohol, fragrance, propylene glycol, purified water, sodium borate decahydrate
Hypochlorous acid (HOCl) 0.009% (rinse)	Safe for use on all animals	OTC; 3 oz	*Vetericyn Plus® Antimicrobial Ear Rinse* Other ingredients: electrolyzed water (H_2O), sodium chloride (NaCl), sodium hypochlorite (NaOCl)
Ketoconazole 0.1% (flush)	Labeled for dogs and cats	OTC; 12 oz	*Curaseb®* Other ingredients: purified water, SDA-40, nonoxynol-9, benzyl alcohol, disodium EDTA, tromethamine technical grade, fragrance
Ketoconazole 0.1%, phytosphingosine HCl 0.01% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*VetraSeb® TRIS* Other ingredients: water, SDA 40B 190 proof, nonoxynol-9, tromethamine, disodium EDTA, methylisothiazolinone
Miconazole 0.1% (flush)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*BioTris®* Other ingredients: disodium EDTA, methylchloroisothiazolinone, methylisothiazolinone, N-octadecanoylphytosphingosine (ceramide Ill), PEG-40 hydrogenated castor oil, propylene glycol, purified water, tromethamine (Tris USP)
Microsilver 0.5%. climbazole 0.5% (concentrate)	Labeled for dogs and cats	OTC; 6 mL	*UltraOtic® Concentrate* Other ingredients: avicel RC-591, benzyl alcohol, glycerin, N-octadecanoylphytosphingosine (ceramide Ill), phenoxyethanol, polysorbate 80, purified water, sodium benzoate, titanium dioxide, triton x100/OP10, xanthan gum

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Microsilver 0.2%, lactic acid 1% (rinse)	Labeled for dogs and cats	OTC; 4 oz	*UltraOtic®* Antiseptic/antibiofilm activity. May be used prior to *UltraOtic® Concentrate* for additional benefit Other ingredients: carboxymethyl cellulose, fragrance, glycerin, magnesium aluminum silicate, methylchloroisothiazolinone, methylisothiazolinone, octoxynol-10, purified water
Salicylic acid 0.15%, ketoconazole 0.1% (ear rinse)	Labeled for dogs and cats	OTC; 4 oz	*PetArmor® Medicated Ear Rinse* Other ingredients: glycerin, potassium sorbate, propyelene glycol, purified water, sodium benzoate

Cerumenolytic Agents, Otic

Indications/Actions

Cerumenolytic agents soften hardened cerumen and make it easier to remove. Cerumenolytic agents can be classified as water-based agents, oil-based agents, or agents that are neither water nor oil based. Docusate sodium (DSS) and triethanolamine polypeptide oleate are water-based agents that act as surfactants to emulsify and disperse cerumen. Oil-based agents, such as propylene glycol, squalane, and various oils, soften and loosen cerumen without disintegrating it.[1] Squalane is the most potent of these agents. Some agents are neither water nor oil based, such as carbamide peroxide (urea-hydrogen peroxide) and glycerin. Carbamide peroxide is a foaming agent that releases oxygen to help break up debris. Glycerin acts by softening and loosening cerumen, similar to oil-based agents.

Docusate sodium, calcium sulfosuccinate (docusate calcium), and carbamide peroxide are considered potent cerumenolytic agents. Squalane and triethanolamine polypeptide oleate are less potent agents. Propylene glycol, glycerin, and oils are considered very mild cerumenolytic agents.

Evidence in humans suggests that no cerumenolytic agent is superior to any other, but any type of cerumenolytic agent tends to be superior to no treatment.[2]

Suggested Dosages/Use

These agents work best when they are applied 10 to 15 minutes prior to cleaning an area with an astringent or drying cleaner. Demonstrate proper cleaning technique in the examination room if cerumenolytic products will be administered at home by an owner. Advise owners to discontinue product application and contact their veterinarian if their animals' ears become redder or more inflamed at any time during the course of cleaning.

Precautions/Adverse Effects

Cerumenolytic agents should not be used with ruptured tympanic membranes (especially in cats) because of the potential for ototoxicity, with the exception of squalane.[3] If cerumenolytic agents are needed during in-hospital ear-flushing procedures, use of these products should be followed by multiple flushes with warm sterile isotonic saline. Alcohol-containing products should be avoided if the status of the tympanic membrane cannot be determined, or if the tympanic membrane is ruptured. Avoid products containing alcohol for eroded, ulcerative, and/or painful and sensitive ears, as the alcohol can be an irritant. The frequency of cleaning varies according to the needs of the individual animal. Some of these products may also be used as antiseptics for the skin. Refer to each product label for details on use and any precautions.

Veterinary-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Squalane 25% in isopropyl myristate and light mineral oil	Labeled for dogs and cats	OTC; 4 oz	*Cerumene®*
Phytosphingosine 0.2% in alcohol, water, and propylene glycol	Labeled for dogs and cats	OTC; 4.2 oz, 8.4 oz	*Douxo® Micellar*
Carbamide peroxide 6.5%, glycerin base	Labeled for dogs and cats	OTC; 4 oz	*Earoxide®*
Propylene glycol, salicylic acid, glycerin	Labeled for dogs and cats	OTC; 8 oz, 12 oz	*EpiKlean®*
Squalane 22%, isopropyl myristate, white mineral oil	Labeled for dogs and cats	OTC; 4 oz	*KlearOtic®*
Methylchloroisothiazolinone, methylisothiazolinone, micellar solution (cetrimonium bromide, disodium EDTA, PEG-6 caprylic/capric glycerides, polysorbate 80, water), N-octadecanoyl-phytosphingosine (ceramide III)	Labeled for dogs and cats	OTC; 8 oz	*Milytic®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Calendula, capryloyl glycine (*Lipacide®* C8G), glycerin, isopropyl alcohol, labrasol, lemon synthetic perfume, polysorbate 80, transcutol V, tromethamine, undecylenoyl glycine (*Lipacide®* UG), water	Labeled for dogs and cats	OTC; 4 oz	*pH•notix®*
Phytosphingosine hydrochloride 0.01%, salicylic acid 0.15%	Labeled for dogs and cats	OTC; 8 oz	*SkinGuard® Cleanse*
Denatured alcohol 10%, lactic acid 1.76%, propylene glycol, docusate sodium, salicylic acid	Labeled for dogs and cats	OTC; 4 oz, 8 oz, and 16 oz	*Vet Solutions®*

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Cleaning/Drying Agents, Otic

Indications/Actions

Cleaning and drying agents are typically used after debris or exudate has been removed from the ear canals with a cerumenolytic agent; however, they may also be used as a sole ear cleanser. The primary ingredients are astringents, such as isopropyl alcohol or aluminum acetate, and acids, such as boric acid or salicylic acid. Some cleaning and drying agents may have antimicrobial effects, and some may also be used as antiseptics for the skin.

Suggested Dosages/Use

The frequency of cleaning varies according to the needs of the individual patient. Cleaning and drying ear solutions can be used on a maintenance regimen to help prevent ear infections and after bathing or swimming to keep the external ear canals dry. It is important to demonstrate the cleaning technique in the examination room.

Precautions/Adverse Effects

Advise the client to discontinue the application and contact the veterinarian if the ears become redder or more inflamed at any time during the course of cleaning.

There is little information regarding the ototoxicity of ear cleaners in dogs and cats. Agents that are believed to be safe in dogs and cats for ear flushing in the presence of ruptured tympanic membrane include sterile water, sterile saline, tris-EDTA, and possibly chlorhexidine at a concentration of less than 0.05%. Propylene glycol- and alcohol-containing products should be avoided if the status of the tympanic membrane cannot be determined or if it is ruptured. Avoid products containing alcohol for eroded, ulcerative, and/or painful and sensitive ears as that alcohol can be irritant.

Veterinary-Labeled Products

Refer to each product label for specific details on use, inactive ingredients, and any precautions.

There is an overwhelming number of these products marketed, and the following are representative; they do not appear to be FDA-approved products. Many products also contain cerumenolytic ingredients. See also ***Cerumenolytic Otic Agents***.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Isopropyl alcohol, glycerin (lotion)	Labeled for dogs and cats	OTC; 4 oz, 6 oz, 8 oz, 12 oz, 32 oz, 1 gallon	*ADL® Ear Flushing Drying Lotion*
Dioctyl sodium sulfosuccinate, glycerin, lactic acid, salicylic acid, SD alcohol (solution)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, 1 gallon	*Aloe vera otic cleanser (formerly AloeClens®); Aurocin®; Clean Ear® with Aloe Vera; Conquer® Hy-Otic*
Lactic acid 1.76%, salicylic acid 0.1%, SD alcohol 10% (solution)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, 16 oz, and 1 gallon	*Ear Cleansing Solution*
Acetic acid, boric acid, ethanol (flush)	Labeled for dogs and cats	OTC; 12 oz, 1 gallon	*EarMed® Boracetic®*
Ethanol (solution)	Labeled for dogs and cats	OTC; 12 oz, 1 gallon	*EarMed® Cleansing Solution and Wash*
Ethanol (wipes)	Labeled for dogs and cats	OTC; 40 wipes 5 x 6, 50 wipes 5 x 8.5, and 160 wipes 6 x 7	*EarMed®*
Propylene glycol 3%, salicylic acid 0.05%, SD alcohol (cleanser)	Labeled for dogs and cats	OTC; 8 oz, 12 oz, 32 oz	*EpiKlean®*
Dioctyl sodium sulfosuccinate, chloroxylenol 0.1%, EDTA 0.5%, salicylic acid 0.2% (solution)	Labeled for dogs and cats	OTC; 4 oz, 8 oz	*Epi-Otic® Advanced*
Eucalyptus oil, malic acid, salicylic acid (solution)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*Eucalyptus Oil (formerly Euclens®)*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Acetic acid 2%, boric acid 2% (cleanser)	Labeled for dogs and cats	OTC; 4 oz, 8 oz, and 16 oz	*MalAcetic®*
Acetic acid 1%, hydrocortisone 1%, ketoconazole 0.15% (cleanser)	Labeled for dogs, cats, and horses	OTC; 2 oz, 8 oz	*MalAcetic ULTRA®*
Boric acid, zinc gluconate (liquid)	Recommended for dogs, cats, and exotics	OTC; 2 oz, 4 oz	*Maxi/Guard® zn 4.5*
Isopropyl alcohol (solution)	Labeled for dogs and cats	OTC; 4 oz, 16 oz	*Nolvasan®*
Malic acid, salicylic acid (solution)	Labeled for dogs and cats	OTC; 4 oz	*Oti-Clens®*
Dioctyl sodium sulfosuccinate, lactic acid, propylene glycol, salicylic acid, SD alcohol (solution)	Labeled for dogs and cats	OTC; 8 oz	*Otiderm®*
Boric acid 2%, glycolic acid 2% (solution)	Labeled for dogs and cats	OTC; 16 oz	*Otiderm® Advanced*
Glycerin, dioctyl sodium sulfosuccinate, lactic acid, salicylic acid (solution)	Labeled for dogs and cats	OTC; 8 oz	*OtiRinse®*
Acetic acid, alcohol (cleanser)	Labeled for dogs and cats	OTC; 4 oz	*Sulfodene®*
Chloroxylenol, SD alcohol (gel)	Labeled for dogs	OTC; 4 oz	*Swimmer's Ear Astringent*
Glycerin, lactic acid 1%, microsilver 0.2% (rinse)	Labeled for dogs and cats	OTC; 4 oz	*UltraOtic®*
Hypochlorous acid (HOCl) 0.009% (rinse)	Safe for all animals	OTC; 3 oz	*Vetericyn Plus®*
Glycerin, lactic acid 2.7%, salicylic acid 0.15%, sodium dioctyl sulfosuccinate, SD alcohol (solution)	Labeled for dogs and cats	OTC; 8 oz, 16 oz	*VetOtic®*
Coco-glucoside, glucose oxidase, glycerin, lactoferrin, lactoperoxidase, lysozyme, zinc gluconate (cleanser)	Labeled for dogs and cats	OTC; 4 oz, 1 gallon	*Zymox®* Ear Cleanser

Corticosteroid Agents, Otic

*See also **Corticosteroid/Antimicrobial Agents, Otic***

Indications/Actions

Corticosteroid-containing ear medications are used in cases of acute or chronic otitis with the goal of reducing inflammation, edema, tissue hyperplasia, pain, and pruritus. In addition, they are helpful in decreasing secretions from sebaceous and apocrine glands, thereby reducing the buildup of debris in ear canals. Otic solutions that contain glucocorticoids without an antibiotic can be used as maintenance therapy, if needed, in cases of allergic otitis.

Suggested Dosages/Uses

The frequency of applications should be according to the patients' needs, usually 1 to 4 times daily.

Precautions/Adverse Effects

When using corticosteroid preparations as maintenance therapy, use the least potent glucocorticoid at the lowest possible frequency to balance efficacy and prevent undesirable adverse effects, including skin atrophy, alopecia, and adrenocortical suppression effects from systemic absorption. Some glucocorticoids, such as dexamethasone and fluocinolone, are believed to be safe for use in the middle ear; however, the combination of fluocinolone and dimethyl sulfoxide (DMSO; *Synotic®*) may not be safe for administration in the ear if the tympanic membrane is ruptured as DMSO may cause ototoxicity.[1]

Veterinary-Labeled Products

Refer to each product's label for inactive ingredients and precautions or contraindications. Some inactive ingredients, including alcohol or alcohol-based solvents such as propylene glycol, can be irritating and cause inflammation, particularly in sensitive, eroded, or ulcerated ears, and should be avoided in ears with ruptured tympanum.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Fluocinolone 0.01%, DMSO 60% (solution)	Labeled for dogs	Rx; 8 mL, 60 mL	*Synotic®*

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Hydrocortisone 0.5% (solution)	Labeled for dogs and cats	OTC; 1.25 oz	*Zymox*®
Hydrocortisone 1% (solution)	Labeled for dogs and cats	OTC; 1.25 oz	*Zymox*®, *Zymox*® *Plus*
Hydrocortisone 1%, Burow's solution 2% (solution)	Labeled for dogs	OTC; 1 oz, 16 oz	*Cort/Astrin*®

Human-Labeled Products

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Fluocinolone acetonide 0.01% (oil)	Labeled for humans	Rx; 20 mL	*DermOtic*®; generic

References

For the complete list of references, see **wiley.com/go/budde/plumb**

Corticosteroid/Antimicrobial Agents, Otic

See also ***Corticosteroid Agents, Otic***

Indications/Actions

Most otic antimicrobial preparations for veterinary use that are commercially available combine an antifungal, an antibiotic, and a glucocorticoid agent. These medications should be used when anti-inflammatory and/or anti-pruritic effects are needed in the presence of concurrent bacteria and yeast infections. Ear cytology should be performed to confirm an infection prior to prescription of ear medications containing antimicrobials, particularly antibiotics, to help prevent antibiotic overuse and bacterial resistance. In chronic cases or when rods are seen on cytology, culture and susceptibility may be recommended to identify multidrug-resistant organisms, such as *Pseudomonas* spp. If only bacteria or yeast are present, more targeted otic antimicrobials, with or without glucocorticoids, may be considered.

Suggested Dosages/Uses

Most of these products should be used 2 to 3 times daily in adequate amounts to coat the entire ear canal. Refer to each product's label for detailed usage instructions. Patients should be rechecked before discontinuing treatment, and treatment must be continued until 1 week after negative cytology is obtained.

Precautions/Adverse Effects

Advise the client to discontinue the medication and contact the veterinarian if the ears become redder or more inflamed at any time during treatment and to avoid contact with the animal's mouth and eyes.

Use with caution when prescribing various topical otic medications in patients with ruptured tympanic membranes due to potential ototoxicity. Cats may be more sensitive to irritation and ototoxicity from topical otic agents.[1] Use of aminoglycosides in the presence of a ruptured tympanic membrane is controversial. One study showed no vestibulotoxic or ototoxic effects from 21 days of otic gentamicin applied every 12 hours to ears with ruptured tympanic membranes. Another study measured brainstem auditory-evoked responses (BAER) and found that although tobramycin significantly impaired hearing, gentamicin did not[2]; however, ototoxicity from topical aminoglycoside administration has been shown in many chinchilla and guinea pig studies.[3–6] In dogs, administration of clotrimazole 2% cream mixed 50:50 with sterile water did not change the BAER score, and vestibular signs were not reported. Some glucocorticoids, such as dexamethasone and fluocinolone, are believed to be safe for use in the middle ear. One dog developed pemphigus vulgaris after 1 week of otic polymyxin B application.[7]

Veterinarians and owners must use caution when handling these medications to prevent potential accidental human exposure, especially direct skin and eye contact. *Claro*® and *Osurnia*® contain specific warnings regarding ocular injuries in humans, including corneal ulcers.[8,9] Eye protection is recommended during administration, and dogs should be restrained to minimize postapplication head shaking.

Veterinary- and Human-Labeled Products

Refer to each product's label for inactive ingredients and precautions or contraindications. Some inactive ingredients, including alcohol or alcohol-based solvents, such as propylene glycol, can be irritating and cause inflammation, particularly in sensitive, eroded, or ulcerated ears, and should be avoided in ears with ruptured tympanum.

Active ingredients (formulation)	Species	Label status; size(s)	Trade names/additional information
Florfenicol 16.6 mg/mL, terbinafine 14.8 mg/mL, mometasone furoate 2.2 mg/mL (solution)	Labeled for dogs	Rx; 1 mL single-dose dropperette	*Claro*® 1 tube per ear; 30-day duration of effect. Shake before use. To be administered in the clinic.
Gentamicin sulfate 0.3%, clotrimazole 1%, betamethasone valerate 0.1% (ointment)	Labeled for dogs	Rx; 7.5 g, 10 g, 15 g, 25 g, 30 g, and 215 g	*Gentizol*®, *MalOtic*®, *Otibiotic*®, *Otomax*®